

substance use. Prolonged use of alcohol or other substances may result in a substance-induced neurocognitive disorder that may be associated with impaired cognitive function. However, in this context the protracted history of substance use and the persistent deficits associated with the neurocognitive disorder would serve to distinguish it from dissociative amnesia, where there is typically no evidence of persistent impairment in intellectual functioning.

Posttraumatic amnesia due to brain injury. Amnesia may occur in the context of a TBI when there has been an impact to the head or other mechanisms of rapid movement or displacement of the brain within the skull. Other characteristics of TBI include loss of consciousness, disorientation and confusion, or, in more severe cases, neurological signs and symptoms (e.g., abnormalities on neuroimaging, a new onset of seizures or a marked worsening of a preexisting seizure disorder, visual field cuts, anosmia). A neurocognitive disorder attributable to TBI must present either immediately after brain injury occurs or immediately after the individual recovers consciousness after the injury, and persist past the

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acute postinjury period. The cognitive presentation of a neurocognitive disorder following TBI is variable and includes difficulties in the domains of complex attention, executive function, learning and memory, as well as slowed speed of information processing and disturbances in social cognition. The patterns of memory deficits are typical of neurocognitive disorders. Mild TBI may precede acute dissociative amnesia presentations, but the dissociative memory deficits are out of proportion to the TBI head trauma and typically follow the dissociative, not the neurocognitive, patterns.

Seizure disorders. Individuals with seizure disorders may exhibit complex behavior during seizures or postictally with subsequent amnesia. Some individuals with a seizure disorder engage in nonpurposive wandering that is limited to the period of seizure activity. Conversely, behavior during a dissociative fugue is usually purposeful, complex, and goal-directed and may last for days, weeks, or longer. Occasionally, individuals with a seizure disorder will report that some autobiographical memories have been “wiped out” as the seizure disorder progresses. Such memory loss is not associated with psychological trauma or adversities and appears to occur randomly. In seizure disorders, serial electroencephalograms usually show abnormalities. Telemetric electroencephalographic monitoring generally shows an association between the episodes of amnesia and seizure activity. Dissociative and epileptic amnesias may coexist.

Memory deficits associated with electroconvulsive therapy. Memory deficits after electroconvulsive therapy (ECT) most commonly occur for the day of ECT administration. More extensive retrograde and even anterograde amnesia after ECT is usually unrelated to stressful or traumatic life epochs, and generally remits after the ECT series concludes. ECT in severely depressed individuals with dissociative disorders does not worsen dissociation, and memory access may improve as depression remits.

Catatonic stupor. Mutism in catatonic stupor may suggest dissociative amnesia, but failure of recall is usually absent. Other catatonic symptoms (e.g., rigidity, posturing, negativism) are usually present. Catatonic symptoms in children can be associated with trauma, abuse, and/or deprivation. Unlike in dissociative amnesia, the pattern of memory loss in catatonia is only for the catatonic episode.

Acute dissociative reactions to stressful events (other specified dissociative disorder). The acute dissociative reactions to stressful events example of other specified dissociative disorder is characterized by a combination of dissociative symptoms that occur together acutely in response to stressful events and typically last less than 1 month. Amnestic episodes that occur as part of these reactions are accompanied by other prominent dissociative symptoms, have a short duration (hours or days), and tend to be circumscribed to limited periods or events in a person's life (micro-amnesias).

Factitious disorder and malingering. There is no test, battery of tests, or set of procedures that invariably distinguishes dissociative amnesia from feigned amnesia. Feigned amnesia is more common in individuals with 1) acute, florid dissociative amnesia; 2) financial, sexual, or legal problems; 3) a wish to escape stressful circumstances; 4) a desire to seem to be a more interesting patient; and/or 5) a plan to engage in litigation for "recovered memories." However, dissociative amnesia can be associated with those same circumstances and can coexist with deliberate feigning. Many individuals who malingering amnesia confess spontaneously or when confronted.

Memory changes with aging or mild neurocognitive disorder. Memory decrements in mild neurocognitive disorder differ from those of dissociative amnesia; in mild neurocognitive disorder, memory changes manifest as difficulty in learning and retaining new information. This is often measured in tests of verbal learning of word lists or a brief story with evaluation of immediate and delayed recall. With normal cognitive aging, individuals may also have similar weaknesses in immediate and delayed recall of new information,

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although normal aging may also affect information processing speed and other complex executive function tasks in addition to memory.

Comorbidity

As is common in individuals with a history of trauma, many comorbidities co-occur with dissociative amnesia, particularly as dissociative amnesia begins to remit. A wide variety of affective phenomena may surface, including dysphoria, grief, rage, shame, guilt, and psychological conflict and turmoil. Individuals may engage in nonsuicidal self-injury and other high-risk behaviors. These individuals may have symptoms that meet diagnostic criteria for persistent depressive disorder, major depressive disorder, or subthreshold depression (other specified depressive disorder). Many individuals with dissociative amnesia develop PTSD at some point during their life, especially when the traumatic antecedents of their amnesia are brought into conscious awareness. Many of these individuals may show symptoms of the dissociative subtype of PTSD. Many individuals with dissociative amnesia have symptoms that meet diagnostic criteria for a comorbid somatic symptom and related disorder (and vice versa), particularly functional neurological symptom disorder (conversion disorder). Substance-related and addictive disorders may be comorbid with dissociative amnesia, as well as feeding and eating disorders and sexual dysfunctions. The most common comorbid personality disorder is other specified personality disorder (with mixed personality disorder features), which often includes avoidant, obsessive-compulsive, dependent, and borderline features.

Depersonalization/Derealization Disorder

Diagnostic Criteria

F48.1

- A. The presence of persistent or recurrent experiences of depersonalization, derealization, or both:
 - 1. **Depersonalization:** Experiences of unreality, detachment, or being an outside observer with respect to one's thoughts, feelings, sensations, body, or actions (e.g., perceptual alterations, distorted sense of time, unreal or absent self, emotional and/or physical numbing).
 - 2. **Derealization:** Experiences of unreality or detachment with respect to surroundings (e.g., individuals or objects are experienced as unreal, dreamlike, foggy, lifeless, or visually distorted).
- B. During the depersonalization or derealization experiences, reality testing remains intact.
- C. The symptoms cause clinically significant distress or impairment in social, occupational, or other important areas of functioning.
- D. The disturbance is not attributable to the physiological effects of a substance (e.g., a drug of abuse, medication) or another medical condition (e.g., seizures).
- E. The disturbance is not better explained by another mental disorder, such as schizophrenia, panic disorder, major depressive disorder, acute stress disorder, posttraumatic stress disorder, or another dissociative disorder.

Diagnostic Features

The essential features of depersonalization/derealization disorder are persistent or recurrent episodes of depersonalization, derealization, or both. Episodes of depersonalization are characterized by a feeling of unreality or detachment from, or unfamiliarity with, the individual's whole self or from aspects of the self (Criterion A1). The individual may feel detached from his or her entire being (e.g., "I am no one," "I have no self"). He or she may

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also feel subjectively detached from aspects of the self, including feelings (e.g., hypoemotionality: "I know I have feelings, but I don't feel them"), thoughts (e.g., "My thoughts don't feel like my own," "head filled with cotton"), whole body or body parts, or sensations (e.g., touch, proprioception, hunger, thirst, libido). There may also be a diminished sense of agency (e.g., feeling robotic, like an automaton; lacking control of speech or movements). The depersonalization experience can sometimes be one of a split self, with one part observing and one participating, known as an "out-of-body experience" in its most extreme form. The unitary symptom of "depersonalization" consists of several symptom factors: anomalous body experiences (i.e., unreality of the self and perceptual alterations); emotional or physical numbing;

and temporal distortions with anomalous subjective recall.

Episodes of derealization are characterized by a feeling of unreality or detachment from, or unfamiliarity with, the world, be it individuals, inanimate objects, or all surroundings (Criterion A2). The individual may feel as if he or she were in a fog, dream, or bubble, or as if there were a veil or a glass wall between the individual and the world around. Surroundings may be experienced as artificial, colorless, or lifeless. Derealization is commonly accompanied by subjective visual distortions, such as blurriness, heightened acuity, widened or narrowed visual field, two-dimensionality or flatness, exaggerated three-dimensionality, or altered distance or size of objects (i.e., macropsia or micropsia). Auditory distortions can also occur, whereby voices or sounds are muted or heightened. In addition, Criterion C requires that the symptoms cause clinically significant distress or impairment in social, occupational, or other important areas of functioning, and Criteria D and E describe exclusionary diagnoses.

Associated Features

Individuals with depersonalization/derealization disorder may have difficulty describing their symptoms and may think they are “crazy” or “going crazy.” Another common experience is the fear of irreversible brain damage. A commonly associated symptom is a subjectively altered sense of time (i.e., too fast or too slow), as well as a subjective difficulty in vividly recalling past memories and owning them as personal and emotional. Vague somatic symptoms, such as head fullness, tingling, or lightheadedness, are not uncommon. Individuals may experience extreme rumination or obsessional preoccupation (e.g., constantly obsessing about whether they really exist, or checking their perceptions to determine whether they appear real). Varying degrees of anxiety and depression are also common associated features. Individuals with the disorder have been found to have physiological hyporeactivity to emotional stimuli. Neural substrates of interest include the hypothalamic-pituitary-adrenocortical axis, inferior parietal lobule, and prefrontal cortical-limbic circuits.

Prevalence

Transient depersonalization/derealization symptoms lasting hours to days are common in the general population. The 12-month prevalence of depersonalization/derealization disorder is thought to be markedly less than for transient symptoms, although precise estimates for the disorder are unavailable. In general, approximately one-half of all adults have experienced at least one lifetime episode of depersonalization/derealization. However, symptomatology that meets full criteria for depersonalization/derealization disorder is markedly less common than transient symptoms. One-month prevalence in the United Kingdom is approximately 1%–2%.

Development and Course

The mean age at onset of depersonalization/derealization disorder is 16 years, although the disorder can start in early or middle childhood; a minority cannot recall ever not having had the symptoms. Less than 20% of individuals experience onset after age 20 years and only 5% after age 25 years. Onset in the fourth decade of life or later is highly unusual.

Onset can range from extremely sudden to gradual. Duration of depersonalization/derealization disorder episodes can vary greatly, from brief (hours or days) to prolonged (weeks, months, or years). Given the rarity of disorder onset after age 40 years, in such cases the individual should be examined more closely for underlying medical conditions (e.g., brain lesions, seizure disorders, sleep apnea). The course of the disorder is often persistent. About one-third of cases involve discrete episodes; another third, continuous symptoms from the start; and still another third, an initially episodic course that eventually becomes continuous.

While in some individuals the intensity of symptoms can wax and wane considerably, others report an unwavering level of intensity that in extreme cases can be constantly present for years or decades. Internal and external factors that affect symptom intensity vary between individuals, yet some typical patterns are reported. Exacerbations can be triggered by stress, worsening mood or anxiety symptoms, novel or overstimulating settings, and physical factors such as lighting or lack of sleep.

Risk and Prognostic Factors

Temperamental. Individuals with depersonalization/derealization disorder are characterized by harm-avoidant temperament, immature defenses, and both disconnection and overconnection schemata. Immature defenses such as idealization/devaluation, projection, and acting out result in denial of reality and poor adaptation. *Cognitive disconnection schemata* reflect defectiveness and emotional inhibition and subsume themes of abuse, neglect, and deprivation. *Overconnection schemata* involve impaired autonomy with themes of dependency, vulnerability, and incompetence.

Environmental. There is a clear association between the disorder and childhood interpersonal traumas in a substantial portion of individuals, although this association is not as prevalent or as extreme in the nature of the traumas as in other dissociative disorders, such as dissociative identity disorder. In particular, emotional abuse and emotional neglect have been most strongly and consistently associated with the disorder. Other stressors can include physical abuse; witnessing domestic violence; growing up with a seriously impaired, mentally ill parent; or unexpected death or suicide of a family member or close friend. Sexual abuse is a much less common antecedent but can be encountered. The most common proximal precipitants of the disorder are severe stress (interpersonal, financial, occupational), depression, anxiety (particularly panic attacks), and illicit drug use. Symptoms may be specifically induced by substances such as tetrahydrocannabinol, hallucinogens, ketamine, MDMA (3,4-methylenedioxymethamphetamine; “ecstasy”), and salvia. Marijuana use may precipitate new-onset panic attacks and depersonalization/derealization symptoms simultaneously.

Culture-Related Diagnostic Issues

Volitionally induced experiences of depersonalization/derealization can be a part of meditative practices that are prevalent in many religious, spiritual, and cultural contexts and should not be diagnosed as a disorder. However, there are individuals who initially induce these states intentionally but over time lose control over them and may develop a fear and aversion for related practices. Cultural frameworks may affect the level of distress or perceived severity associated with uncontrolled depersonalization/derealization experiences by providing explanations for them (e.g., spiritual/supernatural causes), which may alleviate individuals’ fears

that they are “losing their mind.”

Functional Consequences of Depersonalization/Derealization Disorder

Symptoms of depersonalization/derealization disorder are highly distressing and are associated with major morbidity. The affectively flattened and robotic demeanor that these

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individuals often demonstrate may appear incongruent with the extreme emotional pain reported by those with the disorder. Impairment is often experienced in both interpersonal and occupational spheres, largely as a result of the hypoemotionality with others, subjective difficulty in focusing and retaining information, and a general sense of disconnectedness from life.

Differential Diagnosis

Illness anxiety disorder. Although individuals with depersonalization/derealization disorder can present with vague somatic complaints as well as fears of permanent brain damage, the diagnosis of depersonalization/derealization disorder is characterized by the presence of a constellation of typical depersonalization/derealization symptoms and the absence of other manifestations of illness anxiety disorder.

Major depressive disorder. Feelings of numbness, deadness, apathy, and being in a dream are not uncommon in major depressive episodes. However, in depersonalization/derealization disorder, such symptoms are associated with further symptoms of the disorder. If the depersonalization/derealization clearly precedes the onset of a major depressive episode or clearly continues after its resolution, the diagnosis of depersonalization/derealization disorder applies.

Obsessive-compulsive disorder. Some individuals with depersonalization/derealization disorder can become obsessively preoccupied with their subjective experience or develop rituals checking on the status of their symptoms. However, other symptoms of obsessive-compulsive disorder unrelated to depersonalization/derealization are not present.

Other dissociative disorders. In order to diagnose depersonalization/derealization disorder, the symptoms should not occur in the context of another dissociative disorder, such as dissociative identity disorder. Differentiation from dissociative amnesia and functional neurological symptom disorder (conversion disorder) is simpler, as the symptoms of these disorders do not overlap with those of depersonalization/derealization disorder.

Panic attacks. Depersonalization/derealization is one of the symptoms of panic attacks, increasingly common as panic attack severity increases. Therefore, depersonalization/derealization disorder should not be diagnosed when the symptoms occur only during panic attacks that are part of panic disorder, social anxiety disorder, or specific phobia. In addition, it is not uncommon for depersonalization/derealization symptoms to first begin in the context of new-onset panic attacks or as panic disorder progresses and worsens. In such presentations, the diagnosis of depersonalization/derealization disorder can be made if 1) the depersonalization/derealization component of the presentation is very prominent from the start, clearly exceeding in duration and intensity the occurrence of actual panic attacks; or 2) the

depersonalization/derealization continues after panic disorder has remitted or has been successfully treated.

Psychotic disorders. The presence of intact reality testing specifically regarding the depersonalization/derealization symptoms is essential to differentiating depersonalization/derealization disorder from psychotic disorders. Rarely, positive-symptom schizophrenia can pose a diagnostic challenge when nihilistic delusions are present. For example, an individual may complain that he or she is dead or the world is not real; this could be either a subjective experience that the individual knows is not true or a delusional conviction.

Substance/medication-induced disorders. Depersonalization/derealization associated with the physiological effects of substances during acute intoxication or withdrawal is not diagnosed as depersonalization/derealization disorder. The most common precipitating substances are the illicit drugs marijuana, hallucinogens, ketamine, ecstasy, and salvia. In

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about 15% of all cases of depersonalization/derealization disorder, the symptoms are precipitated by ingestion of such substances. If the symptoms persist for some time in the absence of any further substance or medication use, the diagnosis of depersonalization/derealization disorder applies. This diagnosis is usually easy to establish since the vast majority of individuals with this presentation become highly phobic and aversive to the triggering substance and do not use it again.

Traumatic brain injury. Depersonalization/derealization symptoms are typical in traumatic brain injury (TBI) but are distinguished from depersonalization/derealization disorder by onset of symptoms following TBI and the lack of other symptoms of depersonalization/derealization disorder.

Dissociative symptoms due to another medical condition. Features such as onset after age 40 years or the presence of atypical symptoms and course in any individual suggest the possibility of an underlying medical condition. In cases with dissociative symptoms, it is essential to conduct a thorough medical and neurological evaluation, which may include standard laboratory studies, viral titers, an electroencephalogram, vestibular testing, visual testing, sleep studies, and/or brain imaging. When the suspicion of an underlying seizure disorder proves difficult to confirm, an ambulatory electroencephalogram may be indicated; although temporal lobe epilepsy is most commonly implicated, parietal and frontal lobe epilepsy may also be associated.

Comorbidity

In a convenience sample of adults recruited for a number of depersonalization research studies, lifetime comorbidities were high for unipolar depressive disorder and for any anxiety disorder, with a significant proportion of the sample having both disorders. Comorbidity with posttraumatic stress disorder was low. The three most commonly co-occurring personality disorders were avoidant, borderline, and obsessive-compulsive.

Other Specified Dissociative Disorder

This category applies to presentations in which symptoms characteristic of a dissociative disorder that cause clinically significant distress or impairment in social, occupational, or other important areas of functioning predominate but do not meet the full criteria for any of the disorders in the dissociative disorders diagnostic class. The other specified dissociative disorder category is used in situations in which the clinician chooses to communicate the specific reason that the presentation does not meet the criteria for any specific dissociative disorder. This is done by recording “other specified dissociative disorder” followed by the specific reason (e.g., “dissociative trance”).

Examples of presentations that can be specified using the “other specified” designation include the following:

1. **Chronic and recurrent syndromes of mixed dissociative symptoms:** This category includes identity disturbance associated with less-than-marked discontinuities in sense of self and agency, or alterations of identity or episodes of possession in an individual who reports no dissociative amnesia.
2. **Identity disturbance due to prolonged and intense coercive persuasion:** Individuals who have been subjected to intense coercive persuasion (e.g., brainwashing, thought reform, indoctrination while captive, torture, long-term political imprisonment, recruitment by sects/cults or by terror organizations) may present with prolonged changes in, or conscious questioning of, their identity.

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3. **Acute dissociative reactions to stressful events:** This category is for acute, transient conditions that typically last less than 1 month, and sometimes only a few hours or days. These conditions are characterized by constriction of consciousness; depersonalization; derealization; perceptual disturbances (e.g., time slowing, macropsia); microamnesias; transient stupor; and/or alterations in sensory-motor functioning (e.g., analgesia, paralysis).
4. **Dissociative trance:** This condition is characterized by an acute narrowing or complete loss of awareness of immediate surroundings that manifests as profound unresponsiveness or insensitivity to environmental stimuli. The unresponsiveness may be accompanied by minor stereotyped behaviors (e.g., finger movements) of which the individual is unaware and/or that he or she cannot control, as well as transient paralysis or loss of consciousness. The dissociative trance is not a normal part of a broadly accepted collective cultural or religious practice.

Unspecified Dissociative Disorder

F44.9

This category applies to presentations in which symptoms characteristic of a dissociative disorder that cause clinically significant distress or impairment in social, occupational, or other important areas of functioning predominate but do not meet the full criteria for any of the disorders in the dissociative disorders diagnostic class. The unspecified dissociative disorder category is used in situations in which the clinician chooses *not* to specify the reason that the criteria are not met for a specific dissociative disorder and includes presentations for which there is insufficient information to make a more specific diagnosis (e.g., in emergency room settings).

Somatic Symptom and Related Disorders

This chapter includes the diagnoses of somatic symptom disorder, illness anxiety disorder, functional neurological symptom disorder (conversion disorder), psychological factors affecting other medical conditions, factitious disorder, other specified somatic symptom and related disorder, and unspecified somatic symptom and related disorder. All of the disorders in this chapter share a common feature: the prominence of somatic symptoms and/or illness anxiety associated with significant distress and impairment. Individuals with disorders with prominent somatic symptoms or illness anxiety are commonly encountered in primary care and other medical settings but are less commonly encountered in psychiatric and other mental health settings. These reconceptualized diagnoses, based on a reorganization of DSM-IV somatoform disorder diagnoses, are more useful for primary care and other medical (nonpsychiatric) clinicians.

The major diagnosis in this diagnostic class, somatic symptom disorder, emphasizes diagnosis made on the basis of the presence of symptoms and signs (distressing somatic symptoms plus abnormal thoughts, feelings, and behaviors in response to these symptoms) rather than the absence of a medical explanation for somatic symptoms. A distinctive characteristic of many individuals with somatic symptom disorder is not the somatic symptoms per se, but instead the way they present and interpret them. Incorporating affective, cognitive, and behavioral components into the criteria for somatic symptom disorder provides a more comprehensive and accurate reflection of the true clinical picture than can be achieved by assessing the somatic complaints alone.

The principles behind the changes in the somatic symptom and related diagnoses from DSM-IV are crucial in understanding the DSM-5 diagnoses. The DSM-IV term *somatoform disorders* was confusing and was replaced by *somatic symptom and related disorders*. In DSM-IV there was a great deal of overlap across the somatoform disorders and a lack of clarity about the boundaries of diagnoses. Although individuals with these disorders primarily present in medical rather than mental health settings, nonpsychiatric physicians found the DSM-IV somatoform diagnoses difficult to understand and use. The current DSM-5 classification recognizes this overlap by reducing the total number of disorders as well as their subcategories.

The previous criteria overemphasized the centrality of symptoms being unexplained by recognized pathophysiological processes. Such symptoms are present to various degrees, particularly in functional neurological symptom disorder, but somatic symptom disorders can also accompany recognized medical conditions (i.e., those disorders related to clearly recognized pathophysiological processes). The reliability of determining that a somatic symptom is unexplained by a recognized pathophysiological process related to a recognized medical condition is limited, and grounding a diagnosis on the absence of an explanation is problematic and reinforces mind-body dualism. It is not appropriate to give an individual a mental disorder diagnosis solely because a recognized medical condition cannot be demonstrated. Furthermore,

the presence of a recognized medical condition does not exclude the possibility of a comorbid mental disorder, including a somatic symptom and related disorder. Perhaps because of the predominant focus on lack of medical explanation in DSM-IV, individuals regarded these diagnoses as pejorative and demeaning,

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implying that their physical symptoms were not “real.” The DSM-5 classification defines the major diagnosis, somatic symptom disorder, on the basis of positive symptoms (distressing somatic symptoms plus abnormal thoughts, feelings, and behaviors in response to these symptoms). In functional neurological symptom disorder and pseudocyesis (other specified somatic symptom and related disorder), the emphasis is on demonstrating clinical evidence of incompatibility with recognized pathophysiological processes.

It is important to note that some other mental disorders may initially manifest with primarily somatic symptoms (e.g., major depressive disorder, panic disorder). Such diagnoses may account for the somatic symptoms, or they may occur alongside one of the somatic symptom and related disorders in this chapter. There is also considerable medical comorbidity among individuals with somatic symptom and related disorders. Although somatic symptoms are frequently associated with psychological distress and psychopathology, some somatic symptom and related disorders can arise spontaneously, and their causes can remain obscure. Anxiety disorders and depressive disorders may accompany somatic symptom and related disorders. The somatic component adds severity and complexity to depressive and anxiety disorders and results in higher severity, functional impairment, and even refractoriness to traditional treatments. In rare instances, the degree of preoccupation may be so severe as to warrant consideration of a delusional disorder diagnosis.

A number of factors may contribute to somatic symptom and related disorders. These include genetic and biological vulnerability (e.g., increased sensitivity to pain), early traumatic experiences (e.g., violence, abuse, deprivation), medical iatrogenesis (e.g., reinforcement of the sick role, excessive referrals and diagnostic testing), and learning (e.g., lack of reinforcement of nonsomatic expressions of distress), as well as sociocultural norms that minimize or stigmatize psychological suffering as compared with physical suffering. Differences in medical care across cultural contexts affect the presentation, recognition, and management of these somatic presentations. Variations in symptom presentation are likely the result of the interaction of multiple factors within cultural contexts that affect how individuals identify and classify bodily sensations, perceive illness, and seek medical attention for them.

All of these disorders are characterized by the prominent focus on somatic concerns and their initial presentation mainly in medical rather than mental health care settings. Somatic symptom disorder and illness anxiety disorder offer more clinically useful methods of characterizing individuals who may have been considered in the past for a diagnosis of somatization disorder and hypochondriasis. Furthermore, approximately two-thirds to three-fourths of individuals previously diagnosed with hypochondriasis are subsumed under the diagnosis of somatic symptom disorder. However, the remaining one-quarter to one-third of individuals with previously diagnosed hypochondriasis have high health anxiety in the absence of somatic symptoms, and many such individuals’ symptoms would not qualify for an anxiety disorder diagnosis. The DSM-5 diagnosis of illness anxiety disorder is for this latter group of individuals.

Illness anxiety disorder can be considered either a somatic symptom and related disorder or an anxiety disorder. Because of the strong focus on somatic concerns, and because illness anxiety disorder is most often encountered in medical settings, for utility it is listed with the somatic symptom and related disorders. In functional neurological symptom disorder, the key to diagnosis is neurological symptoms that can be demonstrated, on the basis of positive clinical examination features, to be incompatible with recognized pathophysiology. This is now a “rule-in” diagnosis, and not a diagnosis of exclusion, and can be made in the presence of a recognized neurological disorder. It no longer requires the presence of a recent psychological stressor, because such stressors are not always present. Psychological factors affecting other medical conditions is also included in this chapter. Its essential feature is the presence of one or more clinically significant psychological or behavioral factors that adversely affect a medical condition by increasing the risk for suffering, death, or disability. Like the other somatic symptom and related disorders, factitious disorder embodies persistent problems related to illness perception and identity. In the great majority of reported cases of factitious disorder, both

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imposed on self and imposed on another, individuals present with somatic symptoms and expressed medical disease conviction. Consequently, DSM-5 factitious disorder is included among the somatic symptom and related disorders. Other specified somatic symptom and related disorder and unspecified somatic symptom and related disorder include conditions for which some, but not all, of the criteria for somatic symptom disorder or illness anxiety disorder are met, as well as pseudocyesis.

Somatic Symptom Disorder

Diagnostic Criteria

F45.1

- A. One or more somatic symptoms that are distressing or result in significant disruption of daily life.
- B. Excessive thoughts, feelings, or behaviors related to the somatic symptoms or associated health concerns as manifested by at least one of the following:
 - 1. Disproportionate and persistent thoughts about the seriousness of one's symptoms.
 - 2. Persistently high level of anxiety about health or symptoms.
 - 3. Excessive time and energy devoted to these symptoms or health concerns.
- C. Although any one somatic symptom may not be continuously present, the state of being symptomatic is persistent (typically more than 6 months).

Specify if:

With predominant pain (previously pain disorder): This specifier is for individuals whose somatic symptoms predominantly involve pain.

Specify if:

Persistent: A persistent course is characterized by severe symptoms, marked impairment, and long duration (more than 6 months).

Specify current severity:

Mild: Only one of the symptoms specified in Criterion B is fulfilled.

Moderate: Two or more of the symptoms specified in Criterion B are fulfilled.

Severe: Two or more of the symptoms specified in Criterion B are fulfilled, plus there are multiple somatic complaints (or one very severe somatic symptom).

Diagnostic Features

Individuals with somatic symptom disorder typically have multiple, current, somatic symptoms that are distressing or result in significant disruption of daily life (Criterion A), although sometimes only one severe symptom, most commonly pain, is present. Symptoms may be specific (e.g., localized pain) or relatively nonspecific (e.g., fatigue). The symptoms sometimes represent normal bodily sensations or discomfort that does not generally signify serious disease. Somatic symptoms without an evident medical explanation are not sufficient to make this diagnosis. The individual's suffering is authentic, whether or not it is medically explained.

The symptoms may or may not be associated with another medical condition. The diagnoses of somatic symptom disorder and a concurrent medical illness are not mutually exclusive, and these frequently occur together. For example, an individual may become seriously disabled by symptoms of somatic symptom disorder after an uncomplicated myocardial infarction even if the myocardial infarction itself did not result in any disability. If another medical condition or high risk for developing one is present (e.g., strong family history), the thoughts, feelings, and behaviors associated with this condition are excessive (Criterion B).

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Individuals with somatic symptom disorder tend to have very high levels of worry about illness (Criterion B). They appraise their bodily symptoms as unduly threatening, harmful, or troublesome and often think the worst about their health. Even when there is evidence to the contrary, some individuals still fear the medical seriousness of their symptoms. In severe somatic symptom disorder, health concerns may assume a central role in the individual's life, becoming a feature of his or her identity and dominating interpersonal relationships.

Individuals typically experience distress that is principally focused on somatic symptoms and their significance. When asked directly about their distress, some individuals describe it in relation to other aspects of their lives, while others deny any source of distress other than the somatic symptoms. Health-related quality of life is often impaired, both physically and mentally. The diagnosis can further be specified by stating whether complaints predominantly involve pain and/or if complaints are marked by a persistent course.

Additionally, severity of somatic symptom disorder can be specified by the number of fulfilled B criteria. Mild forms of somatic symptom disorder (one symptom as specified in Criterion B is fulfilled) are more prevalent, while moderate (two or more B criteria are present) and severe cases (two or more symptoms as specified in Criterion B are fulfilled in combination

with multiple somatic complaints or one very severe somatic symptom) are marked by higher levels of impairment. In severe somatic symptom disorder, the impairment is marked, and when persistent, the disorder can lead to invalidism.

There is often a high level of medical care utilization, which rarely alleviates the individual's concerns. Consequently, the individual may seek care from multiple doctors for the same symptoms. These individuals often seem unresponsive to medical interventions, and new interventions may only exacerbate the presenting symptoms. Some individuals with the disorder seem unusually sensitive to medication side effects. Some feel that their medical assessment and treatment have been inadequate.

The criteria for somatic symptom disorder appear suitable for use in children and adolescents, but they have been less studied in youth than among adults.

Associated Features

Cognitive features include attention focused on somatic symptoms, attribution of normal bodily sensations to physical illness (possibly with catastrophic interpretations), worry about illness, a self-concept of bodily weakness, and intolerance of bodily complaints. Besides health anxiety, emotional features may include negative affectivity, desperation, and demoralization related to somatic symptoms. The relevant associated behavioral features may include repeated bodily checking for abnormalities, repeated seeking of medical help and reassurance, and avoidance of physical activity. These behavioral features are most pronounced in severe, persistent somatic symptom disorder. These features are usually associated with frequent requests for medical help for different somatic symptoms. This may lead to medical consultations in which individuals are so focused on their concerns about somatic symptom(s) that they cannot be redirected to other matters. Any reassurance by the doctor that the symptoms are not indicative of serious physical illness tends to be short-lived and/or is experienced by the individuals as the doctor not taking their symptoms with due seriousness. As the focus on somatic symptoms is a primary feature of the disorder, individuals with somatic symptom disorder typically present to general medical health services rather than mental health services. The suggestion of referral to a mental health specialist may be met with surprise or even frank refusal by individuals with somatic symptom disorder.

Prevalence

The prevalence of somatic symptom disorder is unclear. Estimates about the prevalence of somatic symptom disorder come from the limited epidemiological literature on

DSM-IV-TR somatoform disorders. However, the prevalence of somatic symptom disorder is expected to be higher than that of the more restrictive DSM-IV-TR somatization disorder (<1%) but lower than that of undifferentiated somatoform disorder (approximately 19%). More recent population-based studies with a questionnaire-based strategy using DSM-5 diagnostic criteria for somatic symptom disorder in adult and adolescent samples report prevalence rates between 6.7% and 17.4%. Based on research conducted in Europe and North America, the prevalence of somatic symptom disorder in the general adult population can be approximated as 4%–6%.

Somatic symptom disorder has a higher frequency in primary care patients than in the general population. Based on reviews and meta-analyses of studies from multiple countries that had still used DSM-IV or ICD-10 criteria, a 12-month prevalence of somatic symptom disorder and related conditions in primary care patients between 10% and 20% appears plausible. Prevalence rates are higher in clinical settings that specialize in psychosomatic or functional disorders, with reported frequencies of somatic symptom disorder between 40% and 60%.

Women tend to report more somatic symptoms than do men, and the prevalence of somatic symptom disorder is consequently likely to be higher in women.

Development and Course

In a study of Danish children ages 5–7 years, functional somatic symptoms were common health complaints, which for a significant minority (roughly one-fifth) of those with complaints were severe enough to cause distress, impairment, school absences, or medical help-seeking. Age at onset does not seem to affect the duration of untreated illness.

The course of somatic symptom disorder is likely to be chronic and fluctuating and influenced by the number of symptoms, individual's age, level of impairment, and any comorbidity. The course is also influenced by personality traits, with less harm avoidance and greater cooperativeness associated with a shorter time to remission.

In children, the most common symptoms are recurrent abdominal pain, headache, fatigue, and nausea. A single prominent symptom is more common in children than in adults. When the diagnosis is being made in younger individuals, it is important to obtain patient, family, and other assessments (e.g., school) of symptom presentation. Patient and caregiver engagement during evaluation and management is fundamental because parents' interpretation of and response to symptoms may determine the level of associated distress, the demands for medical investigations and interventions, and time away from school.

In older individuals, pain localized in several body regions appears to be the most common symptom. Somatic symptoms and concurrent medical illnesses are common as multimorbidity increases with age. Prevalence rates of somatic symptom disorder seem to be stable until age 65 years and might decrease thereafter. For making the diagnosis in older individuals, a focus on the requirement for excessive thoughts, feelings, or behaviors related to the somatic symptoms or associated health concerns (Criterion B) is crucial. Somatic symptom disorder may be underdiagnosed in older adults either because certain somatic symptoms (e.g., pain, fatigue) are considered part of normal aging or because illness worry is considered “understandable” in older adults who have more general medical illnesses and medications than do younger people.

Risk and Prognostic Factors

Temperamental. The personality trait of negative affectivity (neuroticism) has been identified as an independent correlate/risk factor of a high number of somatic symptoms. Comorbid anxiety or depression is common and may exacerbate symptoms and impairment.

Environmental. Somatic symptom disorder is more frequent in individuals with few years of education and low socioeconomic status, and in those who have recently experienced stressful or

health-related life events. Early lifetime adversity such as childhood sexual abuse is also likely a risk factor for somatic symptom disorder in adults.

Course modifiers. Persistent somatic symptoms are associated with demographic features (women, older age, fewer years of education, lower socioeconomic status, unemployment), a reported history of sexual abuse or other childhood adversity, concurrent chronic physical illness or mental disorder (depression, anxiety, persistent depressive disorder, panic), social stress, and reinforcing social factors such as illness benefits. Total somatic symptom severity is probably associated with female gender, anxiety, depression and general medical illness. Cognitive factors that affect clinical course include sensitization to pain, heightened attention to bodily sensations, and attribution of bodily symptoms to a possible medical illness rather than recognizing them as a normal phenomenon or psychological stress.

Culture-Related Diagnostic Issues

High numbers of somatic symptoms are found in population-based and primary care studies around the world, with a similar pattern of the most commonly reported somatic symptoms, impairment, and treatment seeking. The relationship between number of somatic symptoms and illness worry is similar in different cultural contexts, and marked illness worry is associated with impairment and greater treatment seeking cross-culturally. In many cultural contexts, individuals with depression commonly present with somatic symptoms.

Despite these similarities, there are differences in somatic symptoms across cultural contexts and ethnoracial groups. Sociocultural factors, particularly stigma related to mental disorders, may explain differences in somatic symptom reporting across cultural contexts. The description of somatic symptoms varies with linguistic and other local cultural factors. These somatic presentations have been described as “idioms of distress” because somatic symptoms may have special meanings and shape patient-clinician interactions in the particular cultural contexts. For example, sensations of heaviness, complaints of “gas,” too much heat in the body, or burning in the head are common in some cultures or ethnic groups but rare in others. Cultural explanations also vary, and somatic symptoms may be attributed variously to particular family, work (e.g., burnout), or other environmental stresses; general medical illness; the suppression of feelings of anger and resentment; or certain culturally specific attributions, such as semen loss. Certain somatic symptoms may be part of specific explanatory models in a given cultural context; for example, traditional understandings of *shenjing shuairuo* in China link concepts of “weakness of nerves” (neurasthenia) and hot-cold imbalance with prominent symptoms such as fatigue and low energy. There may also be differences in medical treatment seeking and utilization of nonmedical, traditional, alternative and complementary healing practices among cultural groups, in addition to differences due to variable access to medical care services. Cultural beliefs, previous illnesses, insurance status, health literacy, and health care experiences can influence individuals’ perception of somatic symptoms and health care use. Seeking treatment for multiple somatic symptoms in general medical clinics is a worldwide phenomenon.

Sex- and Gender-Related Diagnostic Issues

In population-based studies, women report more somatic symptoms than men, and in one study of primary care patients with chronic pain, women reported more severe somatic symptoms than men. While exposure to sexual trauma, intimate partner violence, and a childhood trauma history

is associated with increased somatic symptom expression in

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both women and men, a childhood history of multiple adverse childhood experiences is especially likely to increase somatic symptom expression in women.

In women, gender is associated with an increased likelihood of developing persistent symptoms of somatic symptom disorder. There appears to be no evidence that gender is associated with the duration of untreated illness and response to psychological or pharmacological treatment.

Association With Suicidal Thoughts or Behavior

Somatic symptom disorder is associated with suicidal thoughts and suicide attempts. It is likely that suicidal thoughts and behaviors are partly explained by the diagnostic overlap and frequent comorbidity of somatic symptom disorder and depressive disorders. In addition, dysfunctional illness perceptions and the severity of somatic symptoms appear to be independently associated with an increased risk of suicidal ideation.

Functional Consequences of Somatic Symptom Disorder

The disorder is associated with marked impairment of health status and high psychological distress. Many individuals with severe somatic symptom disorder are likely to have impaired health status scores more than 2 standard deviations below population norms. Health status is particularly impaired in the presence of multiple or severe symptoms.

Differential Diagnosis

If the somatic symptoms are consistent with another mental disorder (e.g., panic disorder), and the diagnostic criteria for that disorder are fulfilled, then that mental disorder should be considered as an alternative or additional diagnosis. If, as commonly occurs, the criteria for both somatic symptom disorder and another mental disorder diagnosis are fulfilled, then both should be diagnosed, as both may require treatment.

Other medical conditions. The presence of somatic symptoms of unclear etiology is not in itself sufficient to make the diagnosis of somatic symptom disorder. The symptoms of many individuals with disorders like irritable bowel syndrome or fibromyalgia would not satisfy the criterion necessary to diagnose somatic symptom disorder (Criterion B). Conversely, the presence of somatic symptoms of an established medical condition (e.g., diabetes or heart disease) does not exclude the diagnosis of somatic symptom disorder if the criteria are otherwise met. Factors that distinguish individuals with somatic symptom disorder from individuals with general medical conditions alone include the ineffectiveness of analgesics, a history of mental disorders, unclear provocative or palliative factors, persistence without cessation, and stress.

Psychological factors affecting other medical conditions. The diagnosis of somatic symptom disorder requires distressing or impairing somatic symptoms that may or may not be associated with another medical condition but must be accompanied by excessive or disproportionate thoughts, feelings, or behaviors related to the somatic symptoms or associated health concerns. In contrast,

the diagnosis of psychological factors affecting other medical conditions requires the presence of a medical condition, as well as psychological factors that adversely affect its course or interfere with its treatment.

Panic disorder. In panic disorder, somatic symptoms and anxiety about health tend to occur in acute episodes, whereas in somatic symptom disorder, anxiety and somatic symptoms are more persistent.

Generalized anxiety disorder. Individuals with generalized anxiety disorder worry about multiple events, situations, or activities, only one of which may involve their health. The main focus is not usually somatic symptoms or fear of illness as it is in somatic symptom disorder.

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Depressive disorders. Depressive disorders are commonly accompanied by somatic symptoms such as fatigue, headaches, or joint, abdominal, or other pains. However, depressive disorders are differentiated from somatic symptom disorder by the requirement of the presence of depressed mood or, in the case of major depressive disorder, either depressed mood or decreased interest or pleasure in activities. In some cultural contexts, these core symptoms of depression may be initially denied or deemphasized by individuals whose presentations would otherwise meet criteria for a depressive disorder. Such individuals might instead emphasize somatic symptoms that may be idiomatic (e.g., heavy heart) and unfamiliar to clinicians.

Illness anxiety disorder. If the individual has extensive worries about health but no or minimal somatic symptoms, it may be more appropriate to consider illness anxiety disorder.

Functional neurological symptom disorder (conversion disorder). In functional neurological symptom disorder, the presenting symptom is loss of function (e.g., of a limb), whereas in somatic symptom disorder, the focus is on the distress that particular symptoms cause. The features listed under Criterion B of somatic symptom disorder may be helpful in differentiating the two disorders.

Delusional disorder. In somatic symptom disorder, the individual's beliefs that somatic symptoms might reflect serious underlying physical illness are not held with delusional intensity. Nonetheless, the individual's beliefs concerning the somatic symptoms can be firmly held. In contrast, in delusional disorder, somatic type, the individual's conviction that the somatic symptoms are indicative of having a serious underlying illness is stronger than that found in somatic symptom disorder.

Body dysmorphic disorder. In body dysmorphic disorder, the individual is excessively concerned about, and preoccupied by, a perceived defect in his or her physical appearance. In contrast, in somatic symptom disorder, the concern about somatic symptoms reflects fear of underlying illness, not of a defect in appearance.

Obsessive-compulsive disorder. In somatic symptom disorder, the recurrent ideas about somatic symptoms or illness are less intrusive, and individuals with this disorder do not exhibit the associated repetitive behaviors aimed at reducing anxiety that occur in obsessive-compulsive disorder.

Factitious disorder and malingering. In factitious disorder and malingering, individuals present themselves as ill or impaired but have falsified presenting physical signs and symptoms with the

intent to deceive. In contrast, the symptoms of somatic symptom disorder are not simulated or self-induced, and these individuals suffer authentically and seriously from their somatic complaints.

Comorbidity

Somatic symptom disorder is associated with high rates of comorbidity with other mental disorders as well as general medical conditions. The most relevant co-occurring mental disorders are anxiety and depressive disorders, each of which occurs in up to 50% of cases of somatic symptom disorders and significantly contributes to overall functional impairment and poorer quality of life. Other mental disorders that have been found to co-occur with somatic symptom disorder are posttraumatic stress disorder and obsessive-compulsive disorder. Other evidence indicates an association with sexual dysfunction in men.

Elevated levels of the psychological features (Criterion B) of somatic symptom disorder have been found in several general medical conditions. When a concurrent general medical condition is present, the degree of impairment is more marked than would be expected from the physical illness alone. Moreover, somatization in medical illness has been shown to worsen disease and treatment outcomes, adherence, and quality of life and to increase health care utilization.

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Illness Anxiety Disorder

Diagnostic Criteria

F45.21

- A. Preoccupation with having or acquiring a serious illness.
- B. Somatic symptoms are not present or, if present, are only mild in intensity. If another medical condition is present or there is a high risk for developing a medical condition (e.g., strong family history is present), the preoccupation is clearly excessive or disproportionate.
- C. There is a high level of anxiety about health, and the individual is easily alarmed about personal health status.
- D. The individual performs excessive health-related behaviors (e.g., repeatedly checks his or her body for signs of illness) or exhibits maladaptive avoidance (e.g., avoids doctor appointments and hospitals).
- E. Illness preoccupation has been present for at least 6 months, but the specific illness that is feared may change over that period of time.
- F. The illness-related preoccupation is not better explained by another mental disorder, such as somatic symptom disorder, panic disorder, generalized anxiety disorder, body dysmorphic disorder, obsessive-compulsive disorder, or delusional disorder, somatic type.

Specify whether:

Care-seeking type: Medical care, including physician visits or undergoing tests and procedures, is frequently used.

Care-avoidant type: Medical care is rarely used.

Diagnostic Features

Most individuals who previously would have been diagnosed with hypochondriasis in DSM-IV (preoccupation with having a serious disease based on the individual's misinterpretation of bodily symptoms) are now classified as having somatic symptom disorder; however, in one-third of cases, the diagnosis of illness anxiety disorder applies instead.

Illness anxiety disorder entails a preoccupation with having or acquiring a serious, undiagnosed medical illness (Criterion A). Somatic symptoms are not present or, if present, are only mild in intensity (Criterion B). A thorough evaluation fails to identify a serious medical condition that accounts for the individual's concerns. While the concern may be derived from a nonpathological physical sign or sensation, the individual's distress emanates not primarily from the physical complaint itself but rather from his or her anxiety about the meaning, significance, or cause of the complaint (i.e., the suspected medical diagnosis). If a physical sign or symptom is present, it is often a normal physiological sensation (e.g., orthostatic dizziness), a benign and self-limited dysfunction (e.g., transient tinnitus), or a bodily discomfort not generally considered indicative of disease (e.g., belching). If a diagnosable medical condition is present, the individual's anxiety and preoccupation are clearly excessive and disproportionate to the severity of the condition (Criterion B). Most empirical evidence and existing literature pertain to previously defined DSM hypochondriasis and health anxiety, and it is unclear to what extent and how precisely they apply to the description of this new diagnosis.

The preoccupation with the idea that one is sick is accompanied by substantial anxiety about health and disease (Criterion C). Individuals with illness anxiety disorder are easily alarmed about illness, such as by hearing about someone else falling ill or reading a health-related news story. Their concerns about undiagnosed disease do not respond to appropriate medical reassurance, negative diagnostic tests, or benign course. The physician's attempts at reassurance and symptom palliation generally do not alleviate the individual's

concerns and may heighten them. Illness concerns assume a prominent place in the individual's life, affecting daily activities, and may even result in invalidism. Illness becomes a central feature of the individual's identity and self-image, a frequent topic of social discourse, and a characteristic response to stressful life events. Individuals with the disorder often examine themselves repeatedly (e.g., examining one's throat in the mirror) (Criterion D). They research their suspected disease excessively (e.g., on the Internet) and repeatedly seek reassurance from family, friends, or physicians. This incessant worrying often becomes frustrating for others and may result in considerable strain within the family. In some cases, the anxiety leads to maladaptive avoidance of situations (e.g., visiting sick family members) or activities (e.g., exercise) that these individuals fear might jeopardize their health.

Associated Features

Because they believe they are medically ill, individuals with illness anxiety disorder are encountered far more frequently in medical than in mental health settings. The majority of individuals with illness anxiety disorder have extensive yet unsatisfactory medical care. They generally have elevated rates of utilization of medical and mental health services compared with the general population. In a minority of cases of illness anxiety disorder, individuals are too anxious to seek medical attention and avoid medical health care.

They often consult multiple physicians for the same problem and obtain repeatedly negative diagnostic test results. At times, medical attention leads to a paradoxical exacerbation of anxiety or to iatrogenic complications from diagnostic tests and procedures. Individuals with the disorder are generally dissatisfied with their medical care and find it unhelpful, often feeling they are not being taken seriously by physicians. At times, these concerns may be justified, since physicians sometimes are dismissive or respond with frustration or hostility. This response can occasionally result in a failure to diagnose a medical condition that is present.

Prevalence

Prevalence estimates of illness anxiety disorder are based on estimates of the DSM-III and DSM-IV diagnosis *hypochondriasis* and health anxiety. The 1- to 2-year prevalence of health anxiety and/or disease conviction in community surveys and population-based samples from high-income countries such as the United States and Germany ranges from 1.3% to 10%. In ambulatory medical populations, the 6-month/1-year prevalence rates are between 2.2% and 8% across a range of countries, with weighted mean prevalence rates of 3%. By contrast, in a study of patients in specialty clinics, about one-fifth of individuals reported illness anxiety. The prevalence of the disorder is similar in men and women.

Development and Course

The development and course of illness anxiety disorder are unclear. Illness anxiety disorder is generally thought to be a chronic, episodic, and relapsing condition with an age at onset in early and middle adulthood. The disorder is thought to be rare in children, although the onset of health-related anxieties can occur in childhood or adolescence. In some population-based samples, health-related anxiety increases with age, but in others, health anxiety peaks in middle age, before declining in older age. The ages of individuals with high health anxiety in medical settings do not appear to differ from those of other individuals in those settings. In older individuals, health-related anxiety often focuses on memory loss and sensory loss.

Risk and Prognostic Factors

Environmental. Illness anxiety disorder may sometimes be precipitated by a major life stress or a serious but ultimately benign threat to the individual's health. A history of

childhood abuse or of a serious childhood illness, serious illness in a parent, or death of an ill parent during childhood may predispose to development of the disorder in adulthood.

Course modifiers. Approximately one-third to one-half of individuals with illness anxiety disorder have a transient form, which is associated with less psychiatric comorbidity, more medical comorbidity, and less severe illness anxiety disorder.

Culture-Related Diagnostic Issues

The diagnosis should be made with caution in individuals whose ideas about disease are congruent with widely held cultural beliefs. The prevalence appears to be similar across different countries, although little is known about the cross-cultural variation in phenomenology.

Functional Consequences of Illness Anxiety Disorder

Illness anxiety disorder causes substantial role impairment and decrements in physical function and health-related quality of life. Health concerns often interfere with interpersonal relationships, disrupt family life, and damage occupational performance.

Differential Diagnosis

Other medical conditions. The first differential diagnostic consideration is an underlying medical condition, including neurological or endocrine conditions, occult malignancies, and other diseases that affect multiple body systems. The presence of a medical condition does not rule out the possibility of coexisting illness anxiety disorder. If a medical condition is present, the health-related anxiety and disease concerns are clearly disproportionate to its seriousness. Transient preoccupations related to a medical condition do not constitute illness anxiety disorder.

Adjustment disorders. Health-related anxiety is a normal response to serious illness and is not a mental disorder. Such nonpathological health anxiety is clearly related to the medical condition and is typically time-limited. If the health anxiety is severe enough to cause clinically significant distress or impairment in one or more important areas of functioning, an adjustment disorder may be diagnosed. However, if disproportionate health-related anxiety persists for longer than 6 months, a diagnosis of illness anxiety disorder may apply.

Somatic symptom disorder. Both somatic symptom disorder and illness anxiety disorder may be characterized by a high level of anxiety about health and excessive health-related behaviors. They are differentiated by the fact that somatic symptom disorder requires the presence of somatic symptoms that are distressing or result in significant disruption of daily life, whereas in illness anxiety disorder, somatic symptoms either are not present or, if present, are only mild in intensity.

Anxiety disorders. In generalized anxiety disorder, individuals worry about multiple events, situations, or activities, only one of which may involve health. In panic disorder, the individual may be concerned that the panic attacks reflect the presence of a medical illness; however, although these individuals may have health anxiety, their anxiety is typically very acute and episodic. In illness anxiety disorder, the health anxiety and fears are more persistent and enduring. Individuals with illness anxiety disorder may experience panic attacks that are triggered by their illness concerns.

Obsessive-compulsive and related disorders. Individuals with illness anxiety disorder may have intrusive thoughts about having a disease and also may have associated compulsive behaviors

(e.g., seeking reassurance). However, in illness anxiety disorder, the preoccupations are usually focused on having a disease, whereas in obsessive-compulsive disorder (OCD), the thoughts are intrusive and are usually focused on fears of getting a disease in the future. Most individuals with OCD have obsessions or compulsions

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involving other concerns in addition to fears about contracting disease. In body dysmorphic disorder, concerns are limited to the individual's physical appearance, which is viewed as defective or flawed.

Major depressive disorder. Some individuals with a major depressive episode ruminate about their health and worry excessively about illness. A separate diagnosis of illness anxiety disorder is not made if these concerns occur only during major depressive episodes. However, if excessive illness worry persists after remission of an episode of major depressive disorder, the diagnosis of illness anxiety disorder should be considered.

Psychotic disorders. Individuals with illness anxiety disorder are not delusional and can acknowledge the possibility that the feared disease is not present. Their ideas do not attain the rigidity and intensity seen in the somatic delusions occurring in psychotic disorders (e.g., schizophrenia; delusional disorder, somatic type; major depressive disorder, with psychotic features). True somatic delusions are generally more bizarre (e.g., that an organ is rotting or dead) than the concerns seen in illness anxiety disorder. The concerns seen in illness anxiety disorder, though not founded in reality, are plausible.

Comorbidity

Illness anxiety disorder co-occurs with anxiety disorders (in particular, generalized anxiety disorder and panic disorder), OCD, and depressive disorders. Approximately two-thirds of individuals with illness anxiety disorder are likely to have at least one other comorbid major mental disorder. Individuals with illness anxiety disorder may have an elevated risk for personality disorders.

Functional Neurological Symptom Disorder (Conversion Disorder)

Diagnostic Criteria

- A. One or more symptoms of altered voluntary motor or sensory function.
- B. Clinical findings provide evidence of incompatibility between the symptom and recognized neurological or medical conditions.
- C. The symptom or deficit is not better explained by another medical or mental disorder.
- D. The symptom or deficit causes clinically significant distress or impairment in

social, occupational, or other important areas of functioning or warrants medical evaluation.

Coding note: The ICD-10-CM code depends on the symptom type (see below).

Specify symptom type:

F44.4 With weakness or paralysis

F44.4 With abnormal movement (e.g., tremor, dystonia, myoclonus, gait disorder)

F44.4 With swallowing symptoms

F44.4 With speech symptom (e.g., dysphonia, slurred speech)

F44.5 With attacks or seizures

F44.6 With anesthesia or sensory loss

F44.6 With special sensory symptom (e.g., visual, olfactory, or hearing disturbance)

F44.7 With mixed symptoms

Specify if:

Acute episode: Symptoms present for less than 6 months.

Persistent: Symptoms occurring for 6 months or more.

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Specify if:

With psychological stressor (*specify stressor*)

Without psychological stressor

Diagnostic Features

In functional neurological symptom disorder (conversion disorder), there may be one or more neurological symptoms of various types. Motor symptoms include weakness or paralysis; abnormal movements, such as tremor, jerks, or dystonic movements; and gait abnormalities. Sensory symptoms include altered, reduced, or absent skin sensation, vision, or hearing. Episodes of apparent unresponsiveness with or without limb movements may resemble epileptic seizures, syncope, or coma (also called *dissociative*, *psychogenic*, or *nonepileptic seizures or attacks*). Other symptoms include reduced or absent speech volume (dysphonia/aphonia); altered speech articulation, prosody, or fluency; a sensation of a lump in the throat (globus); and diplopia. This disorder has been called “conversion disorder” in prior editions of DSM as well as in much of the psychiatric research literature. The term “conversion” originated in psychoanalytic theory, which proposes that unconscious psychic conflict is “converted” into physical symptoms.

The diagnosis rests on clinical findings that show clear evidence of incompatibility with recognized neurological disease. These should usually be elicited and interpreted in the context of the whole clinical picture by a health care professional with expertise in the diagnosis of

neurological conditions. The diagnosis is not one of exclusion and can be made in individuals who also have neurological diseases like epilepsy or multiple sclerosis. The diagnosis should not be made simply because results from investigations are normal or because the symptom is “bizarre.” Internal inconsistency during examination is one way to demonstrate incompatibility (i.e., demonstrating that physical signs elicited through one examination method are no longer present when tested a different way). There are dozens of examples of such “positive” examination findings. Examples of examination findings that indicate incompatibility with recognized neurological disease include the following:

- For functional limb weakness or paralysis: Hoover’s sign, in which weakness of hip extension returns to normal strength with contralateral hip flexion against resistance; the hip abductor sign, in which weakness of thigh abduction returns to normal with contralateral hip abduction against resistance; or a discrepancy between on-the-bed performance (e.g., weakness of ankle plantar flexion) compared with another task (e.g., ability to walk on tiptoes).
- For functional tremor: the tremor entrainment test, in which a tremor changes when the individual is distracted by copying the examiner in making a rhythmical movement with the contralateral hand or foot. The test is positive when the tremor “entrains” the rhythm of the unaffected hand or foot, the tremor is suppressed, or the individual cannot copy simple rhythmical movements. Other features of functional limb tremor include variability in frequency or direction of the tremor.
- For functional dystonia: individuals typically present with fixed inverted position of the ankle, a clenched fist, or unilateral contraction of platysma, often with sudden onset.
- For attacks resembling epileptic seizures or syncope (also called functional or dissociative [nonepileptic] seizures): features suggestive of functional neurological symptom disorder include persistent eye closure sometimes with resistance to opening, bilateral motor movements with preserved awareness, or a duration longer than 5 minutes. Clinical features usually need to be combined and may be supported with a normal simultaneous ictal electroencephalogram (although this alone does not exclude all forms of epilepsy or syncope).

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- For functional speech symptoms: internal inconsistencies in speech articulation and phonation.
- For functional visual symptoms: a tubular visual field (i.e., tunnel vision) and tests that indicate internal inconsistency in visual acuity, such as the “fogging test” (i.e., while the individual views the eye chart with both eyes open, the “good” eye is subtly fogged so that any useful binocular vision must be a result of “bad” eye function).

It is important to note that the diagnosis of functional neurological symptom disorder should be based on the overall clinical picture and not on a single clinical finding.

Associated Features

Several associated features can support the diagnosis of functional neurological symptom disorder, although none are specific. There may be a history of other functional somatic symptoms or disorders, especially involving pain and fatigue. Onset may be associated with stress or trauma, either psychological or physical in nature. The potential etiological relevance of this stress or trauma may be suggested by a close temporal relationship. However, while assessment for stress and trauma is important, it may be absent in up to 50% of individuals, and the diagnosis should not be withheld if none is found.

Functional neurological symptom disorder is often associated with dissociative symptoms,

such as depersonalization, derealization, and dissociative amnesia, particularly at symptom onset or during attacks.

The phenomenon of *la belle indifférence* (i.e., lack of concern about the nature or implications of the symptom) has been associated with functional neurological symptom disorder, but it is not specific and should not be used to make the diagnosis. Similarly, the concept of *secondary gain* (i.e., when individuals derive external benefits such as money or release from responsibilities) is also not specific to functional neurological symptom disorder.

Prevalence

Transient functional neurological symptoms are common, but the precise prevalence of the disorder is unknown. Based on research in the United States and northern Europe, the incidence of individual persistent functional neurological symptoms is estimated to be 4–12/100,000 per year. Prevalence in specialty clinics appears to be higher, although data are limited. For example, 5% of outpatients ages 9–17 in a Japanese psychiatric clinic and 6% of adult and adolescent admissions to an inpatient psychiatric hospital in Oman received a diagnosis consistent with functional neurological symptom disorder. In neurology clinics, around 5%–15% of individuals have a diagnosis of functional neurological symptom disorder in studies from Scotland and Australia.

Development and Course

Onset has been reported throughout the life course. The mean onset of nonepileptic attacks peaks at ages 20–29 years, and motor symptoms have their mean onset at ages 30–39 years. The symptoms can be transient or persistent. The prognosis may be better in younger children than in adolescents and adults.

Risk and Prognostic Factors

Temperamental. Maladaptive personality traits, especially emotional instability, are commonly associated with functional neurological symptom disorder.

Environmental. There may be a history of childhood abuse and neglect. Stressful life events including physical injury are common but not universal triggering factors.

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Genetic and physiological. The presence of neurological disease that causes similar symptoms is a risk factor (e.g., around one in five individuals with functional [nonepileptic] seizures also have epilepsy).

Course modifiers. Short duration of symptoms and agreement with the diagnosis are positive prognostic factors. Maladaptive personality traits, the presence of comorbid physical disease, and the receipt of disability benefits appear to be negative prognostic factors.

Culture-Related Diagnostic Issues

Episodes of unresponsiveness (including seizures) and motor symptoms are the most common functional neurological symptoms across cultural contexts. High comorbidity between functional

neurological and dissociative symptoms is common cross-culturally, especially in individuals with nonepileptic seizures. Changes resembling functional neurological (and dissociative) symptoms are common in certain culturally sanctioned rituals. If the symptoms are fully explained within the particular cultural context and do not result in clinically significant distress or disability, then the diagnosis of functional neurological symptom disorder is not made.

Sex- and Gender-Related Diagnostic Issues

Functional neurological symptom disorder is two to three times more common in women for most symptom presentations. One large clinical study found higher rates of cognitive impairment and weakness in men and increased past sexual and physical trauma in women.

Association With Suicidal Thoughts or Behavior

Cohort studies of functional neurological symptom disorder mostly show higher rates of suicidal thoughts and attempts. Individuals with functional symptoms in a neurology clinic have a higher rate of suicidal thoughts than individuals with recognized neurological disease. A study in Turkey of 100 consecutive psychiatric outpatients with functional neurological symptom disorder found that a history of suicide attempt was associated with risky use of alcohol, a history of childhood maltreatment, and greater severity of dissociative symptoms as compared with those who did not attempt suicide.

Functional Consequences of Functional Neurological Symptom Disorder

Individuals with functional neurological symptom disorder may have substantial physical disability. The severity of disability can be similar to that experienced by individuals with comparable recognized medical conditions.

Differential Diagnosis

Recognized neurological disease. The main differential diagnosis is recognized neurological disease that might better explain the symptoms. After a thorough neurological assessment, an unexpected neurological disease cause for the symptoms is rarely found at follow-up. However, reassessment may be required if the symptoms appear to be progressive. Functional neurological symptom disorder commonly coexists with recognized neurological disease and may be part of the prodromal state of some progressive neurological diseases.

Somatic symptom disorder. Functional neurological symptom disorder may be diagnosed in addition to somatic symptom disorder. Most of the somatic symptoms

encountered in somatic symptom disorder cannot be demonstrated to be clearly incompatible with recognized neurological or medical disease, whereas in functional neurological symptom disorder, such incompatibility is required for the diagnosis.

Factitious disorder and malingering. Functional neurological symptom disorder describes genuinely experienced symptoms that are not intentionally produced (i.e., not feigned). However, definite

evidence of feigning (e.g., marked discrepancy between reported and observed activities of daily living) would suggest malingering if the individual's apparent aim is to obtain an obvious external reward, or factitious disorder in the absence of such reward.

Dissociative disorders. Dissociative symptoms are common in individuals with functional neurological symptom disorder. If both functional neurological symptom disorder and a dissociative disorder are present, both diagnoses should be made.

Body dysmorphic disorder. Individuals with body dysmorphic disorder are excessively concerned about a perceived defect in their physical appearance but do not complain of symptoms of sensory or motor functioning in the affected body part.

Depressive disorders. In depressive disorders, individuals may report general heaviness of their limbs, whereas the weakness of functional neurological symptom disorder is more focal and prominent. Depressive disorders are also differentiated by the presence of core depressive symptoms.

Panic disorder. Episodic neurological symptoms (e.g., tremor and paresthesia) can occur in both functional neurological symptom disorder and panic attacks. In panic attacks, the neurological symptoms are typically associated with characteristic cardiorespiratory symptoms and retained awareness. Loss of awareness with amnesia for the attack occurs in functional seizures but not in panic attacks.

Comorbidity

Anxiety disorders, especially panic disorder, and depressive disorders commonly co-occur with functional neurological symptom disorder. Somatic symptom disorder may co-occur as well. Personality disorders are more common in individuals with functional neurological symptom disorder than in the general population. Neurological or other medical conditions commonly coexist with functional neurological symptom disorder as well.

Psychological Factors Affecting Other Medical Conditions

Diagnostic Criteria

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- A. A medical symptom or condition (other than a mental disorder) is present.
- B. Psychological or behavioral factors adversely affect the medical condition in one of the following ways:
 - 1. The factors have influenced the course of the medical condition as shown by a close temporal association between the psychological factors and the development or exacerbation of, or delayed recovery from, the medical condition.
 - 2. The factors interfere with the treatment of the medical condition (e.g., poor adherence).

3. The factors constitute additional well-established health risks for the individual.

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4. The factors influence the underlying pathophysiology, precipitating or exacerbating symptoms or necessitating medical attention.
- C. The psychological and behavioral factors in Criterion B are not better explained by another mental disorder (e.g., panic disorder, major depressive disorder, posttraumatic stress disorder).

Specify current severity:

Mild: Increases medical risk (e.g., inconsistent adherence with antihypertension treatment).

Moderate: Aggravates underlying medical condition (e.g., anxiety aggravating asthma).

Severe: Results in medical hospitalization or emergency room visit.

Extreme: Results in severe, life-threatening risk (e.g., ignoring heart attack symptoms).

Diagnostic Features

The essential feature of psychological factors affecting other medical conditions is the presence of one or more clinically significant psychological or behavioral factors that adversely affect a medical condition by increasing the risk for suffering, death, or disability (Criterion B). These factors can adversely affect the medical condition by influencing its course or treatment, by constituting an additional well-established health risk factor, or by influencing the underlying pathophysiology to precipitate or exacerbate symptoms or to necessitate medical attention.

Psychological or behavioral factors include psychological distress, patterns of interpersonal interaction, coping styles, and maladaptive health behaviors, such as denial of symptoms or poor adherence to medical recommendations. Common clinical examples are anxiety-exacerbating asthma, denial of need for treatment for acute chest pain, and manipulation of insulin by an individual with diabetes wishing to lose weight. Many different psychological factors have been demonstrated to adversely influence medical conditions—for example, symptoms of depression or anxiety, stressful life events, relationship style, personality traits, and coping styles. The adverse effects can range from acute, with immediate medical consequences (e.g., Takotsubo cardiomyopathy), to chronic, occurring over a long period of time (e.g., chronic occupational stress increasing risk for hypertension). Affected medical conditions can be those with clear pathophysiology (e.g., diabetes, cancer, coronary disease), functional syndromes (e.g., migraine, irritable bowel syndrome, fibromyalgia), or idiopathic medical symptoms (e.g., pain, fatigue, dizziness).

This diagnosis should be reserved for situations in which the effect of the psychological factor on the medical condition is evident and the psychological factor has clinically significant effects on the course or outcome of the medical condition. Abnormal psychological or behavioral

symptoms that develop in response to a medical condition are more properly coded as an adjustment disorder (a clinically significant psychological response to an identifiable stressor). There must be reasonable evidence to suggest an association between the psychological factors and the medical condition, although it may often not be possible to demonstrate direct causality or the mechanisms underlying the relationship.

Prevalence

The prevalence of psychological factors affecting other medical conditions is unclear. In U.S. private insurance billing data, it was a more common diagnosis than DSM-IV somatic symptom disorders.

Development and Course

Psychological factors affecting other medical conditions can occur across the lifespan. Particularly with young children, corroborative history from parents or school can assist the

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diagnostic evaluation. Some conditions are characteristic of particular life stages (e.g., in older individuals, the stress associated with acting as a caregiver for an ill spouse or partner).

Culture-Related Diagnostic Issues

Many differences between cultural contexts may influence psychological factors and their effects on medical conditions, such as those in language and communication style, idioms of distress, explanatory models of illness, patterns of seeking health care, service availability and organization, doctor-patient relationships and other healing practices, family and gender roles, and attitudes toward pain and death. Psychological factors affecting other medical conditions must be differentiated from culturally specific coping behaviors such as accessing faith, spiritual, or traditional healers or other variations in illness management that are acceptable within cultural contexts and represent an attempt to help heal the medical condition. These local practices may complement rather than obstruct evidence-based interventions. Use of alternative healing practices may delay use of medical services and affect outcomes, but when the intent of the healing practice is to address the problem in a culturally sanctioned way, these practices should not be pathologized as psychological factors affecting other medical conditions.

Functional Consequences of Psychological Factors Affecting Other Medical Conditions

Psychological and behavioral factors have been demonstrated to affect the course of many medical diseases.

Differential Diagnosis

Mental disorder due to another medical condition. A temporal association between symptoms of a mental disorder and those of a medical condition is also characteristic of a mental disorder due to another medical condition, but the presumed causality is in the opposite direction. In a mental

disorder due to another medical condition, the medical condition is judged to be causing the mental disorder through a direct physiological mechanism. In psychological factors affecting other medical conditions, the psychological or behavioral factors are judged to affect the course of the medical condition.

Adjustment disorders. Abnormal psychological or behavioral symptoms that develop in response to a medical condition are more properly coded as an adjustment disorder (a clinically significant psychological response to an identifiable stressor). For example, an individual with angina that is precipitated whenever he becomes enraged would be diagnosed as having psychological factors affecting other medical conditions, whereas an individual with angina who developed maladaptive anticipatory anxiety would be diagnosed as having an adjustment disorder with anxiety. In clinical practice, however, psychological factors and a medical condition are often mutually exacerbating (e.g., anxiety as both a precipitant and a consequence of angina), in which case the distinction is arbitrary. Other mental disorders frequently result in medical complications, most notably substance use disorders (e.g., alcohol use disorder, tobacco use disorder). If an individual has a coexisting major mental disorder that adversely affects or causes another medical condition, diagnoses of the mental disorder and the medical condition are usually sufficient. Psychological factors affecting other medical conditions is diagnosed when the psychological traits or behaviors do not meet criteria for a mental diagnosis.

Somatic symptom disorder. Somatic symptom disorder is characterized by a combination of distressing somatic symptoms and excessive or maladaptive thoughts, feelings, and behavior in response to these symptoms or associated health concerns. The individual may or may not have a diagnosable medical condition. In contrast, in psychological factors

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affecting other medical conditions, the psychological factors adversely affect a medical condition; the individual's thoughts, feelings, and behavior are not necessarily excessive. The difference is one of emphasis, rather than a clear-cut distinction. In psychological factors affecting other medical conditions, the emphasis is on the exacerbation of the medical condition (e.g., an individual with angina that is precipitated whenever he becomes anxious). In somatic symptom disorder, the emphasis is on maladaptive thoughts, feelings, and behavior (e.g., an individual with angina who worries constantly that she will have a heart attack, takes her blood pressure multiple times per day, and restricts her activities).

Illness anxiety disorder. Illness anxiety disorder is characterized by high illness anxiety that is distressing and/or disruptive to daily life with minimal somatic symptoms. The focus of clinical concern is the individual's worry about having a disease; in most cases, no serious disease is present. In psychological factors affecting other medical conditions, anxiety may be a relevant psychological factor affecting a medical condition, but the clinical concern is the adverse effects on the medical condition.

Comorbidity

By definition, the diagnosis of psychological factors affecting other medical conditions entails a relevant psychological or behavioral syndrome or trait and a comorbid medical condition.

Factitious Disorder

Diagnostic Criteria

Factitious Disorder Imposed on Self

F68.10

- A. Falsification of physical or psychological signs or symptoms, or induction of injury or disease, associated with identified deception.
- B. The individual presents himself or herself to others as ill, impaired, or injured.
- C. The deceptive behavior is evident even in the absence of obvious external rewards.
- D. The behavior is not better explained by another mental disorder, such as delusional disorder or another psychotic disorder.

Specify:

Single episode

Recurrent episodes (two or more events of falsification of illness and/or induction of injury)

Factitious Disorder Imposed on Another (Previously Factitious Disorder by Proxy)

F68.A

- A. Falsification of physical or psychological signs or symptoms, or induction of injury or disease, in another, associated with identified deception.
- B. The individual presents another individual (victim) to others as ill, impaired, or injured.
- C. The deceptive behavior is evident even in the absence of obvious external rewards.
- D. The behavior is not better explained by another mental disorder, such as delusional disorder or another psychotic disorder.

Note: The perpetrator, not the victim, receives this diagnosis.

Specify:

Single episode

Recurrent episodes (two or more events of falsification of illness and/or induction of injury)

Recording Procedures

When an individual falsifies illness in another (e.g., children, adults, pets), the diagnosis is factitious disorder imposed on another. The perpetrator, not the victim, is given the diagnosis.

The victim may be given an abuse diagnosis (e.g., T74.12X; see the chapter “Other Conditions That May Be a Focus of Clinical Attention”). If an individual with factitious disorder imposed on another has also deceptively represented his or her own illness or injury, both factitious disorder imposed on self and on another can be diagnosed.

Diagnostic Features

The essential feature of factitious disorder is the falsification of medical or psychological signs and symptoms in the individual or others that are associated with the identified deception. Individuals with factitious disorder can also seek treatment for themselves or another following induction of injury or disease. The diagnosis requires demonstrating that the individual is taking surreptitious actions to misrepresent, simulate, or cause signs or symptoms of illness or injury even in the absence of obvious external rewards. The diagnosis of factitious disorder emphasizes the objective identification of falsification of signs and symptoms of illness and not the individual motivations of the falsifier. Methods of illness falsification can include exaggeration, fabrication, simulation, and induction. While a preexisting medical condition may be present, the deceptive behavior or induction of injury associated with deception causes others to view such individuals (or, in the case of factitious disorder imposed on another, the victim) as more ill or impaired, and this can lead to excessive clinical intervention. Individuals with factitious disorder might, for example, report feelings of depression and suicidal thoughts or behavior following the death of a spouse despite the death not being true or the individual's not having a spouse; deceptively report episodes of neurological symptoms (e.g., seizures, dizziness, or blacking out); manipulate a laboratory test (e.g., by adding blood to urine) to falsely indicate an abnormality; falsify medical records to indicate an illness; ingest a substance (e.g., insulin or warfarin) to induce an abnormal laboratory result or illness; or physically injure themselves or induce illness in themselves or another (e.g., by injecting fecal material to produce an abscess or to induce sepsis). Although individuals with factitious disorder most often present to health care professionals for treatment of their factitious symptoms, some individuals with factitious disorder choose to mislead community members in person or online about illness or injury without necessarily engaging health care professionals.

Associated Features

Individuals with factitious disorder imposed on self or factitious disorder imposed on another are at risk for experiencing great psychological distress or functional impairment by causing harm to themselves and others. Family, friends, faith leaders, and health care professionals are also often adversely affected by their behavior (e.g., devoted time, attention, and resources to provide medical care and emotional support to the falsifier). Individuals with factitious disorder imposed on another sometimes falsely allege the presence of educational deficits or disabilities in their children for which they demand special attention, often at considerable inconvenience to education professionals.

Whereas some aspects of factitious disorders might represent criminal behavior (e.g., factitious disorder imposed on another, in which the parent's actions represent abuse and maltreatment of a child), such criminal behavior and mental illness are not mutually exclusive. Moreover, such behaviors, including the induction of injury or disease, are associated with deception.

Prevalence

The prevalence of factitious disorder is unknown, likely because of the role of deception in this population. Further complicating efforts at determining prevalence is the fact that health care professionals infrequently record the diagnosis, even in recognized cases.

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Based on a study of general hospital inpatients in the United States referred for psychiatric consultation, it is estimated that almost 1% have presentations that meet the criteria for factitious disorder. Factitious disorder imposed on self or another appears to be encountered more frequently in tertiary care settings than at primary care sites.

Development and Course

The course of factitious disorder is usually one of intermittent episodes. Single episodes and episodes that are characterized as persistent and unremitting are both less common. Onset is usually in early adulthood, often after hospitalization for a medical condition or a mental disorder. When imposed on another, the disorder may begin after hospitalization of the individual's child or other dependent. In individuals with recurrent episodes of falsification of signs and symptoms of illness and/or induction of injury, this pattern of successive deceptive contact with medical personnel, including hospitalizations, may become lifelong.

Sex- and Gender-Related Diagnostic Issues

While the prevalence is not known, a pooled analysis of all case series and studies finds that two-thirds of individuals with factitious disorder are women and one-third are men.

Differential Diagnosis

Deception to avoid legal liability. Caregivers who lie about abuse injuries in dependents solely to protect themselves from liability are not diagnosed with factitious disorder imposed on another because protection from liability is an external reward (Criterion C, the deceptive behavior is evident even in the absence of obvious external rewards). Such caregivers who, upon observation, analysis of medical records, and/or interviews with others, are found to lie more extensively than needed for immediate self-protection are diagnosed with factitious disorder imposed on another.

Somatic symptom and related disorders. In somatic symptom disorder and the care-seeking type of illness anxiety disorder, there may be excessive attention and treatment seeking for perceived medical concerns, but there is no evidence that the individual is providing false information or behaving deceptively.

Malingering. Malingering is differentiated from factitious disorder by the intentional reporting of symptoms for personal gain (e.g., money, time off work). In contrast, the diagnosis of factitious disorder requires that the illness falsification is not fully accounted for by external rewards. Factitious disorder and malingering are not mutually exclusive, however. The motives in any single case might be multiple and shifting depending on the circumstances and reactions of others.

Functional neurological symptom disorder (conversion disorder). Functional neurological symptom disorder is characterized by neurological symptoms that are inconsistent with neurological pathophysiology. Factitious disorder with neurological symptoms is distinguished from functional neurological symptom disorder by evidence of deceptive falsification of symptoms.

Borderline personality disorder. Deliberate physical self-harm in the absence of suicidal intent can also occur in association with other mental disorders such as borderline personality disorder. Factitious disorder requires that the induction of injury occur in association with deception.

Medical condition or mental disorder not associated with intentional symptom falsification. Presentation of signs and symptoms of illness that do not conform to an identifiable medical condition or mental disorder increases the likelihood of the presence of a factitious disorder. However, the diagnosis of factitious disorder does not exclude the

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presence of a true medical condition or mental disorder, as comorbid illness often occurs in the individual along with factitious disorder. For example, individuals who might manipulate blood sugar levels to produce symptoms may also have diabetes.

Other Specified Somatic Symptom and Related Disorder

F45.8

This category applies to presentations in which symptoms characteristic of a somatic symptom and related disorder that cause clinically significant distress or impairment in social, occupational, or other important areas of functioning predominate but do not meet the full criteria for any of the disorders in the somatic symptom and related disorders diagnostic class.

Examples of presentations that can be specified using the “other specified” designation include the following:

1. **Brief somatic symptom disorder:** Duration of symptoms is less than 6 months.
2. **Brief illness anxiety disorder:** Duration of symptoms is less than 6 months.
3. **Illness anxiety disorder without excessive health-related behaviors or maladaptive avoidance:** Criterion D for illness anxiety disorder is not met.
4. **Pseudocyesis:** A false belief of being pregnant that is associated with objective signs and reported symptoms of pregnancy.

Unspecified Somatic Symptom and Related Disorder

F45.9

This category applies to presentations in which symptoms characteristic of a somatic

symptom and related disorder that cause clinically significant distress or impairment in social, occupational, or other important areas of functioning predominate but do not meet the full criteria for any of the disorders in the somatic symptom and related disorders diagnostic class. The unspecified somatic symptom and related disorder category should not be used unless there are decidedly unusual situations where there is insufficient information to make a more specific diagnosis.

Feeding and Eating Disorders

Feeding and eating disorders are characterized by a persistent disturbance of eating or eating-related behavior that results in the altered consumption or absorption of food and that significantly impairs physical health or psychosocial functioning. Diagnostic criteria are provided for pica, rumination disorder, avoidant/restrictive food intake disorder, anorexia nervosa, bulimia nervosa, and binge-eating disorder.

The diagnostic criteria for anorexia nervosa, bulimia nervosa, and binge-eating disorder result in a classification scheme that is mutually exclusive, so that during a single episode, only one of these diagnoses can be assigned. The rationale for this approach is that, despite a number of common psychological and behavioral features, the disorders differ substantially in clinical course, outcome, and treatment needs.

Some individuals with disorders described in this chapter report eating-related symptoms resembling those typically endorsed by individuals with substance use disorders, such as craving and patterns of compulsive use. This resemblance may reflect the involvement of the same neural systems, including those implicated in regulatory self-control and reward, in both groups of disorders. However, the relative contributions of shared and distinct factors in the development and perpetuation of eating and substance use disorders remain insufficiently understood.

Finally, obesity is not included in DSM-5 as a mental disorder. Obesity (excess body fat) results from the long-term excess of energy intake relative to energy expenditure. A range of genetic, physiological, behavioral, and environmental factors that vary across individuals contributes to the development of obesity; thus, obesity is not considered a mental disorder. However, there are robust associations between obesity and a number of mental disorders (e.g., binge-eating disorder, depressive and bipolar disorders, schizophrenia). The side effects of some psychotropic medications contribute importantly to the development of obesity, and obesity may be a risk factor for the development of some mental disorders (e.g., depressive disorders).

Pica

Diagnostic Criteria

- A. Persistent eating of nonnutritive, nonfood substances over a period of at least 1 month.
- B. The eating of nonnutritive, nonfood substances is inappropriate to the developmental level of the individual.
- C. The eating behavior is not part of a culturally supported or socially normative practice.

- D. If the eating behavior occurs in the context of another mental disorder (e.g., intellectual developmental disorder [intellectual disability], autism spectrum disorder, schizophrenia) or medical condition (including pregnancy), it is sufficiently severe to warrant additional clinical attention.

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Coding note: The ICD-10-CM codes for pica are **F98.3** in children and **F50.89** in adults.

Specify if:

In remission: After full criteria for pica were previously met, the criteria have not been met for a sustained period of time.

Diagnostic Features

The essential feature of pica is the eating of one or more nonnutritive, nonfood substances on a persistent basis over a period of at least 1 month (Criterion A) that is severe enough to warrant clinical attention. Typical substances ingested tend to vary with age and availability and might include paper, soap, cloth, hair, string, wool, soil, chalk, talcum powder, paint, gum, metal, pebbles, charcoal or coal, ash, clay, starch, or ice. The term *nonfood* is included because the diagnosis of pica does not apply to ingestion of diet products that have minimal nutritional content. There is typically no aversion to food in general. The eating of nonnutritive, nonfood substances must be developmentally inappropriate (Criterion B) and not part of a culturally supported or socially normative practice (Criterion C). A minimum age of 2 years is suggested for a pica diagnosis to exclude developmentally normal mouthing of objects by infants that results in ingestion. The eating of nonnutritive, nonfood substances can be an associated feature of other mental disorders (e.g., intellectual developmental disorder [intellectual disability], autism spectrum disorder, schizophrenia). If the eating behavior occurs exclusively in the context of another mental disorder, a separate diagnosis of pica should be made only if the eating behavior is sufficiently severe to warrant additional clinical attention (Criterion D).

Associated Features

Although deficiencies in vitamins or minerals (e.g., zinc, iron) have been reported, often no specific biological abnormalities are found. In some cases, pica comes to clinical attention only following general medical complications (e.g., mechanical bowel problems; intestinal obstruction, such as that resulting from a bezoar; intestinal perforation; infections such as toxoplasmosis and toxocariasis as a result of ingesting feces or dirt; poisoning, such as by ingestion of lead-based paint).

Prevalence

Limited data suggest that the prevalence of pica is approximately 5% among school-age children. Roughly one-third of pregnant women, especially those with food insecurity (i.e., without reliable access to affordable and nutritious food), engage in pica. Conditions associated with pica include lack of available food and vitamin deficiency.

Development and Course

Onset of pica can occur in childhood, adolescence, or adulthood, although childhood onset is most commonly reported. Pica can occur in otherwise normally developing children, whereas in adults it appears more likely to occur in the context of intellectual developmental disorder or other mental disorders. The course of the disorder can be protracted and can result in medical emergencies (e.g., intestinal obstruction, acute weight loss, poisoning). The disorder can potentially be fatal depending on substances ingested.

Risk and Prognostic Factors

Environmental. Neglect, lack of supervision, and developmental delay can increase the risk for this condition.

Culture-Related Diagnostic Issues

In some populations, the eating of earth or other seemingly nonnutritive substances is believed to be of spiritual, medicinal, or other social value, or may be a culturally supported

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or socially normative practice. Such behavior does not warrant a diagnosis of pica (Criterion C). Pica behavior may be prevalent in some cultural groups, but it should not be assumed to be socially normative without further evaluation.

Sex- and Gender-Related Diagnostic Issues

Pica occurs in both genders. The eating of nonnutritive, nonfood substances may manifest in pregnancy, when specific cravings (e.g., chalk or ice) may occur. The diagnosis of pica during pregnancy is appropriate only if such cravings lead to the ingestion of nonnutritive, nonfood substances to the extent that the eating of these substances poses potential medical risks. A worldwide meta-analysis showed the prevalence rate of pica to be 28% during pregnancy and/or the postpartum period.

Diagnostic Markers

Abdominal flat plate radiography, ultrasound, and other scanning methods may reveal obstructions related to pica. Blood tests and other laboratory tests can be used to ascertain levels of poisoning or the nature of infection.

Functional Consequences of Pica

Pica can significantly impair physical functioning, but it is rarely the sole cause of impairment in social functioning. Pica often occurs with other disorders associated with impaired social functioning.

Differential Diagnosis

Eating of nonnutritive, nonfood substances may occur during the course of other mental

disorders (e.g., autism spectrum disorder, schizophrenia) and in Kleine-Levin syndrome. In any such instance, an additional diagnosis of pica should be given only if the eating behavior is sufficiently persistent and severe to warrant additional clinical attention.

Anorexia nervosa. Pica can usually be distinguished from the other feeding and eating disorders by the consumption of nonnutritive, nonfood substances. It is important to note, however, that some presentations of anorexia nervosa include ingestion of nonnutritive, nonfood substances, such as paper tissues, as a means of attempting to control appetite. In such cases, when the eating of nonnutritive, nonfood substances is primarily used as a means of weight control, anorexia nervosa should be the primary diagnosis.

Factitious disorder. Some individuals with factitious disorder may intentionally ingest foreign objects as part of the pattern of falsification of physical symptoms. In such instances, there is an element of deception that is consistent with deliberate induction of injury or disease.

Nonsuicidal self-injury and nonsuicidal self-injury behaviors in personality disorders. Some individuals may swallow potentially harmful items (e.g., pins, needles, knives) in the context of maladaptive behavior patterns associated with personality disorders or nonsuicidal self-injury.

Comorbidity

Disorders most commonly comorbid with pica are autism spectrum disorder and intellectual developmental disorder (intellectual disability) and, to a lesser degree, schizophrenia and obsessive-compulsive disorder. Pica can be associated with trichotillomania (hair-pulling disorder) and excoriation (skin-picking) disorder. In comorbid presentations, the hair or skin is typically ingested. Pica can also be associated with avoidant/restrictive food intake disorder, particularly in individuals with a strong sensory component to their presentation. When an individual is known to have pica, assessment should include consideration of the possibility of gastrointestinal complications, poisoning, infection, and nutritional deficiency.

Rumination Disorder

Diagnostic Criteria	F98.21
A. Repeated regurgitation of food over a period of at least 1 month. Regurgitated food may be re-chewed, re-swallowed, or spit out. B. The repeated regurgitation is not attributable to an associated gastrointestinal or other medical condition (e.g., gastroesophageal reflux, pyloric stenosis). C. The eating disturbance does not occur exclusively during the course of anorexia nervosa, bulimia nervosa, binge-eating disorder, or avoidant/restrictive food intake disorder. D. If the symptoms occur in the context of another mental disorder (e.g., intellectual developmental disorder [intellectual disability] or another neurodevelopmental	

disorder), they are sufficiently severe to warrant additional clinical attention.

Specify if:

In remission: After full criteria for rumination disorder were previously met, the criteria have not been met for a sustained period of time.

Diagnostic Features

The essential feature of rumination disorder is the repeated regurgitation of food occurring after feeding or eating over a period of at least 1 month (Criterion A). Previously swallowed food that may be partially digested is brought up into the mouth without apparent nausea, involuntary retching, or disgust. The food may be re-chewed and then ejected from the mouth or re-swallowed. Regurgitation in rumination disorder should be frequent, occurring at least several times per week, typically daily. The behavior is not better explained by an associated gastrointestinal or other medical condition (e.g., gastroesophageal reflux, pyloric stenosis) (Criterion B) and does not occur exclusively during the course of anorexia nervosa, bulimia nervosa, binge-eating disorder, or avoidant/restrictive food intake disorder (Criterion C). If the symptoms occur in the context of another mental disorder (e.g., intellectual developmental disorder [intellectual disability]), they must be sufficiently severe to warrant additional clinical attention (Criterion D) and should represent a primary aspect of the individual's presentation requiring intervention. The disorder may be diagnosed across the life span, particularly in individuals who also have intellectual developmental disorder. Many individuals with rumination disorder can be directly observed engaging in the behavior by the clinician. In other instances diagnosis can be made on the basis of self-report or corroborative information from parents or caregivers. Individuals may describe the behavior as habitual or outside of their control.

Associated Features

Infants with rumination disorder display a characteristic position of straining and arching the back with the head held back, making sucking movements with their tongue. They may give the impression of gaining satisfaction from the activity. They may be irritable and hungry between episodes of regurgitation. Weight loss and failure to make expected weight gains are common features in infants with rumination disorder. Malnutrition may occur despite the infant's apparent hunger and the ingestion of relatively large amounts of food, particularly in severe cases, when regurgitation immediately follows each feeding episode and regurgitated food is expelled. Malnutrition might also occur in older children and adults, particularly when the regurgitation is accompanied by restriction of intake. Adolescents and adults may attempt to disguise the regurgitation behavior by placing a hand over the mouth or coughing. Some will avoid eating with others because of the acknowledged social undesirability of the

behavior. This may extend to an avoidance of eating prior to social situations, such as work or school (e.g., avoiding breakfast because it may be followed by regurgitation).

Prevalence

Although rumination disorder was historically described primarily among individuals with intellectual disability, the limited European data available on prevalence suggest that the disorder may occur in approximately 1%–2% of grade-school-age children.

Development and Course

Onset of rumination disorder can occur in infancy, childhood, adolescence, or adulthood. The age at onset in infants is usually between ages 3 and 12 months. In infants, the disorder frequently remits spontaneously, but its course can be protracted and can result in medical emergencies (e.g., severe malnutrition). It can potentially be fatal, particularly in infancy. Rumination disorder can have an episodic course or occur continuously until treated. In infants, as well as in older individuals with intellectual developmental disorder or other neurodevelopmental disorders, the regurgitation and rumination behavior appears to have a self-soothing or self-stimulating function, similar to that of other repetitive motor behaviors such as head banging.

Risk and Prognostic Factors

Environmental. Psychosocial problems such as lack of stimulation, neglect, stressful life situations, and problems in the parent-child relationship may be predisposing factors in infants and young children.

Functional Consequences of Rumination Disorder

Malnutrition secondary to repeated regurgitation may be associated with growth delay and have a negative effect on development and learning potential. Some older individuals with rumination disorder deliberately restrict their food intake because of the social undesirability of regurgitation. They may therefore present with weight loss or low weight. In older children, adolescents, and adults, social functioning is more likely to be adversely affected.

Differential Diagnosis

Gastrointestinal conditions. It is important to differentiate regurgitation in rumination disorder from other conditions characterized by gastroesophageal reflux or vomiting, such as gastroparesis, pyloric stenosis, hiatal hernia, and Sandifer syndrome in infants. These other medical conditions can usually be ruled out on the basis of history and clinical observation.

Anorexia nervosa and bulimia nervosa. Individuals with anorexia nervosa and bulimia nervosa may also engage in regurgitation with subsequent spitting out of food as a means of disposing of ingested calories because of concerns about weight gain.

Comorbidity

Regurgitation with associated rumination can occur in the context of a concurrent medical condition or another mental disorder (e.g., generalized anxiety disorder). When the regurgitation occurs in this context, a diagnosis of rumination disorder is appropriate only when the severity of the disturbance exceeds that routinely associated with such conditions or disorders and warrants additional clinical attention.

Avoidant/Restrictive Food Intake Disorder

Diagnostic Criteria	F50.82
A. An eating or feeding disturbance (e.g., apparent lack of interest in eating or food; avoidance based on the sensory characteristics of food; concern about aversive consequences of eating) associated with one (or more) of the following: 1. Significant weight loss (or failure to achieve expected weight gain or faltering growth in children). 2. Significant nutritional deficiency. 3. Dependence on enteral feeding or oral nutritional supplements. 4. Marked interference with psychosocial functioning. B. The disturbance is not better explained by lack of available food or by an associated culturally sanctioned practice. C. The eating disturbance does not occur exclusively during the course of anorexia nervosa or bulimia nervosa, and there is no evidence of a disturbance in the way in which one's body weight or shape is experienced. D. The eating disturbance is not attributable to a concurrent medical condition or not better explained by another mental disorder. When the eating disturbance occurs in the context of another condition or disorder, the severity of the eating disturbance exceeds that routinely associated with the condition or disorder and warrants additional clinical attention.	
Specify if:	
In remission: After full criteria for avoidant/restrictive food intake disorder were previously met, the criteria have not been met for a sustained period of time.	

Diagnostic Features

Avoidant/restrictive food intake disorder replaces and extends the DSM-IV diagnosis of feeding disorder of infancy or early childhood to include older children, adolescents, and adults. The main diagnostic feature of avoidant/restrictive food intake disorder is avoidance or restriction of food intake that is associated with one or more of the following consequences: significant weight loss, significant nutritional deficiency (or related health impact), dependence on enteral feeding or oral nutritional supplements, or marked interference with psychosocial functioning (Criterion A).

In some individuals, food avoidance or restriction is based on the sensory characteristics of qualities of food, such as extreme sensitivity to appearance, color, smell, texture, temperature, or taste. Such behavior has been described as “restrictive eating,” “selective eating,” “choosy eating,” “perseverant eating,” “chronic food refusal,” and “food neophobia” and may manifest as

refusal to eat particular brands of foods or to tolerate the smell of food being eaten by others. Individuals with heightened sensory sensitivities associated with autism may show similar behaviors.

In other individuals, food avoidance or restriction represents a conditioned negative response associated with food intake following, or in anticipation of, an aversive experience, such as choking; a traumatic procedure, usually involving the gastrointestinal tract (e.g., esophagoscopy); or repeated vomiting. The terms *functional dysphagia* and *globus hystericus* have also been used for such conditions.

In yet other individuals, food avoidance or restriction manifests as a lack of interest in eating or food.

The determination of whether weight loss is significant (Criterion A1) is a clinical judgment; instead of losing weight, children and adolescents who have not completed growth may not maintain weight or height increases along their developmental trajectory.

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Determination of significant nutritional deficiency (Criterion A2) is also based on clinical assessment (e.g., assessment of dietary intake, physical examination, and laboratory testing), and related impact on physical health can be of a similar severity to that seen in anorexia nervosa (e.g., hypothermia, bradycardia, anemia). In severe cases, particularly in infants, malnutrition can be life-threatening. “Dependence” on enteral feeding or oral nutritional supplements (Criterion A3) means that supplementary feeding is required to sustain adequate intake. Examples of individuals requiring supplementary feeding include infants with failure to thrive who require nasogastric tube feeding, children with neurodevelopmental disorders who are dependent on nutritionally complete supplements, and individuals who rely on gastrostomy tube feeding or complete oral nutrition supplements in the absence of an underlying medical condition. Inability to participate in normal social activities, such as eating with others, attending school or work, or sustaining relationships as a result of the disturbance would indicate marked interference with psychosocial functioning (Criterion A4). Substantial disruption of family functioning (e.g., marked restriction of foods permitted in the home, inordinate accommodations to provide foods from specific grocery stores or restaurants) may also satisfy Criterion A4.

Avoidant/restrictive food intake disorder does not include avoidance or restriction of food intake related to lack of availability of food (e.g., food insecurity) or to cultural practices (e.g., religious fasting or normal dieting) (Criterion B). The disturbance is not better explained by excessive concern about body weight or shape (Criterion C) or by concurrent medical factors or mental disorders (Criterion D).

Associated Features

Several features may be associated with food avoidance or reduced food intake, and these features may differ across ages. Very young infants may present with food refusal, gagging, or vomiting. Infants and young children may not engage with a primary caregiver during feeding or communicate hunger in favor of other activities. In older children and adolescents, food avoidance or restriction may be associated with more generalized emotional difficulties that do not meet diagnostic criteria for an anxiety, depressive, or bipolar disorder, sometimes called

“food avoidance emotional disorder.”

Prevalence

Little information is available on the prevalence of avoidant/restrictive food intake disorder. A study in Australia reported a frequency of 0.3% among individuals age 15 years or older.

Development and Course

Food avoidance or restriction associated with insufficient intake or lack of interest in eating most commonly develops in infancy or early childhood and may persist in adulthood. Likewise, avoidance based on sensory characteristics of food tends to arise in the first decade of life but may persist into adulthood. Avoidance related to aversive consequences can arise at any age. The scant literature regarding long-term outcomes suggests that food avoidance or restriction based on sensory aspects is relatively stable and long-standing, but when persisting into adulthood, such avoidance/restriction can be associated with relatively normal functioning. There is currently insufficient evidence directly linking avoidant/restrictive food intake disorder and subsequent onset of an eating disorder.

Infants with avoidant/restrictive food intake disorder may be irritable and difficult to console during feeding, or may appear apathetic and withdrawn. In some instances, caregiver-child interaction may contribute to the infant’s feeding problem (e.g., presenting food inappropriately, or interpreting the infant’s behavior as an act of aggression or rejection). Inadequate nutritional intake may exacerbate the associated features (e.g., irritability, developmental lags) and further contribute to feeding difficulties. Associated factors

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include infant temperament or developmental impairments that reduce an infant’s responsiveness to feeding. Coexisting parental psychopathology, or child abuse or neglect, is suggested if feeding and weight improve in response to changing caregivers. In infants, children, and prepubertal adolescents, avoidant/restrictive food intake disorder may be associated with growth delay, and the resulting malnutrition negatively affects development and learning potential. In older children, adolescents, and adults, social functioning tends to be adversely affected. Regardless of the age, family function may be affected, with heightened stress at mealtimes and in other feeding or eating contexts involving friends and relatives.

Avoidant/restrictive food intake disorder manifests more commonly in children and adolescents than in adults, and there may be a long delay between onset and clinical presentation. Triggers for presentation vary considerably and include physical, social, and emotional difficulties.

Risk and Prognostic Factors

Temperamental. Anxiety disorders, autism spectrum disorder, obsessive-compulsive disorder, and attention-deficit/hyperactivity disorder may increase risk for avoidant or restrictive feeding or eating behavior characteristic of the disorder.

Environmental. Environmental risk factors for avoidant/restrictive food intake disorder include familial anxiety. Higher rates of feeding disturbances may occur in children of mothers with

eating disorders.

Genetic and physiological. History of gastrointestinal conditions, gastroesophageal reflux disease, vomiting, and a range of other medical problems has been associated with feeding and eating behaviors characteristic of avoidant/restrictive food intake disorder.

Culture-Related Diagnostic Issues

Presentations similar to avoidant/restrictive food intake disorder occur in various populations, including in the United States, Canada, Australia, Europe, Japan, and China. Avoidant/restrictive food intake disorder should not be diagnosed when avoidance of food intake is solely related to specific religious or cultural practices.

Sex- and Gender-Related Diagnostic Issues

Avoidant/restrictive food intake disorder appears to be approximately equally common in boys and girls, but avoidant/restrictive food intake disorder comorbid with autism spectrum disorder has a male predominance. Food avoidance or restriction related to altered sensory sensitivities can occur in some physiological conditions, most notably pregnancy, but is not usually extreme and does not meet full criteria for the disorder.

Functional Consequences of Avoidant/Restrictive Food Intake Disorder

Associated developmental and functional limitations include impairment of physical development and social difficulties that can have a significant negative impact on family function.

Differential Diagnosis

Restriction of food intake is a nonspecific symptom that can accompany a number of mental disorders and medical conditions and that can also be developmentally appropriate. Avoidant/restrictive food intake disorder can be diagnosed concurrently with the disorders below if all criteria are met, and the eating disturbance requires specific clinical attention.

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Other medical conditions (e.g., gastrointestinal disease, food allergies and intolerances, occult malignancies).

Restriction of food intake may occur in other medical conditions, especially those with ongoing symptoms such as vomiting, loss of appetite, nausea, abdominal pain, or diarrhea. A diagnosis of avoidant/restrictive food intake disorder requires that the disturbance of intake is beyond that directly accounted for by physical symptoms consistent with a medical condition; the eating disturbance may also persist after being triggered by a medical condition and following resolution of the medical condition.

Underlying medical or comorbid mental conditions may complicate feeding and eating. Because older individuals, postsurgical patients, and individuals receiving chemotherapy often lose their appetite, an additional diagnosis of avoidant/restrictive food intake disorder requires that the eating disturbance is a primary focus for intervention.

Obsessive-compulsive and related disorder due to pediatric acute-onset neuropsychiatric

syndrome.

Acute-onset symptoms, late age at onset, or atypical symptoms suggest the need for a thorough assessment to rule out the diagnosis of obsessive-compulsive and related disorder due to pediatric acute-onset neuropsychiatric syndrome (PANS). PANS is characterized by abrupt, dramatic onset of obsessive-compulsive symptoms or severely restricted food intake, together with a range of additional neuropsychiatric symptoms.

Specific neurological/neuromuscular, structural, or congenital disorders and conditions associated with feeding difficulties.

Feeding difficulties are common in a number of congenital and neurological conditions often related to problems with oral/esophageal/pharyngeal structure and function, such as hypotonia of musculature, tongue protrusion, and unsafe swallowing. Avoidant/restrictive food intake disorder can be diagnosed in individuals with such presentations as long as all diagnostic criteria are met.

Reactive attachment disorder. Some degree of withdrawal from caregivers is characteristic of reactive attachment disorder and can lead to a disturbance in the caregiver-child relationship that can affect feeding and the child's intake. Avoidant/restrictive food intake disorder should be diagnosed concurrently only if all criteria are met for both disorders and the feeding disturbance is a primary focus for intervention.

Autism spectrum disorder. Individuals with autism spectrum disorder often present with rigid eating behaviors and heightened sensory sensitivities. However, these features do not always result in the level of impairment that would be required for a diagnosis of avoidant/restrictive food intake disorder. Avoidant/restrictive food intake disorder should be diagnosed concurrently only if all criteria are met for both disorders and when the eating disturbance requires specific treatment.

Specific phobia, social anxiety disorder, and other anxiety disorders. Specific phobia, other type, includes as an example "situations that may lead to choking or vomiting" and can represent the primary trigger for the fear, anxiety, or avoidance required for diagnosis. Distinguishing specific phobia from avoidant/restrictive food intake disorder can be difficult when a fear of choking or vomiting has resulted in food avoidance. Although avoidance or restriction of food intake secondary to a pronounced fear of choking or vomiting can be conceptualized as specific phobia, in situations when the eating problem becomes the primary focus of clinical attention, avoidant/restrictive food intake disorder becomes the appropriate diagnosis. In social anxiety disorder, the individual may present with a fear of being observed by others while eating, which can also occur in avoidant/restrictive food intake disorder.

Anorexia nervosa. Restriction of energy intake relative to requirements leading to significantly low body weight is a core feature of anorexia nervosa. However, individuals with anorexia nervosa also display a fear of gaining weight or of becoming fat, or persistent behavior that interferes with weight gain, as well as specific disturbances in

relation to perception and experience of their own body weight and shape. These features are not present in avoidant/restrictive food intake disorder, and the two disorders should not be diagnosed concurrently. Differential diagnosis between avoidant/restrictive food intake disorder and anorexia nervosa may be difficult, especially in late childhood and early adolescence, because these disorders may share a number of common symptoms (e.g., food avoidance, low

weight). Differential diagnosis is also potentially difficult in individuals with anorexia nervosa who deny any fear of fatness but nonetheless engage in persistent behaviors that prevent weight gain and who do not recognize the medical seriousness of their low weight—a presentation sometimes termed “non-fat phobic anorexia nervosa.” Full consideration of symptoms, course, and family history is advised, and diagnosis may be best made in the context of a clinical relationship over time. In some individuals, avoidant/restrictive food intake disorder might precede the onset of anorexia nervosa.

Obsessive-compulsive disorder. Individuals with obsessive-compulsive disorder may present with avoidance or restriction of intake in relation to preoccupations with food or ritualized eating behavior. Avoidant/restrictive food intake disorder should be diagnosed concurrently only if all criteria are met for both disorders and when the aberrant eating is a major aspect of the clinical presentation requiring specific intervention.

Major depressive disorder. In major depressive disorder, appetite might be affected to such an extent that individuals present with significantly restricted food intake, usually in relation to overall energy intake and often associated with weight loss. Usually appetite loss and related reduction of intake abate with resolution of mood problems. Avoidant/restrictive food intake disorder should only be used concurrently if full criteria are met for both disorders and when the eating disturbance requires specific treatment.

Schizophrenia spectrum disorders. Individuals with schizophrenia, delusional disorder, or other psychotic disorders may exhibit odd eating behaviors, avoidance of specific foods because of delusional beliefs, or other manifestations of avoidant or restrictive intake. In some cases, delusional beliefs may contribute to a concern about negative consequences of ingesting certain foods. Avoidant/restrictive food intake disorder should be used concurrently only if all criteria are met for both disorders and when the eating disturbance requires specific treatment.

Factitious disorder or factitious disorder imposed on another. Avoidant/restrictive food intake disorder should be differentiated from factitious disorder or factitious disorder imposed on another. In order to assume the sick role, some individuals with factitious disorder may intentionally describe diets that are much more restrictive than those they are actually able to consume, as well as complications of such behavior, such as a need for enteral feedings or nutritional supplements, an inability to tolerate a normal range of foods, and/or an inability to participate normally in age-appropriate situations involving food. The presentation may be impressively dramatic and engaging, and the symptoms reported inconsistently. In factitious disorder imposed on another, the caregiver describes symptoms consistent with avoidant/restrictive food intake disorder and may induce physical symptoms such as failure to gain weight. As with any diagnosis of factitious disorder imposed on another, the caregiver receives the diagnosis rather than the affected individual, and diagnosis should be made only on the basis of a careful, comprehensive assessment of the affected individual, the caregiver, and their interaction.

Developmentally normal behavior. During normal development, some toddlers and children transiently narrow the variety of foods they are willing to eat. This phenomenon, sometimes referred to as “picky eating,” usually resolves spontaneously without intervention. Avoidant/restrictive food intake disorder does not include such developmentally normal behaviors unless they become sufficiently severe to lead to failure to meet appropriate nutritional

needs or produce significant impairment in functioning (Criterion A).

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Comorbidity

The most commonly observed disorders comorbid with avoidant/restrictive food intake disorder are anxiety disorders, obsessive-compulsive disorder, and neurodevelopmental disorders (specifically autism spectrum disorder, attention-deficit/hyperactivity disorder, and intellectual developmental disorder [intellectual disability]).

Anorexia Nervosa

Diagnostic Criteria

- A. Restriction of energy intake relative to requirements, leading to a significantly low body weight in the context of age, sex, developmental trajectory, and physical health. *Significantly low weight* is defined as a weight that is less than minimally normal or, for children and adolescents, less than that minimally expected.
- B. Intense fear of gaining weight or of becoming fat, or persistent behavior that interferes with weight gain, even though at a significantly low weight.
- C. Disturbance in the way in which one's body weight or shape is experienced, undue influence of body weight or shape on self-evaluation, or persistent lack of recognition of the seriousness of the current low body weight.

Coding note: The ICD-10-CM code depends on the subtype (see below).

Specify whether:

F50.01 Restricting type: During the last 3 months, the individual has not engaged in recurrent episodes of binge-eating or purging behavior (i.e., self-induced vomiting or the misuse of laxatives, diuretics, or enemas). This subtype describes presentations in which weight loss is accomplished primarily through dieting, fasting, and/or excessive exercise.

F50.02 Binge-eating/purging type: During the last 3 months, the individual has engaged in recurrent episodes of binge-eating or purging behavior (i.e., self-induced vomiting or the misuse of laxatives, diuretics, or enemas).

Specify if:

In partial remission: After full criteria for anorexia nervosa were previously met, Criterion A (low body weight) has not been met for a sustained period, but either Criterion B (intense fear of gaining weight or becoming fat or behavior that interferes with weight gain) or Criterion C (disturbances in self-perception of weight and shape) is still met.

In full remission: After full criteria for anorexia nervosa were previously met, none of the criteria have been met for a sustained period of time.

Specify current severity:

The minimum level of severity is based, for adults, on current body mass index (BMI) (see below) or, for children and adolescents, on BMI percentile. The ranges below are derived from World Health Organization categories for thinness in adults; for children and adolescents, corresponding BMI percentiles should be used. The level of severity may be increased to reflect clinical symptoms, the degree of functional disability, and the need for supervision.

Mild: BMI ≥ 17 kg/m².

Moderate: BMI 16–16.99 kg/m².

Severe: BMI 15–15.99 kg/m².

Extreme: BMI < 15 kg/m².

Subtypes

Most individuals with the binge-eating/purging type of anorexia nervosa who binge eat also purge through self-induced vomiting or the misuse of laxatives, diuretics, or enemas. Some individuals with this subtype of anorexia nervosa do not binge eat but do regularly purge after the consumption of small amounts of food.

Crossover between the subtypes over the course of the disorder is not uncommon; therefore, subtype description should be used to describe current symptoms rather than longitudinal course.

Diagnostic Features

There are three essential features of anorexia nervosa: persistent energy intake restriction; intense fear of gaining weight or of becoming fat, or persistent behavior that interferes with weight gain; and a disturbance in self-perceived weight or shape. The individual maintains a body weight that is below a minimally normal level for age, sex, developmental trajectory, and physical health (Criterion A). Individuals' body weights frequently meet this criterion following a significant weight loss, but among children and adolescents, there may alternatively be failure to make expected weight gain or to maintain a normal developmental trajectory (i.e., while growing in height) instead of weight loss.

Criterion A requires that the individual's weight be significantly low (i.e., less than minimally normal or, for children and adolescents, less than that minimally expected). Weight assessment can be challenging because normal weight range differs among individuals, and different thresholds have been published defining thinness or underweight status. Body mass index (BMI; calculated as weight in kilograms/height in meters²) is a useful measure to assess body weight for height. For adults, a BMI of 18.5 kg/m² has been employed by the Centers for Disease Control and Prevention (CDC) and the World Health Organization (WHO) as the lower limit of normal body weight. Therefore, most adults with a BMI greater than or equal to 18.5 kg/m² would not be considered to have a significantly low body weight. On the other hand, a BMI of lower than 17.0 kg/m² has been considered by the WHO to indicate moderate or severe thinness; therefore, an individual with a BMI less than 17.0 kg/m² would likely be considered to

have a significantly low weight. An adult with a BMI between 17.0 and 18.5 kg/m², or even slightly above 18.5 kg/m², might be considered to have a significantly low weight if clinical history or other physiological information supports this judgment. Adults who are not underweight by population-based standards—for example, adults with BMIs of 19.0 kg/m² or greater—should not be assigned a diagnosis of anorexia nervosa; a diagnosis of other specified feeding or eating disorder (atypical anorexia nervosa) may be considered for such individuals.

For children and adolescents, determining a BMI-for-age percentile is useful (see, e.g., the CDC BMI percentile calculator for children and teenagers; <https://www.cdc.gov/healthyweight/bmi/calculator.html>). As for adults, it is not possible to provide definitive standards for judging whether a child's or an adolescent's weight is significantly low, and variations in developmental trajectories among youth limit the utility of simple numerical guidelines. The CDC has used a BMI-for-age below the 5th percentile as suggesting underweight; children and adolescents with a BMI above this benchmark may be judged to be significantly underweight in light of failure to maintain their expected growth trajectory. However, such individuals whose BMI nonetheless remains greater than the median BMI for age should not be assigned a diagnosis of anorexia nervosa; a diagnosis of other specified feeding or eating disorder (atypical anorexia nervosa) may be considered for such individuals.

Individuals with this disorder typically display an intense fear of gaining weight or of becoming fat (Criterion B). This intense fear of becoming fat is usually not alleviated by weight loss. In fact, concern about weight gain may increase even as weight falls. Younger individuals with anorexia nervosa, as well as some adults, may not recognize or

acknowledge a fear of weight gain. In the absence of another explanation for the significantly low weight, clinician inference drawn from collateral history, observational data, physical and laboratory findings, or longitudinal course either indicating a fear of weight gain or supporting persistent behaviors that prevent it may be used to establish Criterion B.

The experience and significance of body weight and shape are distorted in these individuals (Criterion C). Some individuals feel globally overweight. Others realize that they are thin but are still concerned that certain body parts, particularly the abdomen, buttocks, and thighs, are “too fat.” They may employ a variety of techniques to evaluate their body size or weight, including frequent weighing, obsessive measuring of body parts, and persistent use of a mirror to check for perceived areas of “fat.” The self-esteem of individuals with anorexia nervosa is highly dependent on their perceptions of body shape and weight. Weight loss is often viewed as an impressive achievement and a sign of extraordinary self-discipline, whereas weight gain is perceived as an unacceptable failure of self-control. Although some individuals with this disorder may acknowledge being thin, they often do not recognize the serious medical implications of their malnourished state.

Often, the individual is brought to professional attention by family members after marked weight loss (or failure to make expected weight gains) has occurred. If individuals seek help on their own, it is usually because of distress over the somatic and psychological sequelae of starvation. It is rare for an individual with anorexia nervosa to complain of weight loss per se. In fact, individuals with anorexia nervosa frequently either lack insight into or deny the problem. It

is therefore often important to obtain information from family members or other sources to evaluate the history of weight loss and other features of the illness.

Associated Features

The semistarvation of anorexia nervosa, and the purging behaviors sometimes associated with it, can result in significant and potentially life-threatening medical conditions. The nutritional compromise associated with this disorder affects most major organ systems and can produce a variety of disturbances. Physiological disturbances, including amenorrhea and vital sign abnormalities, are common. While most of the physiological disturbances associated with malnutrition are reversible with nutritional rehabilitation, some, including loss of bone mineral density, are often not completely reversible. Behaviors such as self-induced vomiting and misuse of laxatives, diuretics, and enemas may cause a number of disturbances that lead to abnormal laboratory findings; however, some individuals with anorexia nervosa exhibit no laboratory abnormalities.

When seriously underweight, many individuals with anorexia nervosa have depressive signs and symptoms such as depressed mood, social withdrawal, irritability, insomnia, and diminished interest in sex. Because these features are also observed in individuals without anorexia nervosa who are significantly undernourished, many of the depressive features may be secondary to the physiological sequelae of semistarvation, although they may also be sufficiently severe to warrant an additional diagnosis of major depressive disorder.

Obsessive-compulsive features, both related and unrelated to food, are often prominent. Most individuals with anorexia nervosa are preoccupied with thoughts of food. Some collect recipes or hoard food. Observations of behaviors associated with other forms of starvation suggest that obsessions and compulsions related to food may be exacerbated by undernutrition. When individuals with anorexia nervosa exhibit obsessions and compulsions that are not related to food, body shape, or weight, an additional diagnosis of obsessive-compulsive disorder (OCD) may be warranted.

Other features sometimes associated with anorexia nervosa include concerns about eating in public, feelings of ineffectiveness, a strong desire to control one's environment, inflexible thinking, limited social spontaneity, and overly restrained emotional expression.

Compared with individuals with anorexia nervosa, restricting type, those with binge-eating/purging type have higher rates of impulsivity and are more likely to abuse alcohol and other drugs.

Some individuals with anorexia nervosa show excessive levels of physical activity. Increases in physical activity often precede onset of the disorder, and over the course of the disorder increased activity accelerates weight loss. During treatment, excessive activity may be difficult to control, thereby jeopardizing weight recovery.

Individuals with anorexia nervosa may misuse medications, such as by manipulating dosage, in order to achieve weight loss or avoid weight gain. Individuals with diabetes mellitus may omit or reduce insulin doses in order to minimize carbohydrate utilization.

Prevalence

According to two U.S. epidemiological studies conducted in community samples, the 12-month prevalence of anorexia nervosa ranges from 0.0% to 0.05% with much higher rates in women than in men (0% to 0.08% in women; 0% to 0.01% in men), and the lifetime prevalence ranges from 0.60% to 0.80% (0.9% to 1.42% in women; 0.12% to 0.3% in men). By contrast, one study of adolescents found similar rates in both genders.

Anorexia nervosa appears to be most prevalent in postindustrialized, high-income countries such as in the United States, many European countries, Australia, New Zealand, and Japan. Although the prevalence of anorexia nervosa is uncertain in most low- and middle-income countries, it appears to be increasing in many countries in the Global South, including in Asia and the Middle East. Anorexia nervosa occurs across U.S. ethnoracial groups; however, its prevalence seems to be lower among Latinx and non-Latinx Black Americans than among non-Latinx Whites.

Development and Course

Anorexia nervosa commonly begins during adolescence or young adulthood. It rarely begins before puberty or after age 40, but cases of both early and late onset have been described. The onset of this disorder is often associated with a stressful life event, such as leaving home for college. The course and outcome of anorexia nervosa are highly variable. Younger individuals may manifest atypical features, including denying “fear of fat.” Older individuals more likely have a longer duration of illness, and their clinical presentation may include more signs and symptoms of long-standing disorder. Clinicians should not exclude anorexia nervosa from the differential diagnosis solely on the basis of older age.

Many individuals have a period of changed eating behavior prior to full criteria for the disorder being met. Some individuals with anorexia nervosa recover fully after a single episode, with some exhibiting a fluctuating pattern of weight gain followed by relapse, and others experiencing a chronic course over many years. Hospitalization may be required to restore weight and to address medical complications. Most individuals with anorexia nervosa experience remission within 5 years of presentation. Among individuals admitted to hospitals, overall remission rates may be lower. The crude mortality rate for anorexia nervosa is approximately 5% per decade. Death most commonly results from medical complications associated with the disorder itself or from suicide.

Risk and Prognostic Factors

Temperamental. Individuals who develop anxiety disorders or display obsessional traits in childhood are at increased risk for developing anorexia nervosa.

Environmental. Historical and cross-cultural variability in the prevalence of anorexia nervosa supports its association with cultures and settings in which thinness is valued. Occupations and avocations that encourage thinness, such as modeling and elite athletics, are also associated with increased risk.

Genetic and physiological. There is an increased risk for anorexia nervosa and for other eating and psychiatric disorders among biological relatives of individuals with anorexia nervosa. Genome-wide association studies have begun to identify specific risk loci, including loci associated with other psychiatric disorders and with metabolic traits such as insulin resistance and lipid profile. A range of brain abnormalities, many suggesting abnormal processing of reward, has been described in anorexia nervosa using functional imaging technologies such as functional magnetic resonance imaging and positron emission tomography. The degree to which these findings reflect changes associated with malnutrition versus primary abnormalities associated with the disorder is unclear.

Culture-Related Diagnostic Issues

Anorexia nervosa occurs across culturally and socially diverse populations, although available evidence suggests cross-cultural variation in its occurrence and presentation. The presentation of weight concerns among individuals with feeding and eating disorders varies substantially across cultural contexts. The absence of an expressed intense fear of weight gain, sometimes referred to as “fat phobia,” appears to be relatively more common in populations in Asia, where the rationale for dietary restriction is commonly related to a more culturally sanctioned complaint such as gastrointestinal discomfort. Mental health service utilization in the United States among individuals with an eating disorder is significantly lower among underserved ethnic and racialized groups.

Diagnostic Markers

The following laboratory abnormalities may be observed in anorexia nervosa; their presence may serve to increase diagnostic confidence.

Hematology. Leukopenia is common, with the loss of all cell types but usually with apparent lymphocytosis. Mild anemia can occur, as well as thrombocytopenia and, rarely, bleeding problems.

Serum chemistry. Dehydration may be reflected by an elevated blood urea nitrogen level. Hypercholesterolemia is common. Hepatic enzyme levels may be elevated. Hypomagnesemia, hypozincemia, hypophosphatemia, and hyperamylasemia are occasionally observed. Self-induced vomiting may lead to metabolic alkalosis (elevated serum bicarbonate), hyponatremia, and hypokalemia; laxative abuse may cause a mild metabolic acidosis.

Endocrine. Serum thyroxine (T_4) levels are usually in the low-normal range; triiodothyronine (T_3) levels are decreased, while reverse T_3 levels are elevated. Females have low serum estrogen levels, whereas males have low levels of serum testosterone.

Electrocardiography. Sinus bradycardia is common, and, rarely, arrhythmias are noted. Significant prolongation of the QTc interval is observed in some individuals.

Bone mass. Low bone mineral density, with specific areas of osteopenia or osteoporosis, is often seen. The risk of fracture is significantly elevated.

Electroencephalography. Diffuse abnormalities, reflecting a metabolic encephalopathy, may result from significant fluid and electrolyte disturbances.

Resting energy expenditure. There is often a significant reduction in resting energy expenditure.

Physical signs and symptoms. Many of the physical signs and symptoms of anorexia nervosa are attributable to starvation. Amenorrhea is commonly present and appears to be an indicator of physiological dysfunction. If present, amenorrhea is usually a consequence of the weight loss, but in a minority of individuals it may actually precede the weight loss. In prepubertal females, menarche may be delayed. In addition to amenorrhea, there may be complaints of constipation, abdominal pain, cold intolerance, lethargy, and excess energy.

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The most remarkable finding on physical examination is emaciation. Commonly, there is also significant hypotension, hypothermia, and bradycardia. Some individuals develop lanugo, a fine downy body hair. Some develop peripheral edema, especially during weight restoration or upon cessation of laxative and diuretic abuse. Rarely, petechiae or ecchymoses, usually on the extremities, may indicate a bleeding diathesis. Some individuals evidence a yellowing of the skin associated with hypercarotenemia. As may be seen in individuals with bulimia nervosa, individuals with anorexia nervosa who self-induce vomiting may have hypertrophy of the salivary glands, particularly the parotid glands, as well as dental enamel erosion. Some individuals may have scars or calluses on the dorsal surface of the hand from repeated contact with the teeth while inducing vomiting.

Association With Suicidal Thoughts or Behavior

Suicide risk is elevated in anorexia nervosa, with rates reported to be 18 times greater than in an age- and gender-matched comparison group. A systematic review found that suicide is the second leading cause of death in anorexia nervosa. Another review found that one-quarter to one-third of individuals with anorexia nervosa have suicidal ideation, and approximately 9%–25% of individuals with anorexia nervosa have attempted suicide. Likely contributors to the increased risk for suicide in those with eating disorders include greater exposure to sexual abuse; impaired decision-making; high rates of nonsuicidal self-injury, a known risk factor for suicide attempts; and comorbidity with mood disorders.

Functional Consequences of Anorexia Nervosa

Individuals with anorexia nervosa may exhibit a range of functional limitations associated with the disorder. While some individuals remain active in social and professional functioning, others demonstrate significant social isolation and/or failure to fulfill academic or career potential.

Differential Diagnosis

In addition to the general consideration of differential diagnosis for all cases, it is especially important to consider other possible causes of either significantly low body weight or significant weight loss when the presenting features are atypical (e.g., onset after age 40 years).

Medical conditions (e.g., gastrointestinal disease, hyperthyroidism, occult malignancies, and acquired immunodeficiency syndrome [AIDS]).

Serious weight loss may occur in medical conditions, but individuals with these disorders usually do not also manifest a disturbance in the way their body weight or shape is experienced or an intense fear of weight gain or persist in behaviors that interfere with appropriate weight gain.

Acute weight loss associated with a medical condition can occasionally be followed by the onset or recurrence of anorexia nervosa, which can initially be masked by the comorbid medical condition. Rarely, anorexia nervosa develops after bariatric surgery.

Major depressive disorder. In major depressive disorder, severe weight loss may occur, but most individuals with major depressive disorder do not have either a desire for excessive weight loss or an intense fear of gaining weight.

Schizophrenia. Individuals with schizophrenia may exhibit odd eating behavior and occasionally experience significant weight loss, but they rarely show the fear of gaining weight and the body image disturbance required for a diagnosis of anorexia nervosa.

Substance use disorders. Individuals with substance use disorders may experience low weight because of poor nutritional intake but generally do not fear gaining weight and do not manifest body image disturbance. Individuals who abuse substances that reduce appetite (e.g., cocaine, stimulants) and who also endorse fear of weight gain should be carefully evaluated for the possibility of comorbid anorexia nervosa, given that the substance use may represent a persistent behavior that interferes with weight gain (Criterion B).

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Social anxiety disorder, obsessive-compulsive disorder, and body dysmorphic disorder. Some of the features of anorexia nervosa overlap with the criteria for social anxiety disorder, OCD, and body dysmorphic disorder. Specifically, individuals may feel humiliated or embarrassed to be seen eating in public, as in social anxiety disorder; may exhibit obsessions and compulsions related to food, as in OCD; or may be preoccupied with an imagined defect in bodily appearance, as in body dysmorphic disorder. If the individual with anorexia nervosa has social fears that are limited to eating behavior alone, the diagnosis of social anxiety disorder should not be made, but social fears unrelated to eating behavior (e.g., excessive fear of speaking in public) may warrant an additional diagnosis of social anxiety disorder. Similarly, an additional diagnosis of OCD should be considered only if the individual exhibits obsessions and compulsions unrelated to food (e.g., an excessive fear of contamination), and an additional diagnosis of body dysmorphic disorder should be considered only if the distortion is unrelated to body shape and size (e.g., preoccupation that one's nose is too big).

Bulimia nervosa. Individuals with bulimia nervosa exhibit recurrent episodes of binge eating, engage in inappropriate behavior to avoid weight gain (e.g., self-induced vomiting), and are overly concerned with body shape and weight. However, unlike individuals with anorexia nervosa, binge-eating/purging type, individuals with bulimia nervosa maintain body weight at or above a minimally normal level.

Avoidant/restrictive food intake disorder. Individuals with this disorder may exhibit significant weight loss or significant nutritional deficiency, but they do not have a fear of gaining weight or of becoming fat, nor do they have a disturbance in the way they experience their body shape and weight.

Comorbidity

Bipolar, depressive, and anxiety disorders commonly co-occur with anorexia nervosa. Many

individuals with anorexia nervosa report the presence of either an anxiety disorder or symptoms of anxiety prior to onset of their eating disorder. OCD is described in some individuals with anorexia nervosa, especially those with the restricting type. Alcohol use disorder and other substance use disorders may also be comorbid with anorexia nervosa, especially among those with the binge-eating/purging type.

Bulimia Nervosa

Diagnostic Criteria

F50.2

- A. Recurrent episodes of binge eating. An episode of binge eating is characterized by both of the following:
 - 1. Eating, in a discrete period of time (e.g., within any 2-hour period), an amount of food that is definitely larger than what most individuals would eat in a similar period of time under similar circumstances.
 - 2. A sense of lack of control over eating during the episode (e.g., a feeling that one cannot stop eating or control what or how much one is eating).
- B. Recurrent inappropriate compensatory behaviors in order to prevent weight gain, such as self-induced vomiting; misuse of laxatives, diuretics, or other medications; fasting; or excessive exercise.
- C. The binge eating and inappropriate compensatory behaviors both occur, on average, at least once a week for 3 months.
- D. Self-evaluation is unduly influenced by body shape and weight.
- E. The disturbance does not occur exclusively during episodes of anorexia nervosa.

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Specify if:

In partial remission: After full criteria for bulimia nervosa were previously met, some, but not all, of the criteria have been met for a sustained period of time.

In full remission: After full criteria for bulimia nervosa were previously met, none of the criteria have been met for a sustained period of time.

Specify current severity:

The minimum level of severity is based on the frequency of inappropriate compensatory behaviors (see below). The level of severity may be increased to reflect other symptoms and the degree of functional disability.

Mild: An average of 1–3 episodes of inappropriate compensatory behaviors per week.

Moderate: An average of 4–7 episodes of inappropriate compensatory behaviors per week.

Severe: An average of 8–13 episodes of inappropriate compensatory behaviors per week.

Extreme: An average of 14 or more episodes of inappropriate compensatory behaviors per week.

Diagnostic Features

There are three essential features of bulimia nervosa: recurrent episodes of binge eating (Criterion A), recurrent inappropriate compensatory behaviors to prevent weight gain (Criterion B), and self-evaluation that is unduly influenced by body shape and weight (Criterion D). To qualify for the diagnosis, the binge eating and inappropriate compensatory behaviors must occur, on average, at least once per week for 3 months (Criterion C).

An “episode of binge eating” is defined as eating, in a discrete period of time, an amount of food that is definitely larger than most individuals would eat in a similar period of time under similar circumstances (Criterion A1). The context in which the eating occurs may affect the clinician’s estimation of whether the intake is excessive. For example, a quantity of food that might be regarded as excessive for a typical meal might be considered normal during a celebration or holiday meal. A “discrete period of time” refers to a limited period, usually less than 2 hours. A single episode of binge eating need not be restricted to one setting. For example, an individual may begin a binge in a restaurant and then continue to eat on returning home. Continual snacking on small amounts of food throughout the day would not be considered an eating binge.

An occurrence of excessive food consumption must be accompanied by a sense of lack of control (Criterion A2) to be considered an episode of binge eating. An indicator of loss of control is the inability to refrain from eating or to stop eating once started. Some individuals describe a dissociative quality during, or following, the binge-eating episodes. The impairment in control associated with binge eating may not be absolute; for example, an individual may continue binge eating while the telephone is ringing but will cease if a roommate or spouse unexpectedly enters the room. Some individuals report that their binge-eating episodes are no longer characterized by an acute feeling of loss of control but rather by a more generalized pattern of uncontrolled eating. If individuals report that they have abandoned efforts to control their eating, loss of control should be considered as present. Binge eating can also be planned in some instances.

The type of food consumed during binges varies both across individuals and for a given individual. Binge eating appears to be characterized more by an abnormality in the amount of food consumed than by a craving for a specific nutrient. However, during binges, individuals tend to eat foods they would otherwise avoid.

Individuals with bulimia nervosa are typically ashamed of their eating problems and attempt to conceal their symptoms. Binge eating usually occurs in secrecy or as inconspicuously as possible. The binge eating often continues until the individual is uncomfortably, or even painfully, full. The most common antecedent of binge eating is negative affect.

Other triggers include interpersonal stressors; dietary restraint; negative feelings related to body weight, body shape, and food; and boredom. Binge eating may minimize or mitigate factors

that precipitated the episode in the short-term, but negative self-evaluation and dysphoria often are the delayed consequences.

Another essential feature of bulimia nervosa is the recurrent use of inappropriate compensatory behaviors to prevent weight gain (Criterion B). Many individuals with bulimia nervosa employ several methods to compensate for binge eating. Self-induced vomiting, a type of purging behavior, is the most common inappropriate compensatory behavior. The immediate effects of vomiting include relief from physical discomfort and reduction of fear of gaining weight. In some cases, vomiting becomes a goal in itself, and the individual will binge eat in order to vomit or will vomit after eating a small amount of food. Individuals with bulimia nervosa may use a variety of methods to induce vomiting, including the use of fingers or instruments to stimulate the gag reflex. Individuals generally become adept at inducing vomiting and are eventually able to vomit at will. Rarely, individuals consume syrup of ipecac to induce vomiting. Other purging behaviors include the misuse of laxatives and diuretics and, in rare cases, the misuse of enemas following episodes of binge eating, although this is seldom the sole compensatory method employed. A number of compensatory methods other than purging may also be used in rare cases. Some individuals may take thyroid hormone in an attempt to avoid weight gain. Individuals with diabetes mellitus and bulimia nervosa may omit or reduce insulin doses in order to reduce the metabolism of food consumed during eating binges. Individuals with bulimia nervosa may fast for a day or more or exercise excessively in an attempt to prevent weight gain. Exercise may be considered excessive when it significantly interferes with important activities, when it occurs at inappropriate times or in inappropriate settings, or when the individual continues to exercise despite injury or other medical complications.

Individuals with bulimia nervosa place an excessive emphasis on body shape or weight in their self-evaluation, and these factors are typically extremely important in determining self-esteem (Criterion D). Individuals with this disorder may closely resemble those with anorexia nervosa in their fear of gaining weight, in their desire to lose weight, and in the level of dissatisfaction with their bodies. However, a diagnosis of bulimia nervosa should not be given when the disturbance occurs only during episodes of anorexia nervosa (Criterion E).

Associated Features

Individuals with bulimia nervosa typically are within the normal weight or overweight range (body mass index [BMI] ≥ 18.5 and < 30 in adults). The disorder occurs but is uncommon among obese individuals. Between eating binges, individuals with bulimia nervosa typically restrict their total caloric consumption and preferentially select low-calorie (“diet”) foods while avoiding foods that they perceive to be fattening or likely to trigger a binge.

Menstrual irregularity or amenorrhea often occurs among females with bulimia nervosa; it is uncertain whether such disturbances are related to weight fluctuations, to nutritional deficiencies, or to emotional distress. The fluid and electrolyte disturbances resulting from the purging behavior are sometimes sufficiently severe to constitute medically serious problems. Rare but potentially fatal complications include esophageal tears, gastric rupture, and cardiac arrhythmias. Serious cardiac and skeletal myopathies have been reported among individuals following repeated use of syrup of ipecac to induce vomiting. Individuals who chronically abuse laxatives may become dependent on their use to stimulate bowel movements. Gastrointestinal symptoms are commonly associated with bulimia nervosa, and rectal prolapse has also been reported among

individuals with this disorder.

Prevalence

According to two U.S. epidemiological studies conducted in adult community samples, the 12-month prevalence of bulimia nervosa ranges from 0.14% to 0.3%, with much higher

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rates in women than in men (0.22% to 0.5% in women; 0.05% to 0.1% in men), and the lifetime prevalence ranges from 0.28% to 1.0% (0.46% to 1.5% in women; 0.05% to 0.08% in men). In one study of adolescents ages 13–18, lifetime prevalence rates were 1.3% and 0.5% in girls and boys, respectively.

In the United States, the prevalence of bulimia nervosa is similar across U.S. ethnoracial groups. The reported prevalence of bulimia nervosa is highest in populations residing in high-income industrialized countries, such as the United States, Canada, Australia, New Zealand, and many European countries; in most of these countries the prevalence of bulimia nervosa is roughly comparable.

The prevalence of bulimia nervosa in some regions of Latin America and the Middle East is similar to the prevalence in most high-income countries. The prevalence of bulimia nervosa appears to be gradually increasing in many low- and middle-income countries.

Development and Course

Bulimia nervosa commonly begins in adolescence or young adulthood. Onset before puberty or after age 40 is uncommon. The binge eating frequently begins during or after an episode of dieting to lose weight. Experiencing multiple stressful life events also can precipitate onset of bulimia nervosa.

Disturbed eating behavior persists for at least several years in a high percentage of clinical samples. The course may be chronic or intermittent, with periods of remission alternating with recurrences of binge eating. However, over longer-term follow-up, the symptoms of many individuals appear to diminish with or without treatment, although treatment clearly impacts outcome. Periods of remission longer than 1 year are associated with better long-term outcome.

Significantly elevated risk for mortality (all-cause and suicide) has been reported for individuals with bulimia nervosa. The crude mortality rate (ratio of the number of deaths during the year to the average population in that year) for bulimia nervosa is nearly 2% per decade.

Diagnostic crossover from initial bulimia nervosa to anorexia nervosa occurs in a minority of cases (10%–15%). Individuals who do experience crossover to anorexia nervosa commonly will revert back to bulimia nervosa or have multiple occurrences of crossovers between these disorders. A subset of individuals with bulimia nervosa continue to binge eat but no longer engage in inappropriate compensatory behaviors, and therefore their symptoms meet criteria for binge-eating disorder or other specified eating disorder. Diagnosis should be based on the current (i.e., past 3 months) clinical presentation.

Risk and Prognostic Factors

Temperamental. Weight concerns, low self-esteem, depressive symptoms, social anxiety disorder,

and childhood generalized anxiety disorder are associated with increased risk for the development of bulimia nervosa.

Environmental. Internalization of a thin body ideal has been found to increase risk for developing weight concerns, which in turn increases risk for the development of bulimia nervosa. Individuals who experienced childhood sexual or physical abuse are at increased risk for developing bulimia nervosa.

Genetic and physiological. Childhood obesity and early pubertal maturation increase risk for bulimia nervosa. Familial transmission of bulimia nervosa may be present, as well as genetic vulnerabilities for the disorder.

Course modifiers. Severity of psychiatric comorbidity predicts worse long-term outcome of bulimia nervosa.

Culture-Related Diagnostic Issues

Although data show that community-based prevalence of bulimia nervosa does not differ significantly across U.S. ethnoracial groups, treatment utilization for bulimia nervosa is lower among underserved U.S. ethnic and racialized groups than among the non-Latinx White population.

Sex- and Gender-Related Diagnostic Issues

Bulimia nervosa is much more common in girls and women than in boys and men. Boys and men are especially underrepresented in treatment-seeking samples, for reasons that have not yet been systematically examined.

Diagnostic Markers

No specific diagnostic test for bulimia nervosa currently exists. However, several laboratory abnormalities may occur as a consequence of purging and may increase diagnostic certainty. These include fluid and electrolyte abnormalities, such as hypokalemia (which can provoke cardiac arrhythmias), hypochloremia, and hyponatremia. The loss of gastric acid through vomiting may produce a metabolic alkalosis (elevated serum bicarbonate), and the frequent induction of diarrhea or dehydration through laxative and diuretic abuse can cause metabolic acidosis. Some individuals with bulimia nervosa exhibit mildly elevated levels of serum amylase, probably reflecting an increase in the salivary isoenzyme.

Physical examination usually yields no physical findings. However, inspection of the mouth may reveal significant and permanent loss of dental enamel, especially from lingual surfaces of the front teeth because of recurrent vomiting. These teeth may become chipped and appear ragged and “moth-eaten.” There may also be an increased frequency of dental caries. In some individuals, the salivary glands, particularly the parotid glands, may become notably enlarged. Individuals who induce vomiting by manually stimulating the gag reflex may develop calluses or scars on the dorsal surface of the hand from repeated contact with the teeth. Serious cardiac and skeletal myopathies have been reported among individuals following repeated use of syrup of ipecac to induce vomiting.

Association With Suicidal Thoughts or Behavior

Suicide risk is elevated in bulimia nervosa. A review found that approximately one-quarter to one-third of individuals with bulimia nervosa have had suicidal ideation, and a similar proportion have attempted suicide.

Functional Consequences of Bulimia Nervosa

Individuals with bulimia nervosa may exhibit a range of functional limitations associated with the disorder and have reduced health-related quality of life. A minority of individuals report severe role impairment, with the social-life domain most likely to be adversely affected by bulimia nervosa.

Differential Diagnosis

Anorexia nervosa, binge-eating/purging type. Individuals whose binge-eating behavior occurs only during episodes of anorexia nervosa are given the diagnosis anorexia nervosa, binge-eating/purging type, and should not be given the additional diagnosis of bulimia nervosa. For individuals with an initial diagnosis of anorexia nervosa who binge and purge but whose presentation no longer meets the full criteria for anorexia nervosa, binge-eating/purging type (e.g., when weight is normal), a diagnosis of bulimia nervosa should be given only when all criteria for bulimia nervosa have been met for at least 3 months.

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Binge-eating disorder. Some individuals binge eat but do not engage in regular inappropriate compensatory behaviors. In these cases, the diagnosis of binge-eating disorder should be considered.

Kleine-Levin syndrome. In certain neurological or other medical conditions, such as Kleine-Levin syndrome, there is disturbed eating behavior, but the characteristic psychological features of bulimia nervosa, such as overconcern with body shape and weight, are not present.

Major depressive disorder, with atypical features. Overeating is common in major depressive disorder, with atypical features, but individuals with this disorder do not engage in inappropriate compensatory behaviors and do not exhibit the excessive concern with body shape and weight characteristic of bulimia nervosa. If criteria for both disorders are met, both diagnoses should be given.

Borderline personality disorder. Binge-eating behavior is included in the impulsive behavior criterion that is part of the definition of borderline personality disorder. If the criteria for both borderline personality disorder and bulimia nervosa are met, both diagnoses should be given.

Comorbidity

Comorbidity with mental disorders is common in individuals with bulimia nervosa, with most experiencing at least one other mental disorder and many experiencing multiple comorbidities. Comorbidity is not limited to any particular subset but rather occurs across a wide range of mental disorders. There is an increased frequency of depressive symptoms (e.g., low self-esteem) and bipolar and depressive disorders (particularly depressive disorders) in individuals with

bulimia nervosa. In many individuals, the mood disturbance begins at the same time as or following the development of bulimia nervosa, and individuals often ascribe their mood disturbances to the bulimia nervosa. However, in some individuals, the mood disturbance clearly precedes the development of bulimia nervosa. There may also be an increased frequency of anxiety symptoms (e.g., fear of social situations) or anxiety disorders. These mood and anxiety disturbances frequently remit following effective treatment of the bulimia nervosa. The lifetime prevalence of substance use disorder, particularly alcohol use disorder or stimulant use disorder, is at least 30% among individuals with bulimia nervosa. Stimulant use often begins in an attempt to control appetite and weight. A substantial percentage of individuals with bulimia nervosa also have personality features that meet criteria for one or more personality disorders, most frequently borderline personality disorder.

Binge-Eating Disorder

Diagnostic Criteria

F50.81

- A. Recurrent episodes of binge eating. An episode of binge eating is characterized by both of the following:
 - 1. Eating, in a discrete period of time (e.g., within any 2-hour period), an amount of food that is definitely larger than what most people would eat in a similar period of time under similar circumstances.
 - 2. A sense of lack of control over eating during the episode (e.g., a feeling that one cannot stop eating or control what or how much one is eating).
- B. The binge-eating episodes are associated with three (or more) of the following:
 - 1. Eating much more rapidly than normal.
 - 2. Eating until feeling uncomfortably full.
 - 3. Eating large amounts of food when not feeling physically hungry.
 - 4. Eating alone because of feeling embarrassed by how much one is eating.
 - 5. Feeling disgusted with oneself, depressed, or very guilty afterward.
- C. Marked distress regarding binge eating is present.
- D. The binge eating occurs, on average, at least once a week for 3 months.
- E. The binge eating is not associated with the recurrent use of inappropriate compensatory behavior as in bulimia nervosa and does not occur exclusively during the course of bulimia nervosa or anorexia nervosa.

Specify if:

In partial remission: After full criteria for binge-eating disorder were previously met, binge eating occurs at an average frequency of less than one episode per

week for a sustained period of time.

In full remission: After full criteria for binge-eating disorder were previously met, none of the criteria have been met for a sustained period of time.

Specify current severity:

The minimum level of severity is based on the frequency of episodes of binge eating (see below). The level of severity may be increased to reflect other symptoms and the degree of functional disability.

Mild: 1–3 binge-eating episodes per week.

Moderate: 4–7 binge-eating episodes per week.

Severe: 8–13 binge-eating episodes per week.

Extreme: 14 or more binge-eating episodes per week.

Diagnostic Features

The essential feature of binge-eating disorder is recurrent episodes of binge eating that must occur, on average, at least once per week for 3 months (Criterion D). An “episode of binge eating” is defined as eating, in a discrete period of time, an amount of food that is definitely larger than most people would eat in a similar period of time under similar circumstances (Criterion A1). The context in which the eating occurs may affect the clinician’s estimation of whether the intake is excessive. For example, a quantity of food that might be regarded as excessive for a typical meal might be considered normal during a celebration or holiday meal. A “discrete period of time” refers to a limited period, usually less than 2 hours. A single episode of binge eating need not be restricted to one setting. For example, an individual may begin a binge in a restaurant and then continue to eat on returning home. Continual snacking on small amounts of food throughout the day would not be considered an eating binge.

An occurrence of excessive food consumption must be accompanied by a sense of lack of control (Criterion A2) to be considered an episode of binge eating. An indicator of loss of control is the inability to refrain from eating or to stop eating once started. Some individuals describe a dissociative quality during, or following, the binge-eating episodes. The impairment in control associated with binge eating may not be absolute; for example, an individual may continue binge eating while the telephone is ringing but will cease if a roommate or spouse unexpectedly enters the room. Some individuals report that their binge-eating episodes are no longer characterized by an acute feeling of loss of control but rather by a more generalized pattern of uncontrolled eating. If individuals report that they have abandoned efforts to control their eating, loss of control may still be considered as present. Binge eating can also be planned in some instances.

The type of food consumed during binges varies both across individuals and for a given individual. Binge eating appears to be characterized more by an abnormality in the amount of food consumed than by a craving for a specific nutrient.

Binge eating must be characterized by marked distress (Criterion C) and at least three of the following features: eating much more rapidly than normal; eating until feeling uncomfortably

full; eating large amounts of food when not feeling physically hungry; eating alone because of feeling embarrassed by how much one is eating; and feeling disgusted with oneself, depressed, or very guilty afterward (Criterion B).

Individuals with binge-eating disorder are typically ashamed of their eating problems and attempt to conceal their symptoms. Binge eating usually occurs in secrecy or as inconspicuously as possible. The most common antecedent of binge eating is negative affect. Other triggers include interpersonal stressors; dietary restraint; negative feelings related to body weight, body shape, and food; and boredom. Binge eating may minimize or mitigate factors that precipitated the episode in the short-term, but negative self-evaluation and dysphoria often are the delayed consequences.

Associated Features

Binge-eating disorder occurs in normal-weight/overweight and obese individuals. It is reliably associated with overweight and obesity in treatment-seeking individuals. Nevertheless, binge-eating disorder is distinct from obesity. Most obese individuals do not engage in recurrent binge eating. In addition, compared with weight-matched obese individuals without binge-eating disorder, those with the disorder consume more calories in laboratory studies of eating behavior and have greater functional impairment, lower quality of life, more subjective distress, and greater psychiatric comorbidity.

Prevalence

According to two U.S. epidemiological studies conducted in community samples, the 12-month prevalence of binge-eating disorder ranges from 0.44% to 1.2%, with rates two to three times higher in women than in men (0.6% to 1.6% in women; 0.26% to 0.8% in men), and the lifetime prevalence ranges from 0.85% to 2.8% (1.25% to 3.5% in women; 0.42% to 2.0% in men).

In the United States, the prevalence of binge-eating disorder appears comparable across ethnoracial groups.

Binge-eating disorder has a roughly similar prevalence in most high-income industrialized countries, including the United States, Canada, many European countries, Australia, and New Zealand, with 12-month prevalence in high-income countries ranging from 0.1% to 1.2%. Although fewer data are available from populations in low- and middle-income countries, binge-eating disorder prevalence in some regions of Latin America appears to be at least as high as in the United States and Europe. Mexican Americans in the United States have a higher prevalence of binge-eating disorder than their counterparts in Mexico.

Development and Course

Little is known about the development of binge-eating disorder. Both binge eating and loss-of-control eating without objectively excessive consumption occur in children and are associated with increased body fat, weight gain, and increases in psychological symptoms. Binge eating is common in adolescent and college-age samples. Loss-of-control eating or episodic binge eating may represent a prodromal phase of eating disorders for some individuals.

Dieting follows the development of binge eating in many individuals with binge-eating disorder. (This is in contrast to bulimia nervosa, in which dysfunctional dieting usually precedes

the onset of binge eating.) Binge-eating disorder typically begins in adolescence or young adulthood but can begin in later adulthood. Individuals with binge-eating disorder who seek treatment usually are older than individuals with either bulimia nervosa or anorexia nervosa who seek treatment.

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Remission rates in both natural course and treatment outcome studies are higher for binge-eating disorder than for bulimia nervosa or anorexia nervosa. The course of binge-eating disorder is variable and as yet incompletely understood, with at least some affected individuals showing a relatively persistent, at times relapsing and remitting, symptom trajectory comparable to that of bulimia nervosa in terms of severity and duration. Crossover from binge-eating disorder to other eating disorders is uncommon.

Risk and Prognostic Factors

Genetic and physiological. Binge-eating disorder appears to run in families, which may reflect additive genetic influences.

Culture-Related Diagnostic Issues

Clinical presentations of binge-eating disorder differ across ethnoracial groups in the United States. Black individuals may report fewer symptoms of distress associated with binge eating and present for treatment with higher frequency of binge eating compared with White individuals.

Association With Suicidal Thoughts or Behavior

Suicidal ideation has been reported to occur in approximately 25% of individuals with binge-eating disorder.

Functional Consequences of Binge-Eating Disorder

Binge-eating disorder is associated with a range of functional consequences, including social role adjustment problems, impaired health-related quality of life and life satisfaction, increased medical morbidity and mortality, and associated increased health care utilization compared with body mass index (BMI)-matched control subjects. It may also be associated with an increased risk for weight gain and the development of obesity.

Differential Diagnosis

Bulimia nervosa. Binge-eating disorder has recurrent binge eating in common with bulimia nervosa but differs from the latter disorder in some fundamental respects. In terms of clinical presentation, the recurrent inappropriate compensatory behavior (e.g., purging, driven exercise) seen in bulimia nervosa is absent in binge-eating disorder. Unlike individuals with bulimia nervosa, individuals with binge-eating disorder typically do not show marked or sustained dietary restriction designed to influence body weight and shape between binge-eating episodes. They may, however, report frequent attempts at dieting. Binge-eating disorder also differs from bulimia nervosa in terms of response to treatment. Rates of improvement are consistently higher

among individuals with binge-eating disorder than among those with bulimia nervosa.

Obesity. Binge-eating disorder is associated with overweight and obesity but has several key features that are distinct from obesity. First, levels of overvaluation of body weight and shape are higher in obese individuals with the disorder than in those without the disorder. Second, rates of psychiatric comorbidity are significantly higher among obese individuals with the disorder compared with those without the disorder. Third, the outcome of evidence-based psychological treatments for binge-eating disorder is more often successful than the treatment of obesity in individuals with comorbid obesity and binge-eating disorder.

Bipolar and depressive disorders. Increases in appetite and weight gain are included in the criteria for major depressive episode and in the atypical features specifiers for

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depressive and bipolar disorders. Increased eating in the context of a major depressive episode may or may not be associated with loss of control. If the full criteria for both disorders are met, both diagnoses can be given. Binge eating and other symptoms of disordered eating are seen in association with bipolar disorder. If the full criteria for both disorders are met, both diagnoses should be given.

Borderline personality disorder. Binge eating is included in the impulsive behavior criterion that is part of the definition of borderline personality disorder. If the full criteria for both disorders are met, both diagnoses should be given.

Comorbidity

Binge-eating disorder is associated with significant psychiatric comorbidity that is comparable to that of bulimia nervosa and anorexia nervosa. The most common comorbid disorders are major depressive disorder and alcohol use disorder. The psychiatric comorbidity is linked to the severity of binge eating and not to the degree of obesity.

Other Specified Feeding or Eating Disorder

F50.89

This category applies to presentations in which symptoms characteristic of a feeding and eating disorder that cause clinically significant distress or impairment in social, occupational, or other important areas of functioning predominate but do not meet the full criteria for any of the disorders in the feeding and eating disorders diagnostic class. The other specified feeding or eating disorder category is used in situations in which the clinician chooses to communicate the specific reason that the presentation does not meet the criteria for any specific feeding and eating disorder. This is done by recording “other specified feeding or eating disorder” followed by the specific reason (e.g., “bulimia nervosa of low frequency”).

Examples of presentations that can be specified using the “other specified” designation include the following:

1. **Atypical anorexia nervosa:** All of the criteria for anorexia nervosa are met, except that despite significant weight loss, the individual's weight is within or above the normal range. Individuals with atypical anorexia nervosa may experience many of the physiological complications associated with anorexia nervosa.
2. **Bulimia nervosa (of low frequency and/or limited duration):** All of the criteria for bulimia nervosa are met, except that the binge eating and inappropriate compensatory behaviors occur, on average, less than once a week and/or for less than 3 months.
3. **Binge-eating disorder (of low frequency and/or limited duration):** All of the criteria for binge-eating disorder are met, except that the binge eating occurs, on average, less than once a week and/or for less than 3 months.
4. **Purging disorder:** Recurrent purging behavior to influence weight or shape (e.g., self-induced vomiting; misuse of laxatives, diuretics, or other medications) in the absence of binge eating.
5. **Night eating syndrome:** Recurrent episodes of night eating, as manifested by eating after awakening from sleep or by excessive food consumption after the evening meal. There is awareness and recall of the eating. The night eating is not better explained by external influences such as changes in the individual's sleep-wake cycle or by local social norms. The night eating causes significant distress and/or impairment in functioning. The disordered pattern of eating is not better explained by binge-eating disorder or another mental disorder, including substance use, and is not attributable to another medical condition or to an effect of medication.

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Unspecified Feeding or Eating Disorder

F50.9

This category applies to presentations in which symptoms characteristic of a feeding and eating disorder that cause clinically significant distress or impairment in social, occupational, or other important areas of functioning predominate but do not meet the full criteria for any of the disorders in the feeding and eating disorders diagnostic class. The unspecified feeding or eating disorder category is used in situations in which the clinician chooses *not* to specify the reason that the criteria are not met for a specific feeding and eating disorder and includes presentations in which there is insufficient information to make a more specific diagnosis (e.g., in emergency room settings).

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Elimination Disorders

Elimination disorders all involve the inappropriate elimination of urine or feces and are usually first diagnosed in childhood or adolescence. This group of disorders includes *enuresis*, the repeated voiding of urine into inappropriate places, and *encopresis*, the repeated passage of feces into inappropriate places. Subtypes are provided to differentiate nocturnal from diurnal (i.e., during waking hours) voiding for enuresis and the presence or absence of constipation and overflow incontinence for encopresis. Although there are minimum age requirements for diagnosis of both disorders, these are based on developmental age and not solely on chronological age. Both disorders may be voluntary or involuntary. Although these disorders typically occur separately, co-occurrence may also be observed.

Enuresis

Diagnostic Criteria

F98.0

- A. Repeated voiding of urine into bed or clothes, whether involuntary or intentional.
- B. The behavior is clinically significant as manifested by either a frequency of at least twice a week for at least 3 consecutive months or the presence of clinically significant distress or impairment in social, academic (occupational), or other important areas of functioning.
- C. Chronological age is at least 5 years (or equivalent developmental level).
- D. The behavior is not attributable to the physiological effects of a substance (e.g., a diuretic, an antipsychotic medication) or another medical condition (e.g., diabetes, spina bifida, a seizure disorder).

Specify whether:

Nocturnal only: Passage of urine only during nighttime sleep.

Diurnal only: Passage of urine during waking hours.

Nocturnal and diurnal: A combination of the two subtypes above.

Subtypes

The nocturnal-only subtype of enuresis, sometimes referred to as *monosymptomatic enuresis*, is the most common subtype and involves incontinence only during nighttime sleep, typically during the first one-third of the night. The diurnal-only subtype occurs in the absence of nocturnal enuresis and may be referred to simply as *urinary incontinence*. Individuals with this

subtype can be divided into two groups. Individuals with “urge incontinence” have sudden urge symptoms and detrusor instability, whereas individuals with “voiding postponement” consciously defer micturition urges until incontinence results. The nocturnal-and-diurnal subtype is also known as *nonmonosymptomatic enuresis*.

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Diagnostic Features

The essential feature of enuresis is repeated voiding of urine during the day or at night into bed or clothes (Criterion A). Most often the voiding is involuntary, but occasionally it may be intentional. To qualify for a diagnosis of enuresis, the voiding of urine must occur at least twice a week for at least 3 consecutive months or must cause clinically significant distress or impairment in social, academic (occupational), or other important areas of functioning (Criterion B). The individual must have reached an age at which continence is expected (i.e., a chronological age of at least 5 years or, for children with developmental delays, a mental age of at least 5 years) (Criterion C). The urinary incontinence is not attributable to the physiological effects of a substance (e.g., a diuretic, an antipsychotic medication) or another medical condition (e.g., diabetes, spina bifida, ectopic ureter in a female, posterior urethral valves in a male, tethered cord, a seizure disorder) (Criterion D).

Associated Features

During nocturnal enuresis, occasionally the voiding takes place during rapid eye movement (REM) sleep, and the child may recall a dream that involved the act of urinating. During daytime (diurnal) enuresis, the child defers voiding until incontinence occurs, sometimes because of a reluctance to use the toilet as a result of social anxiety or a preoccupation with school or play activity. The enuretic events most commonly occur in the early afternoon on school days or after returning from school. Children with executive functioning problems and other neurological problems that may be associated with symptoms of disruptive behavior may be at high risk for urinary incontinence without sensory awareness. It is not uncommon for children with daytime urinary incontinence and the nocturnal-and-diurnal subtype of enuresis to have persistence of incontinence after appropriate treatment of an associated infection.

Prevalence

The prevalence of daytime incontinence ranges from 3.2% to 9.0% in children age 7 years, from 1.1% to 4.2% in youth ages 11–13 years, and from 1.2% to 3.0% in adolescents ages 15–17 years.

The prevalence of nocturnal enuresis in the community decreases with age; in several geographical settings, including the United States, the Netherlands, and Hong Kong, the range is around 5%–10% among 5-year-olds, 3%–5% among 10-year-olds, and around 1% among individuals 15 years or older. Boys and members of socially oppressed groups may have higher prevalence as found in African American children in the United States and Turkish or Moroccan children in the Netherlands. The disorder may also have higher prevalence in youth with learning disabilities or attention-deficit/hyperactivity disorder.

Development and Course

Enuresis can follow two courses: a “primary” type, in which the individual has never established urinary continence, and a “secondary” type, in which the disturbance develops after a period of established urinary continence. There are no differences in prevalence of comorbid mental disorders between the two types. By definition, primary enuresis begins at age 5 years. The most common time for the onset of secondary enuresis is between ages 5 and 8 years, but it may occur at any time. After age 5 years, the rate of spontaneous remission is 5%–10% per year. Most children with the disorder become continent by adolescence, but in approximately 1% of cases the disorder continues into adulthood. Diurnal enuresis is uncommon after age 9 years. While occasional diurnal incontinence is not uncommon in middle childhood, it is substantially more common in those who also have other co-occurring mental health problems, including cognitive and behavioral problems. When enuresis persists into late childhood or adolescence, the incontinence may resolve, but urinary frequency generally persists and incontinence can recur later in adulthood in women.

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Risk and Prognostic Factors

A number of predisposing factors for bladder dysfunction have been suggested, including developmental delays and neuropsychiatric problems.

Environmental. Factors recognized to be associated with bladder dysfunction include delayed toileting and psychosocial stress.

Genetic and physiological. Nocturnal enuresis has been associated with a mismatch between nocturnal urine production, nocturnal bladder storage capacity, and the ability to arouse from sleep. Underlying these mechanisms are possibly disorders of central nervous system signal processing and the default mode network. The increased arousal thresholds do not, however, mean that these children sleep well; in fact, sleep quality of enuretic children is often poor. Nocturnal enuresis is a genetically heterogeneous disorder. Heritability has been shown in family, twin, and segregation analyses. Risk for childhood nocturnal enuresis is approximately 3.6 times higher in offspring of enuretic mothers and 10.1 times higher in the presence of paternal urinary incontinence. The risk magnitudes for nocturnal enuresis and diurnal incontinence are similar.

Culture-Related Diagnostic Issues

Enuresis has been reported in a variety of European, African, and Asian countries as well as in the United States. At a national level, prevalence rates are remarkably similar, and there is great similarity in the developmental trajectories found in different countries. Local school-based surveys, however, show wide prevalence variation of nocturnal enuresis across settings in Africa, South Asia, Europe, and the Caribbean (4%–50%), at least in part due to methodological variation. The very high rates of enuresis in orphanages and other residential institutions are not explained by the mode or early timing of toilet training.

Cultural contexts affect both the diagnosis and the perceived etiology of enuresis. For example, traditional Chinese medicine attributes enuresis to long-term kidney yang (masculine

energy) deficiency. Heightened impact on parents of children's enuresis has been reported in societies with economic limitations in obtaining care for the child or in the context of social policies that restrict the number of children (e.g., China's one-child policy), possibly leading to higher risk of parental emotional disorders.

Sex- and Gender-Related Diagnostic Issues

Nocturnal enuresis is more common in males than in females (almost 2:1). This male preponderance is particularly true in younger age groups, cases with milder severity, and cases involving enuresis occurring only at night. Urinary tract infections are frequently associated with daytime wetting, especially in females. Diurnal incontinence is more common in females than in males, and the ratio increases with age. The relative risk of having a child who develops enuresis is greater for previously enuretic fathers than for previously enuretic mothers.

Functional Consequences of Enuresis

The amount of impairment associated with enuresis is a function of the limitation on the child's social activities (e.g., ineligibility for sleep-away camp) or its effect on the child's self-esteem, the degree of social ostracism by peers, and the anger, punishment, and rejection on the part of caregivers.

Differential Diagnosis

Neurogenic bladder or another medical condition. The diagnosis of enuresis is not made in the presence of a neurogenic bladder or any other structural condition (such as a posterior urethral valve or ectopic ureter) or another medical condition that causes polyuria or

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urgency (e.g., untreated diabetes mellitus or diabetes insipidus) or during an acute urinary tract infection. However, a diagnosis is compatible with such medical conditions if urinary incontinence was regularly present prior to the development of another medical condition or if it persists after the institution of appropriate treatment of the medical condition.

Medication side effects. Enuresis may occur during treatment with antipsychotic medications, diuretics, or other medications that may induce constipation, polyuria, or alterations in executive functioning, all of which may lead to incontinence. In this case, the diagnosis should not be made in isolation but may be noted as a medication side effect. However, a diagnosis of enuresis may be made if urinary incontinence was regularly present prior to treatment with the medication.

Comorbidity

Although most children with enuresis do not have a comorbid mental disorder, the prevalence of comorbid behavioral and developmental symptoms is higher in children with both diurnal and nocturnal enuresis than in children without incontinence. Developmental delays, including speech, language, learning, and motor skills delays, are also present in a portion of children with enuresis. Encopresis and constipation are present in both day and night incontinence. Restless legs syndrome and parasomnias such as non-rapid eye movement sleep arousal disorders

(sleepwalking and sleep terror types) are associated with nocturnal enuresis. Additionally, there is a link between nocturnal enuresis and heavy snoring or sleep apneas. Approximately 50% of enuretic children with proven sleep-disordered breathing will become dry by undergoing adenotonsillectomy. Urinary tract infections are more common in children with daytime urinary incontinence and nonmonosymptomatic nocturnal enuresis, especially the diurnal subtype, than in those who are continent.

Encopresis

Diagnostic Criteria	F98.1
A. Repeated passage of feces into inappropriate places (e.g., clothing, floor), whether involuntary or intentional. B. At least one such event occurs each month for at least 3 months. C. Chronological age is at least 4 years (or equivalent developmental level). D. The behavior is not attributable to the physiological effects of a substance (e.g., laxatives) or another medical condition except through a mechanism involving constipation. <i>Specify whether:</i> With constipation and overflow incontinence: There is evidence of constipation on physical examination or by history. Without constipation and overflow incontinence: There is no evidence of constipation on physical examination or by history.	

Subtypes

Feces in encopresis, “with constipation and overflow incontinence” subtype, are characteristically (but not invariably) poorly formed, and leakage can be infrequent to continuous, occurring throughout the day and at times during sleep. Only part of the feces is passed during toileting, and the incontinence resolves after treatment of the constipation.

In the “without constipation and overflow incontinence” subtype, feces are likely to be of normal form and consistency, and soiling is intermittent. Feces may be deposited in a

prominent location. This is usually associated with the presence of oppositional defiant disorder or conduct disorder or may be the consequence of anal masturbation. Soiling without constipation is less common than soiling with constipation.

Diagnostic Features

The essential feature of encopresis is repeated passage of feces into inappropriate places (e.g., clothing or floor) (Criterion A). Most often the passage is involuntary but occasionally may be

intentional. The event must occur at least once a month for at least 3 months (Criterion B), and the chronological age of the child must be at least 4 years (or for children with developmental delays, the mental age must be at least 4 years) (Criterion C). The fecal incontinence must not be exclusively attributable to the physiological effects of a substance (e.g., laxatives) or another medical condition except through a mechanism involving constipation (Criterion D).

When the passage of feces is involuntary rather than intentional, it is often related to constipation, impaction, and retention with subsequent overflow. The constipation may develop for psychological reasons (e.g., anxiety about defecating in a particular place, a more general pattern of anxious or oppositional behavior), leading to avoidance of defecation and excessive volitional stool retention. Physiological predispositions to constipation include ineffectual straining or paradoxical defecation dynamics, with contraction rather than relaxation of the external sphincter or pelvic floor during straining for defecation. Dietary habits (such as insufficient fluid intake), celiac disease, hypothyroidism, or a medication side effect can also induce constipation. Once constipation has developed, it may be complicated by an anal fissure, painful defecation, and further fecal retention. The consistency of the stool may vary. In some individuals the stool may be of normal or near-normal consistency. In other individuals—such as those with overflow incontinence secondary to fecal retention—it may be liquid.

Associated Features

The child with encopresis often feels ashamed and may wish to avoid situations (e.g., camp, school) that might lead to embarrassment. The amount of impairment is a function of the effect on the child's self-esteem, the degree of social ostracism by peers, and the anger, punishment, and rejection on the part of caregivers. Smearing feces may be deliberate or accidental, resulting from the child's attempt to clean or hide feces that were passed involuntarily. When the incontinence is clearly deliberate, features of oppositional defiant disorder or conduct disorder may also be present. Many children with encopresis and chronic constipation also have enuresis symptoms and may have associated urinary reflux in the bladder or ureters that may lead to chronic urinary infections, the symptoms of which may remit with treatment of the constipation.

Prevalence

It is estimated that the majority of children older than 4 years with encopresis have the subtype “with constipation and overflow incontinence.” Encopresis affects 1%–4% of children in high-income countries, while in some Asian countries (Iran, South Korea, Sri Lanka) a prevalence of 2%–8% has been reported. Encopresis is more prevalent among children ages 4–6 years (> 4%) than among children ages 10–12 years (< 2%); prevalence is also higher among children who experience early abuse or neglect and low-income youth.

Development and Course

Encopresis is not diagnosed until a child has reached a chronological age of at least 4 years (or for children with developmental delays, a mental age of at least 4 years). Inadequate,

inconsistent toilet training and psychosocial stress (e.g., entering school, the birth of a sibling)

may be predisposing factors. Two types of course have been described: a “primary” type, in which the individual has never established fecal continence, and a “secondary” type, in which the disturbance develops after a period of established fecal continence. Encopresis can persist, with intermittent exacerbations, for years.

Risk and Prognostic Factors

Painful defecation can lead to constipation and a cycle of withholding behaviors that make encopresis more likely. Male gender and age prior to adolescence are risk factors for encopresis. A number of factors are thought to contribute to the development of fecal incontinence, including anxiety, depression, behavioral disorders, psychological stressors (e.g., bullying, poor school performance), and lower socioeconomic status.

Culture-Related Diagnostic Issues

Differences in food and beverage intake in different cultures, hot climatic conditions in tropical countries, and psychosocial adversity may influence the incidence of constipation, unexplained abdominal pain, and fecal retention that lead to encopresis. Parents in some societies may not seek health services for encopresis because of sociocultural reasons. For example, Turkish and Moroccan parents in the Netherlands may not report encopresis because of religious concerns about the impurity of urine and feces.

Sex- and Gender-Related Diagnostic Issues

In children younger than 5 years, the gender ratio appears to be equal, but encopresis tends to be more common in boys than in girls among older children, with a ratio that varies globally (in community and hospital-based studies) from 2:1 (in the United States) to 6:1.

Diagnostic Markers

The diagnosis of encopresis is a clinical diagnosis based on history and physical examination and generally does not require any diagnostic testing. Detection of a rectal fecal impaction by digital rectal examination would support the diagnosis of encopresis, with constipation and overflow incontinence. Although not indicated for the diagnosis of encopresis, an abdominal radiograph demonstrating a fecal impaction would also support the diagnosis of encopresis, with constipation and overflow incontinence. Colonic transit testing, which typically involves ingestion of radiopaque markers followed by abdominal imaging to evaluate colonic transit time, can help differentiate between encopresis with or without constipation and overflow incontinence. Abdominal imaging demonstrating retention of radiopaque markers would suggest encopresis, with constipation and overflow incontinence, while prompt evacuation of radiopaque markers would support the diagnosis of encopresis, without constipation and overflow incontinence. In certain children, anorectal manometry testing may be helpful for better understanding physiological factors that may be contributing to encopresis. Anorectal manometry allows for evaluation of anorectal function and sensation. In the child with refractory symptoms or signs suggesting the presence of an underlying medical condition leading to fecal incontinence, further evaluation may be indicated. Such evaluation is designed to exclude other medical conditions.

Functional Consequences of Encopresis

Encopresis is associated with a significant decrease in health-related quality of life and family functioning, particularly in older children.

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Differential Diagnosis

A diagnosis of encopresis in the presence of another medical condition is appropriate only if the mechanism involves constipation that cannot be explained by other medical conditions. Fecal incontinence related to other medical conditions (e.g., chronic diarrhea, spina bifida, anal stenosis) would not warrant a DSM-5 diagnosis of encopresis.

Comorbidity

Enuresis is often present in children with encopresis, particularly in children with encopresis, without constipation and overflow incontinence.

Other Specified Elimination Disorder

This category applies to presentations in which symptoms characteristic of an elimination disorder that cause clinically significant distress or impairment in social, occupational, or other important areas of functioning predominate but do not meet the full criteria for any of the disorders in the elimination disorders diagnostic class. The other specified elimination disorder category is used in situations in which the clinician chooses to communicate the specific reason that the presentation does not meet the criteria for any specific elimination disorder. This is done by recording “other specified elimination disorder” followed by the specific reason (e.g., “low-frequency enuresis”).

Coding note: Code **N39.498** for other specified elimination disorder with urinary symptoms; **R15.9** for other specified elimination disorder with fecal symptoms.

Unspecified Elimination Disorder

This category applies to presentations in which symptoms characteristic of an elimination disorder that cause clinically significant distress or impairment in social, occupational, or other important areas of functioning predominate but do not meet the full criteria for any of the disorders in the elimination disorders diagnostic class. The unspecified elimination disorder category is used in situations in which the

clinician chooses *not* to specify the reason that the criteria are not met for a specific elimination disorder and includes presentations in which there is insufficient information to make a more specific diagnosis (e.g., in emergency room settings).

Coding note: Code **R32** for unspecified elimination disorder with urinary symptoms; **R15.9** for unspecified elimination disorder with fecal symptoms.

Sleep-Wake Disorders

The DSM-5 classification of sleep-wake disorders is intended for use by mental health and general medical clinicians (those caring for adult, geriatric, and pediatric individuals). Sleep-wake disorders encompass 10 disorders or disorder groups: insomnia disorder, hypersomnolence disorder, narcolepsy, breathing-related sleep disorders, circadian rhythm sleep-wake disorders, non-rapid eye movement (NREM) sleep arousal disorders, nightmare disorder, rapid eye movement (REM) sleep behavior disorder, restless legs syndrome, and substance/medication-induced sleep disorder. Individuals with these disorders typically present with sleep-wake complaints of dissatisfaction regarding the quality, timing, and amount of sleep. Resulting daytime distress and impairment are core features shared by all of these sleep-wake disorders.

The organization of this chapter is designed to facilitate differential diagnosis of sleep-wake complaints and to clarify when referral to a sleep specialist is appropriate for further assessment and treatment planning. The DSM-5 sleep disorders nosology uses a simple, clinically useful approach, while also reflecting scientific advances in epidemiology, genetics, pathophysiology, assessment, and interventions research since DSM-IV. The approach taken to the classification of sleep-wake disorders in DSM-5 can be understood within the context of “lumping versus splitting.” For example, in some categories (e.g., insomnia disorder), a “lumping” approach has been adopted (i.e., three categories that were separate in DSM-IV—insomnia with other mental disorders, insomnia with other medical conditions, and insomnia with other sleep disorders—are all included in the single insomnia category as specifiers), whereas in other categories (e.g., narcolepsy), a “splitting” approach has been taken (i.e., there are four separately coded subtypes of narcolepsy, such as type 1 with cataplexy or hypocretin deficiency, and type 2 without cataplexy and either without hypocretin deficiency or hypocretin unmeasured), reflecting the availability of validators derived from epidemiological, neurobiological, and interventions research.

Because DSM-5 is intended for use by mental health and general medical clinicians who are not experts in sleep medicine, DSM-5 presents an effort to simplify sleep-wake disorders classification and thus aggregates diagnoses under broader, less differentiated labels. In contrast, the *International Classification of Sleep Disorders*, 3rd Edition (ICSD-3), elaborates numerous diagnostic subtypes, reflects the science and opinions of the sleep specialist community, and has been prepared by and for sleep specialists.

The simpler, less-differentiated approach to the diagnosis of sleep-wake disorders in DSM-5 shows superior interrater reliability, as well as convergent, discriminant, and face validity. The text accompanying each set of diagnostic criteria provides linkages to the corresponding disorders included in ICSD-3.

The field of sleep disorders medicine has progressed in this direction since the publication of DSM-IV. The use of biological validators is now embodied in the DSM-5 classification of sleep-

wake disorders, particularly for disorders of excessive sleepiness, such as narcolepsy, for which cerebrospinal fluid hypocretin-1 immunoreactivity values can be diagnostic; for breathing-related sleep disorders, for which formal sleep studies (i.e., polysomnography) are indicated; and for restless legs syndrome, which can often coexist with periodic limb movements during sleep, detectable via polysomnography.

Co-Occurring Disorders and Differential Diagnosis

Sleep disorders are often accompanied by depression, anxiety, and cognitive changes that must be addressed in treatment planning and management. Furthermore, persistent sleep disturbances (both insomnia and excessive sleepiness) are established risk factors for the subsequent development of mental illnesses (including substance use and non-substance use disorders) and other medical conditions. They may also represent a prodromal expression of an episode of mental illness, allowing the possibility of early intervention to preempt or to attenuate a full-blown episode.

The differential diagnosis of sleep-wake complaints necessitates a multidimensional approach, with consideration of possibly coexisting clinical conditions, which are the rule and not the exception. Sleep disturbances furnish a clinically useful indicator of clinical conditions that often coexist with depression and other common mental disorders. Prominent among these comorbidities are breathing-related sleep disorders, cardiac and pulmonary conditions (e.g., congestive heart failure, chronic obstructive pulmonary disease), neurodegenerative disorders (e.g., Alzheimer's disease), and disorders of the musculoskeletal system (e.g., osteoarthritis). These disorders not only may disturb sleep but also may themselves be worsened during sleep (e.g., prolonged apneas or electrocardiographic arrhythmias during REM sleep; confusional arousals in individuals with major neurocognitive disorder; seizures in individuals with complex partial seizures). REM sleep behavior disorder is often an early indicator of neurodegenerative disorders (alpha synucleinopathies) like Parkinson's disease. For all of these reasons—related to differential diagnosis, clinical comorbidity, and facilitation of treatment planning—sleep disorders are included in DSM-5.

Key Concepts and Terms

Four distinct sleep stages can be measured by polysomnography: REM sleep and three stages of NREM sleep (N1, N2, and N3).

- REM sleep, during which the majority of typical story-like dreams occur, occupies about 20%–25% of total sleep.
- NREM sleep stage 1 (N1) is a transition from wakefulness to sleep and occupies about 5% of time spent asleep in healthy adults.
- NREM sleep stage 2 (N2), which is characterized by specific electroencephalographic waveforms (sleep spindles and K complexes), occupies about 50% of time spent asleep.
- NREM sleep stage 3 (N3) (also known as slow-wave sleep) is the deepest level of sleep and occupies about 20% of sleep time in healthy, younger adults.

These sleep stages have a characteristic temporal organization across the night. N3 tends to occur in the first one-third to one-half of the night and increases in duration in response to sleep deprivation. REM sleep occurs cyclically throughout the night, alternating with NREM sleep about every 80–100 minutes. REM sleep periods increase in duration toward the morning.

Human sleep also varies characteristically across the life span. After relative stability with large amounts of slow-wave sleep in childhood and early adolescence, sleep continuity and depth deteriorate across the adult age range. This deterioration is reflected by increased wakefulness and N1 sleep and decreased N3 sleep. Because of this, age must be considered in the diagnosis of a sleep disorder in any individual.

Polysomnography is the monitoring of multiple electrophysiological parameters during sleep. Most polysomnographic studies are conducted during the individual's usual sleeping hours—that is, at night. However, daytime polysomnographic studies also are used to quantify daytime sleepiness. The most common daytime procedure is the multiple sleep

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latency test, in which the individual is instructed to lie down in a dark room and not resist falling asleep; this protocol is repeated five times during the day. The amount of time required to fall asleep (sleep latency) is measured on each trial and is used as an index of physiological sleepiness.

The following standard terminology for polysomnographic measures is used throughout the text in this chapter, and other terms provide context for the chapter discussion:

- *Sleep continuity* refers to the overall balance of sleep and wakefulness during a night of sleep. “Better” sleep continuity indicates consolidated sleep with little wakefulness or fragmentation; “worse” sleep continuity indicates disrupted sleep with more wakefulness and fragmentation.

Specific sleep continuity measures include *sleep latency*—the amount of time required to fall asleep (expressed in minutes); *wake after sleep onset*—the amount of awake time between initial sleep onset and final awakening (expressed in minutes); the *number of awakenings*; and *sleep efficiency*—the ratio of actual time spent asleep to time spent in bed (expressed as a percentage, with higher numbers indicating better sleep continuity).

- *Sleep architecture* refers to the amount and distribution of specific sleep stages. Sleep architecture measures include absolute amounts of REM sleep and each NREM sleep stage (in minutes), relative amount of REM sleep and NREM sleep stages (expressed as a percentage of total sleep time), and latency between sleep onset and the first REM period (*REM latency*). When the latency to onset of REM sleep is < 15 minutes, the terms *sleep-onset REM* and *sleep-onset REM period* are employed.

Association With Suicidal Thoughts or Behavior

A review of multiple studies found that the symptom of insomnia may increase the risk for suicidal thoughts, suicidal behavior, and death, even after adjustment for depression, and that nightmares increase risk for suicidal thoughts and behavior. In one study of college students, 31.3% of those with sleep problems had suicidal thoughts, and conversely, nearly all (82.7%) individuals with suicidal thoughts had sleep problems. A review and consensus statement of the American Academy of Sleep Medicine concluded that in teenagers, < 8 hours of sleep is

associated with increased risk of self-harm, suicidal thoughts, and suicidal behavior.

Insomnia Disorder

Diagnostic Criteria

F51.01

- A. A predominant complaint of dissatisfaction with sleep quantity or quality, associated with one (or more) of the following symptoms:
 - 1. Difficulty initiating sleep. (In children, this may manifest as difficulty initiating sleep without caregiver intervention.)
 - 2. Difficulty maintaining sleep, characterized by frequent awakenings or problems returning to sleep after awakenings. (In children, this may manifest as difficulty returning to sleep without caregiver intervention.)
 - 3. Early-morning awakening with inability to return to sleep.
- B. The sleep disturbance causes clinically significant distress or impairment in social, occupational, educational, academic, behavioral, or other important areas of functioning.
- C. The sleep difficulty occurs at least 3 nights per week.

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- D. The sleep difficulty is present for at least 3 months.
- E. The sleep difficulty occurs despite adequate opportunity for sleep.
- F. The insomnia is not better explained by and does not occur exclusively during the course of another sleep-wake disorder (e.g., narcolepsy, a breathing-related sleep disorder, a circadian rhythm sleep-wake disorder, a parasomnia).
- G. The insomnia is not attributable to the physiological effects of a substance (e.g., a drug of abuse, a medication).
- H. Coexisting mental disorders and medical conditions do not adequately explain the predominant complaint of insomnia.

Specify if:

With mental disorder, including substance use disorders

With medical condition

With another sleep disorder

Coding note: The code F51.01 applies to all three specifiers. Code also the relevant associated mental disorder, medical condition, or other sleep disorder immediately after the code for insomnia disorder in order to indicate the association.

Specify if:

Episodic: Symptoms last at least 1 month but less than 3 months.

Persistent: Symptoms last 3 months or longer.

Recurrent: Two (or more) episodes within the space of 1 year.

Note: Acute and short-term insomnia (i.e., symptoms lasting less than 3 months but otherwise meeting all criteria with regard to frequency, intensity, distress, and/or impairment) should be coded as an other specified insomnia disorder.

Note: The diagnosis of insomnia disorder is given whether it occurs as an independent condition or is comorbid with another mental disorder (e.g., major depressive disorder), medical condition (e.g., pain), or another sleep disorder (e.g., a breathing-related sleep disorder). For instance, insomnia may develop its own course with some anxiety and depressive features without those features meeting criteria for any one mental disorder. Insomnia may also manifest as a clinical feature of a more predominant mental disorder. Persistent insomnia is a risk factor for depression, anxiety disorders, and alcohol use disorder and is a common residual symptom after treatment for these conditions. When insomnia is comorbid with a mental disorder, treatment may need to target both conditions. Given these different courses, it is often impossible to establish the precise nature of the relationship between these clinical entities, and this relationship may change over time. Therefore, in the presence of insomnia and a comorbid disorder, it is not necessary to make a causal attribution between the two conditions. Rather, the diagnosis of insomnia disorder is made with concurrent specification of the comorbid conditions. A concurrent insomnia diagnosis should only be considered when the insomnia is sufficiently severe to warrant independent clinical attention; otherwise, no separate diagnosis is necessary.

Recording Procedures

The specifiers “with mental disorder, including substance use disorders”; “with medical condition”; and “with another sleep disorder” are available to allow the clinician to note clinically relevant comorbidities. In such cases, record F51.01 insomnia disorder, with [name of comorbid condition(s) or disorder(s)] followed by the diagnostic code(s) for the comorbid conditions or disorders (e.g., F51.01 insomnia disorder, with moderate cocaine use disorder and trigeminal neuralgia; F14.20 moderate cocaine use disorder; G50.0 trigeminal neuralgia).

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Diagnostic Features

The essential feature of insomnia disorder is dissatisfaction with sleep quantity or quality with complaints of difficulty initiating or maintaining sleep. The sleep complaints are accompanied by clinically significant distress or impairment in social, occupational, or other important areas of functioning. The sleep disturbance may occur during the course of another mental disorder or medical condition, or it may occur independently.

Different manifestations of insomnia can occur at different times of the sleep period. *Sleep-onset insomnia* (or *initial insomnia*) involves difficulty initiating sleep at bedtime. *Sleep-maintenance insomnia* (or *middle insomnia*) involves frequent or prolonged awakenings

throughout the night. *Late insomnia* involves early-morning awakening with an inability to return to sleep. Difficulty maintaining sleep is the most common single symptom of insomnia, affecting about 60% of those with insomnia, followed by early awakening and difficulty falling asleep, according to a U.S. national sample of health care plan members. A combination of these symptoms is the most common presentation overall. The specific type of sleep complaint often varies over time. Individuals who complain of difficulty falling asleep at one time may later complain of difficulty maintaining sleep, and vice versa. Symptoms of difficulty falling asleep and difficulty maintaining sleep can be quantified by the individual's retrospective self-report, sleep diaries in which information is collected prospectively, or other methods, such as actigraphy or polysomnography. However, the diagnosis of insomnia disorder is based on the individual's subjective perception of sleep or a caretaker's report. Subjective reports from individuals with insomnia disorder frequently indicate longer sleep latencies, greater time awake during the night, and less total sleep time than objective (e.g. polysomnographic) data demonstrate. The reasons for this discrepancy are not well understood, but disturbances in the underlying neurophysiology reflective of hyperarousal or cortical activation are believed to play a role.

Nonrestorative sleep, a complaint of poor sleep quality that does not leave the individual rested upon awakening despite adequate duration, is a common sleep complaint usually occurring in association with difficulty initiating or maintaining sleep, or less frequently in isolation. The precise relationship of isolated nonrestorative sleep to insomnia disorder remains unclear. The prevalence of isolated nonrestorative sleep has been estimated at about 5% and, unlike insomnia complaints, is reported more commonly in younger individuals. This complaint can also be reported in association with another sleep disorder (e.g., breathing-related sleep disorder). When a complaint of nonrestorative sleep occurs in isolation (i.e., in the absence of difficulty initiating and/or maintaining sleep or other sleep-wake disorders), a diagnosis of other specified sleep-wake disorder is made.

Aside from the frequency and duration criteria required to make the diagnosis, additional guidelines are useful to quantify insomnia severity. These quantitative guidelines, while arbitrary, are provided for illustrative purpose only. For instance, difficulty initiating sleep is defined by a subjective sleep latency > 20–30 minutes, and difficulty maintaining sleep is defined by a subjective time awake after sleep onset > 20–30 minutes. Although there is no standard definition of early-morning awakening, this symptom involves awakening at least 1 hour before the scheduled time and before total sleep time reaches 6½ hours. It is essential to take into account not only the final awakening time but also the bedtime on the previous evening. Awakening at 4:00 A.M. does not have the same clinical significance in those who go to bed at 9:00 P.M. as in those who go to bed at 11:00 P.M. Such a symptom may also reflect an age-dependent decrease in the ability to sustain sleep or an age-dependent shift in the timing of the main sleep period. Although these quantitative criteria are frequently employed in research designs, they do not in their own right reliably distinguish individuals with insomnia from normal sleepers. Moreover, individuals whose presentations no longer meet subjective diagnostic criteria for insomnia disorder may continue to show objective disturbance by these parameters, as well as daytime impairment.

Insomnia disorder involves daytime impairments as well as nighttime sleep difficulties. These include fatigue or, less commonly, daytime sleepiness; the latter is more common among older individuals and when insomnia is comorbid with another medical condition (e.g., chronic pain) or sleep disorder (e.g., sleep apnea). Impairment in cognitive performance may include difficulties with attention, concentration and memory, and performing complex manual skills. Associated mood disturbances are typically described as irritability or mood lability and less commonly as depressive or anxiety symptoms. Not all individuals with nighttime sleep disturbances are distressed or have functional impairment. For example, sleep continuity is often interrupted in healthy older adults who nevertheless identify themselves as good sleepers. A diagnosis of insomnia disorder should be reserved for those individuals with significant daytime distress or impairment related to their nighttime sleep difficulties.

Associated Features

Insomnia is often associated with physiological and cognitive arousal and conditioning factors that interfere with sleep. A preoccupation with sleep and distress attributable to the inability to sleep may lead to a vicious cycle: the more the individual strives to sleep, the more frustration builds and further impairs sleep. Thus, excessive attention and efforts to sleep, which override normal sleep-onset mechanisms, may contribute to the development of insomnia. Individuals with persistent insomnia may also acquire maladaptive sleep habits (e.g., spending excessive time in bed; following an erratic sleep schedule; napping) and cognitions (e.g., fear of sleeplessness; apprehensions of daytime impairments; clock monitoring) during the course of the disorder. Engaging in such activities in an environment in which the individual has frequently spent sleepless nights may further compound the conditioned arousal and perpetuate sleep difficulties. Conversely, the individual may fall asleep more easily when not trying to do so. Some individuals also report better sleep when away from their own bedrooms and their usual routines.

Insomnia may be accompanied by a variety of daytime complaints and symptoms, including fatigue, decreased energy, and mood disturbances. Individuals with insomnia disorder may appear either fatigued or haggard or, conversely, overaroused and “wired.” There may be an increased incidence of stress-related psychophysiological symptoms (e.g., tension headache, muscle tension or pain, gastrointestinal symptoms); however, there are no consistent or characteristic abnormalities on physical examination. Symptoms of anxiety or depression that do not meet criteria for a specific mental disorder may be present, as well as an excessive focus on the perceived effects of sleep loss on daytime functioning.

Individuals with insomnia may have elevated scores on self-report psychological or personality inventories with profiles indicating mild depression and anxiety, a worrisome cognitive style, an emotion-focused and internalizing style of conflict resolution, and a somatic focus. Patterns of neurocognitive impairment among individuals with insomnia disorder are inconsistent, although there may be impairments in performing tasks of higher complexity and those requiring frequent changes in performance strategy. Individuals with insomnia often require more effort to maintain cognitive performance.

Prevalence

Population-based estimates vary, depending on the sample and the criteria employed, but

indicate that, across multiple countries, about one-third of adults report insomnia symptoms, 10%–15% experience associated daytime impairments, and 4%–22% have symptoms that meet criteria for insomnia disorder, with an average of about 10%. Insomnia disorder is the most prevalent of all sleep disorders. In primary care settings cross-nationally, approximately 20%–40% of individuals complain of significant insomnia symptoms. Prevalence rates for medical and psychiatric populations are significantly higher than those in the general population,

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especially among individuals with mood, anxiety, and substance use disorders. Forty to fifty percent of individuals with an insomnia disorder have a comorbid mental disorder. Insomnia is a more prevalent complaint among women than among men, with a gender ratio of about 1.3:1 in multinational samples. The gender ratio rises to 1.7:1 after age 45. The prevalence in Norway among older adolescents (16–18 years) is nearly double in girls compared with boys. Although insomnia can be a symptom or an independent disorder, it is most frequently observed as a comorbid condition with another medical condition or mental disorder.

Development and Course

The onset of insomnia symptoms can occur at any time during life, but the first episode is more common in young adulthood. Less frequently, insomnia begins in childhood or adolescence. In women, the incidence of new-onset insomnia increases with menopause and may persist even after other symptoms (e.g., hot flashes) have resolved. Insomnia may have a late-life onset, which is often associated with the onset of other health-related conditions.

Insomnia can be situational, persistent, or recurrent. Situational or acute insomnia usually lasts a few days or a few weeks and is often associated with life events or rapid changes in sleep schedules or environment. It usually resolves once the initial precipitating event subsides. For some individuals, perhaps those more vulnerable to sleep disturbances, insomnia may persist long after the initial triggering event, possibly because of conditioning factors and heightened arousal. The factors that precipitate insomnia may differ from those that perpetuate it. For example, an individual who is bedridden with a painful injury and has difficulty sleeping may then develop negative associations for sleep. Conditioned arousal may then persist and lead to persistent insomnia. A similar course may develop in the context of an acute psychological stress or a mental disorder. For instance, insomnia that occurs during an episode of major depressive disorder can become a focus of attention, with consequent negative conditioning, and persist even after resolution of the depressive episode in at least 40%–50% of individuals. In some cases, insomnia may also have an insidious onset without any identifiable precipitating factor.

The course of insomnia may also be episodic, with recurrent episodes of sleep difficulties associated with the occurrence of stressful events. Chronicity rates range from 45% to 75% for follow-ups of 1–7 years. Even when the course of the insomnia has become chronic, there is night-to-night variability in sleep patterns, with an occasional restful night's sleep interspersed with several nights of poor sleep. The characteristics of insomnia may also change over time. Many individuals with insomnia have a history of “light” or easily disturbed sleep prior to onset of more persistent sleep problems.

Insomnia complaints are more prevalent among middle-age and older adults. The type of

insomnia symptom changes as a function of age, with difficulties initiating sleep being more common among young adults and problems maintaining sleep occurring more frequently among middle-age and older individuals.

Difficulties initiating and maintaining sleep can also occur in children and adolescents, but there are more limited data on prevalence, risk factors, and comorbidity during these developmental phases of the life span. Sleep difficulties in childhood can result from conditioning factors (e.g., a child who does not learn to fall asleep or return to sleep without the presence of a parent) or from the absence of consistent sleep schedules and bedtime routines. Insomnia in adolescence is often triggered or exacerbated by irregular sleep schedules, especially phase delay. In both children and adolescents, psychological and medical factors can contribute to insomnia.

The increased prevalence of insomnia in older adults is partly explained by the higher incidence of physical health problems with aging. Changes in sleep patterns associated with the normal developmental process must be differentiated from those exceeding age-related changes. Older individuals may experience significant delays in sleep onset or frequent awakenings that are not associated with complaints or daytime consequences. Although

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polysomnography is of limited value in the routine evaluation of insomnia, it may be more useful in the differential diagnosis among older adults because comorbid conditions associated with insomnia (e.g., sleep apnea) are more common in older individuals.

Risk and Prognostic Factors

While the risk and prognostic factors discussed in this section increase vulnerability to insomnia, sleep disturbances are more likely to occur when predisposed individuals are exposed to precipitating events, such as major life events (e.g., illness, separation) or less severe but more chronic daily stress. Most individuals resume normal sleep patterns after the initial triggering event has disappeared, but others—perhaps those more vulnerable to insomnia—continue experiencing persistent sleep difficulties. Perpetuating factors such as poor sleep habits, irregular sleep scheduling, and the fear of not sleeping feed into the insomnia problem and may contribute to a vicious cycle that may induce persistent insomnia.

Temperamental. Anxiety or worry-prone personality or cognitive styles, increased arousal predisposition, higher stress reactivity, and tendency to repress emotions can increase vulnerability to insomnia.

Environmental. Noise, light, or uncomfortably high or low temperature may increase vulnerability to insomnia. High altitude may also predispose to insomnia attributable to periodic breathing difficulties during sleep.

Genetic and physiological. Female sex and advancing age are associated with increased vulnerability to insomnia. Disrupted sleep and insomnia display a familial disposition. Thirty-five percent to seventy percent of those with insomnia disorder report one or more first-degree relatives (most commonly, the mother) with a history of insomnia. Heritability may be highest for insomnia disorder without comorbidities. The prevalence of insomnia is higher among monozygotic twins relative to dizygotic twins; it is also higher in first-degree family members

compared with the general population. The extent to which this link is inherited through a genetic predisposition, learned by observations of parental models, or established as a by-product of another psychopathology remains undetermined, although sleep reactivity to stress appears to play some role.

Course modifiers. Deleterious course modifiers include poor sleep hygiene practices (e.g., excessive caffeine use, irregular sleep schedules).

Culture-Related Diagnostic Issues

Insomnia is a universal human experience. Identification of insomnia as a problem, the explanatory models for the condition, and associated help-seeking choices are affected by culture. Insomnia may be understood as a normal part of aging or of stress response, leading to low help-seeking or to coping via social support and traditional activities such as prayer. Explanatory models of insomnia vary greatly, including attributions to the effect of the environment (e.g., humidity) and bodily processes (e.g., poor blood circulation, internal heat) among others, and may be associated with nonbiomedical treatment-seeking.

Sex- and Gender-Related Diagnostic Issues

First onset in females is often associated with the birth of a new child or with menopause. Despite higher prevalence among perimenopausal and postmenopausal females, polysomnographic studies suggest better preservation of sleep continuity and slow-wave sleep in older females than in older males.

Diagnostic Markers

Polysomnography usually shows impairments of sleep continuity (e.g., increased sleep latency and time awake after sleep onset and decreased sleep efficiency [percentage of time

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in bed asleep]) and may show increased stage 1 sleep and decreased stage 3 sleep. The severity of these sleep impairments does not always match the individual's clinical presentation or subjective complaint of poor sleep, as individuals with insomnia often underestimate sleep duration and overestimate wakefulness relative to polysomnography. Quantitative electroencephalographic analyses may indicate that individuals with insomnia have greater high-frequency electroencephalography power relative to good sleepers both around the sleep-onset period and during non-rapid eye movement sleep, although findings vary according to age and gender. This feature is consistent with increased cortical arousal. Neuroimaging studies have suggested altered regional brain function consistent with hyperarousal in insomnia, although interpretation of these findings is complex. Individuals with insomnia disorder may have a lower sleep propensity and typically do not show increased daytime sleepiness on objective sleep laboratory measures compared with individuals without sleep disorders.

Other laboratory measures show evidence, although not consistently, of increased arousal and a generalized activation of the hypothalamic-pituitary-adrenal axis (e.g., increased cortisol levels, heart rate variability, reactivity to stress, increased metabolic rate). In general, findings are consistent with the hypothesis that increased physiological and cognitive arousal plays a

significant role in insomnia disorder.

Association With Suicidal Thoughts or Behavior

The symptom of insomnia has been identified as an independent risk factor for suicidal thoughts and behavior.

Functional Consequences of Insomnia Disorder

Interpersonal, social, and occupational problems may develop as a result of insomnia or excessive concern with sleep, increased daytime irritability, and poor concentration. Decreased attention and concentration are common and may be related to higher rates of accidents observed in individuals with insomnia. Persistent insomnia is also associated with long-term consequences, including twofold or greater increased risk of new-onset major depressive disorder, anxiety disorders, and substance use disorders. Insomnia symptoms may also be a risk factor for relapse of major depressive disorder. Insomnia disorder, especially with objectively demonstrated short-sleep duration (<6 hours), is a significant risk factor for numerous cardiovascular diseases, including hypertension, coronary artery disease/myocardial infarction, congestive heart failure, and cerebrovascular disease. Increased absenteeism and reduced productivity at work, reduced quality of life, and increased economic burden are also significant functional consequences of insomnia disorder.

Differential Diagnosis

Normal sleep variations. Normal sleep duration varies considerably across persons. Some individuals who require little sleep (“short sleepers”) may be concerned about their sleep duration. Short sleepers differ from individuals with insomnia disorder by the lack of difficulty falling or staying asleep and by the absence of characteristic daytime symptoms (e.g., fatigue, concentration problems, irritability). However, some short sleepers may desire or attempt to sleep for a longer period of time and, by prolonging time in bed, may create an insomnia-like sleep pattern. Clinical insomnia also should be distinguished from normal, age-related sleep changes. Insomnia must also be distinguished from sleep deprivation attributable to inadequate opportunity or circumstance for sleep resulting, for example, from an emergency or from professional or family obligations forcing the individual to stay awake.

Situational/acute insomnia. *Situational/acute insomnia* is a condition lasting a few days to several weeks, often associated with acute stress due to life events or with changes in

sleep schedules. These acute or short-term insomnia symptoms may also produce significant distress and interfere with social, personal, and occupational functioning. When such symptoms are frequent enough and meet all other criteria except for the 3-month duration, a diagnosis of other specified insomnia disorder or unspecified insomnia disorder is made. While the disorder often remits with subsidence of the stress or adjustment to the change in sleep schedule, some individuals will develop maladaptive patterns of thought and behavior that result in the development of a chronic insomnia disorder.

Delayed sleep phase and shift work types of circadian rhythm sleep-wake disorder. Individuals with the delayed sleep phase type of circadian rhythm sleep-wake disorder report sleep-onset insomnia only when they try to sleep at socially normal times, but they do not report difficulty falling asleep or staying asleep when their bed and rising times are delayed and coincide with their endogenous circadian rhythm. This pattern is observed particularly among adolescents and younger adults. Shift work type differs from insomnia disorder by the history of recent shift work.

Restless legs syndrome. Restless legs syndrome often produces difficulties initiating and maintaining sleep. However, an urge to move the legs and any accompanying unpleasant leg sensations are features that differentiate this disorder from insomnia disorder.

Breathing-related sleep disorders. Most individuals with a breathing-related sleep disorder have a history of loud snoring, breathing pauses during sleep, and excessive daytime sleepiness. Nonetheless, as many as 50% of individuals with sleep apnea may also report insomnia symptoms, a feature that is more common among women and older adults.

Narcolepsy. Narcolepsy may cause insomnia complaints but is distinguished from insomnia disorder by the predominance of symptoms of excessive daytime sleepiness, cataplexy, sleep paralysis, and sleep-related hallucinations.

Parasomnias. Parasomnias are characterized by a complaint of unusual behavior or events during sleep that may lead to intermittent awakenings and difficulty resuming sleep. However, it is these behavioral events, rather than the insomnia per se, that dominate the clinical picture.

Substance/medication-induced sleep disorder, insomnia type. Substance/medication-induced sleep disorder, insomnia type, is distinguished from insomnia disorder by the fact that a substance (i.e., a drug of abuse, a medication, or exposure to a toxin) is judged to be etiologically related to the insomnia (see “Substance/Medication-Induced Sleep Disorder” later in this chapter). For example, insomnia occurring only in the context of heavy coffee consumption would be diagnosed as caffeine-induced sleep disorder, insomnia type, with onset during intoxication.

Comorbidity

Insomnia is a common comorbidity of many medical conditions, including but not limited to cancer, diabetes, coronary heart disease, chronic obstructive pulmonary disease, arthritis, fibromyalgia, other chronic pain conditions, degenerative brain diseases, and traumatic brain injury. The risk relationship appears to be bidirectional: insomnia increases the risk of many of these medical conditions, and medical problems increase the risk of insomnia. The direction of the relationship is not always clear and may change over time; for this reason, comorbid insomnia is the preferred terminology when insomnia coexists with another medical condition (or mental disorder). Insomnia disorder also coexists with numerous other sleep disorders. Approximately one in seven individuals with insomnia disorder has moderate to severe obstructive sleep apnea. Rates of insomnia complaints among individuals with narcolepsy are estimated to be about 50%.

Individuals with insomnia disorder frequently have a comorbid mental disorder, particularly

bipolar, depressive, and anxiety disorders. Persistent insomnia represents a risk factor or an early symptom of subsequent bipolar, depressive, anxiety, and substance use disorders. Individuals with insomnia may misuse medications or alcohol to help with nighttime sleep, anxiolytics to combat tension or anxiety, and caffeine or other stimulants to combat excessive daytime fatigue. In addition to worsening the insomnia, this type of substance use may in some cases progress to a substance use disorder.

Relationship to International Classification of Sleep Disorders

The *International Classification of Sleep Disorders*, 3rd Edition (ICSD-3), recognizes three insomnia diagnoses: *chronic insomnia disorder*, *short-term insomnia disorder*, and *other insomnia disorder*. DSM-5 insomnia disorder and ICSD-3 chronic insomnia disorder closely parallel each other with respect to symptom, duration, and frequency criteria; however, unlike DSM-5, ICSD-3 does not include a separate designation for substance/medication-induced sleep disorder, insomnia type.

Hypersomnolence Disorder

Diagnostic Criteria

F51.11

- A. Self-reported excessive sleepiness (hypersomnolence) despite a main sleep period lasting at least 7 hours, with at least one of the following symptoms:
 - 1. Recurrent periods of sleep or lapses into sleep within the same day.
 - 2. A prolonged main sleep episode of more than 9 hours per day that is nonrestorative (i.e., unrefreshing).
 - 3. Difficulty being fully awake after abrupt awakening.
- B. The hypersomnolence occurs at least three times per week, for at least 3 months.
- C. The hypersomnolence is accompanied by significant distress or impairment in cognitive, social, occupational, or other important areas of functioning.
- D. The hypersomnolence is not better explained by and does not occur exclusively during the course of another sleep disorder (e.g., narcolepsy, breathing-related sleep disorder, circadian rhythm sleep-wake disorder, or a parasomnia).
- E. The hypersomnolence is not attributable to the physiological effects of a substance (e.g., a drug of abuse, a medication).
- F. Coexisting mental and medical disorders do not adequately explain the predominant complaint of hypersomnolence.

Specify if:

With mental disorder, including substance use disorders

With medical condition

With another sleep disorder

Coding note: The code F51.11 applies to all three specifiers. Code also the relevant associated mental disorder, medical condition, or other sleep disorder immediately after the code for hypersomnolence disorder in order to indicate the association.

Specify if:

Acute: Duration of less than 1 month.

Subacute: Duration of 1–3 months.

Persistent: Duration of more than 3 months.

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Specify current severity:

Specify severity based on degree of difficulty maintaining daytime alertness as manifested by the occurrence of multiple attacks of irresistible sleepiness within any given day occurring, for example, while sedentary, driving, visiting with friends, or working.

Mild: Difficulty maintaining daytime alertness 1–2 days/week.

Moderate: Difficulty maintaining daytime alertness 3–4 days/week.

Severe: Difficulty maintaining daytime alertness 5–7 days/week.

Recording Procedures

The specifiers “with mental disorder, including substance use disorders”; “with medical condition”; and “with another sleep disorder” are available to allow the clinician to note clinically relevant comorbidities. In such cases, record F51.11 hypersomnolence disorder, with [name of comorbid condition(s) or disorder(s)] followed by the diagnostic code(s) for the comorbid conditions or disorders (e.g., F51.11 hypersomnolence disorder, with major depressive disorder; F33.1 major depressive disorder, recurrent, moderate).

Diagnostic Features

Hypersomnolence disorder includes symptoms of excessive quantity of sleep (e.g., extended nocturnal sleep or long naps), sleepiness, and *sleep inertia* (i.e., a period of impaired performance and reduced vigilance following awakening from the regular sleep episode or from a nap) (Criterion A). Individuals with this disorder generally fall asleep quickly and have a good sleep efficiency (> 90%). Individuals typically feel sleepiness developing over a period of time, rather than experiencing a sudden sleep “attack.” Unintentional sleep episodes typically occur in sedentary situations (e.g., while attending lectures, reading, watching television, or driving long distances), but in more severe cases they can manifest in high-attention situations such as at work, in meetings, or at social gatherings. The persistent need for sleep can lead to automatic behavior (usually of a very routine, low-complexity type) that the individual carries out with little or no subsequent recall. For example, individuals may find themselves having driven several miles from where they thought they were, unaware of the “automatic” driving they did in the preceding minutes.

About 40% of individuals with hypersomnolence disorder may have sleep inertia (also referred to as “sleep drunkenness”), and this symptom may help differentiate hypersomnolence disorder from other causes of sleepiness. They may have difficulty waking up in the morning, sometimes appearing confused, combative, or ataxic. Individuals may set multiple alarm clocks or rely on others to help get them out of bed. Sleep inertia can also occur upon awakening from a daytime nap. During that period, the individual appears awake, but motor coordination is impaired, behavior may be inappropriate, and memory deficits, disorientation in time and space, and feelings of grogginess may occur. This period may last some minutes to hours.

For some individuals with hypersomnolence disorder, the major sleep episode (for most individuals, nocturnal sleep) has a duration of 9 hours or more. In the most extreme cases, sleep episodes can last up to 20 hours. However, the sleep is often nonrestorative and is followed by difficulty awakening in the morning. For other individuals with hypersomnolence disorder, the major sleep episode is of normal nocturnal sleep duration (7–9 hours), and they take relatively long daytime naps (>1 hour) that do not improve alertness. Most individuals with hypersomnolence disorder take daytime naps nearly every day regardless of the nocturnal sleep duration. While many individuals with hypersomnolence are able to reduce their sleep time during working days, weekend and holiday sleep is greatly increased (by up to 3 hours).

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Associated Features

Approximately 80% of individuals with hypersomnolence disorder report that their sleep is nonrestorative, but this symptom is nonspecific and can occur with disorders that disrupt sleep, such as obstructive sleep apnea. Naps are often long (>1 hour) and unrefreshing. Short naps (i.e., duration of <30 minutes) are often unrefreshing. Individuals with hypersomnolence often appear sleepy and may even fall asleep in the clinician’s waiting area.

A subset of individuals with hypersomnolence disorder have a family history of hypersomnolence and also have symptoms of autonomic nervous system dysfunction, including recurrent vascular-type headaches, reactivity of the peripheral vascular system (Raynaud’s phenomenon), and fainting.

Prevalence

Approximately 5%–10% of individuals in the United States who consult in sleep disorder clinics with complaints of daytime sleepiness are diagnosed as having hypersomnolence disorder. It is estimated that about 1% of the European and U.S. general population has episodes of sleep inertia. Hypersomnolence occurs with relatively equal frequency in men and women.

Development and Course

Hypersomnolence disorder usually begins in late adolescence or early adulthood, with a mean age at onset of 17–24 years and a gradual progression over weeks to months. Little is known of the natural history, but for most individuals, the symptoms are persistent and stable, unless treatment is initiated. Spontaneous remission occurs in about 11%–25% of individuals after 5–7 years. Individuals with hypersomnolence disorder are diagnosed, on average, 10–15 years after the appearance of the first symptoms. Pediatric cases are rare. The development of other sleep

disorders (e.g., breathing-related sleep disorder) may worsen the degree of sleepiness.

Risk and Prognostic Factors

Environmental. Hypersomnolence can be increased temporarily by psychological stress and alcohol use, but they have not been documented as environmental precipitating factors. Viral infections have been reported to have preceded or accompanied hypersomnolence in about 10% of cases. Hypersomnolence is common in the months after traumatic brain injury.

Genetic and physiological. Hypersomnolence may be familial, with an autosomal-dominant mode of inheritance.

Diagnostic Markers

Nocturnal polysomnography demonstrates a normal to prolonged sleep duration, short sleep latency, and normal to increased sleep continuity. The nocturnal distribution of rapid eye movement (REM) sleep is also normal. Sleep efficiency is typically > 90%. The multiple sleep latency test documents sleep tendency, typically indicated by mean sleep latency values of < 8 minutes. In hypersomnolence disorder, the mean sleep latency is typically < 10 minutes and frequently 8 minutes or less. Sleep-onset REM periods (i.e., the occurrence of REM sleep within 20 minutes of sleep onset) may be present but occur infrequently. Unfortunately, the multiple sleep latency test has poor test-retest reliability, and it does not distinguish well between hypersomnolence disorder and narcolepsy type 2.

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A 2-week sleep diary can help document amounts and timing of sleep, and actigraphy provides more accurate data on habitual sleep patterns. In a 32-hour bed rest protocol in which subjects were encouraged to sleep ad lib, individuals with hypersomnolence disorder slept > 4 hours more than control subjects.

Functional Consequences of Hypersomnolence Disorder

The low level of alertness that occurs while an individual fights the need for sleep can lead to reduced efficiency, diminished concentration, and poor memory during daytime activities. Hypersomnolence can lead to significant distress and dysfunction in work and social relationships. Prolonged nocturnal sleep and difficulty awakening can result in difficulty in meeting morning obligations, such as arriving at work on time. Unintentional daytime sleep episodes can be embarrassing and even dangerous, if, for instance, the individual is driving or operating machinery when the episode occurs.

Differential Diagnosis

Normative variation in sleep. “Normal” sleep duration varies considerably in the general population. “Long sleepers” (i.e., persons who require a greater than average amount of sleep) do not have excessive sleepiness, sleep inertia, or automatic behavior when they obtain their required amount of nocturnal sleep. Sleep is reported to be refreshing. If social or occupational demands lead to shorter nocturnal sleep, daytime symptoms may appear. In hypersomnolence

disorder, by contrast, symptoms of excessive sleepiness occur regardless of nocturnal sleep duration.

An inadequate amount of nocturnal sleep, or *behaviorally induced insufficient sleep syndrome*, can produce symptoms of daytime sleepiness very similar to those of hypersomnolence disorder. An average sleep duration of fewer than 7 hours per night strongly suggests inadequate nocturnal sleep, yet in the United States, the average adult obtains only 6.75 hours of sleep on typical weeknights. Individuals with inadequate nocturnal sleep typically “catch up” with longer sleep durations on days when they are free from social or occupational demands or on vacations. A diagnosis of hypersomnolence disorder should not be made if there is a question regarding the adequacy of nocturnal sleep duration. A diagnostic and therapeutic trial of sleep extension for 10–14 days can often clarify the diagnosis.

Narcolepsy. As in hypersomnolence disorder, individuals with narcolepsy have chronic sleepiness, but several clinical and laboratory findings help distinguish the disorders. In contrast to those with hypersomnolence disorder, individuals with narcolepsy tend to sleep 7–8 hours each day and generally feel refreshed on waking in the morning. Individuals with narcolepsy generally feel more alert after a 15- to 20-minute nap, whereas those with hypersomnolence disorder tend to take longer naps, have trouble waking from naps, and do not feel alert afterward. Individuals with narcolepsy also have varying amounts of cataplexy, hypnagogic hallucinations, sleep paralysis, and fragmented nocturnal sleep, whereas cataplexy never occurs in hypersomnolence disorder and the other symptoms are uncommon. The multiple sleep latency test typically shows more than two sleep-onset REM periods in narcolepsy.

Fatigue as a symptom of another mental disorder or medical condition. Hypersomnolence disorder should be distinguished from tiredness related to fatigue that may be a symptom of another mental disorder (e.g., generalized anxiety disorder) or medical condition (e.g., chronic fatigue syndrome). Unlike hypersomnolence, tiredness is not necessarily relieved by increased sleep and is unrelated to sleep quantity or quality.

Breathing-related sleep disorders. Chronic sleepiness is common in breathing-related sleep disorders. Individuals with hypersomnolence and breathing-related sleep disorders may have similar patterns of excessive sleepiness. Breathing-related sleep disorders are suggested by a history of loud snoring, pauses in breathing during sleep, and nonrefreshing sleep. Examination often reveals obesity, a small airway, and large neck diameter. Hypertension is common, and some individuals may demonstrate signs of heart failure. Polysomnographic studies can confirm the presence of apneic events in breathing-related sleep disorder (and their absence in hypersomnolence disorder).

Circadian rhythm sleep-wake disorders. In contrast to individuals with hypersomnolence disorder, individuals with specific subtypes of circadian rhythm sleep-wake disorder show specific temporal patterns of symptoms. For example, individuals with delayed sleep phase type often have sleep inertia and sleepiness in the morning and feel most alert in the evening and night, with habitually late bedtimes. In contrast, those with advanced sleep phase type become sleepy and go to bed early in the evening but are alert and wake easily in the early morning.

Parasomnias. Parasomnias such as non-REM sleep arousal disorders (sleepwalking/sleep terrors)

or REM sleep behavior disorder rarely produce the prolonged, undisturbed nocturnal sleep or daytime sleepiness characteristic of hypersomnolence disorder. However, parasomnias such as nightmare disorder, which may result in significant curtailment of total sleep time, may conceivably manifest with daytime sleepiness.

Hypersomnolence in other mental disorders and medical conditions. Hypersomnolence disorder must be distinguished from hypersomnolence occurring as a symptom of another mental disorder (e.g., major depressive episode, especially episodes with atypical features) or medical condition (e.g., certain cancers, multiple sclerosis). If the predominant complaint of excessive sleepiness is adequately explained by another mental disorder or medical condition, then an additional diagnosis of hypersomnolence disorder is not warranted. However, if the hypersomnolence is not adequately explained by a comorbid mental disorder or medical condition (e.g., the severity and nature of the hypersomnolence far exceed what would be expected with the mental disorder or medical condition), an additional diagnosis of hypersomnolence disorder is warranted.

Comorbidity

Many individuals with hypersomnolence disorder have symptoms of depression that may meet criteria for a depressive disorder. This presentation may be related to the psychosocial consequences of persistent increased sleep need. More than half of individuals with hypersomnolence disorder have attention-deficit/hyperactivity disorder symptoms. Individuals with hypersomnolence disorder are also at risk for substance-related disorders, particularly related to self-medication with stimulants. This general lack of specificity may contribute to very heterogeneous profiles among individuals whose symptoms meet the diagnostic criteria for hypersomnolence disorder. Neurodegenerative conditions, such as Alzheimer's disease, Parkinson's disease, and multiple system atrophy, may also be associated with hypersomnolence.

Relationship to International Classification of Sleep Disorders

The *International Classification of Sleep Disorders*, 3rd Edition (ICSD-3), differentiates nine subtypes of "central disorders of hypersomnolence," including disorders not covered in DSM such as Kleine-Levin syndrome (recurrent episodes of hypersomnia), hypersomnolence due to a medical/neurological condition or substance use, and insufficient sleep syndrome.

Narcolepsy

Diagnostic Criteria

- A. Recurrent periods of an irrepressible need to sleep, lapsing into sleep, or napping occurring within the same day. These must have been occurring at least three times per week over the past 3 months.
- B. The presence of at least one of the following:

1. Episodes of cataplexy, defined as either (a) or (b), occurring at least a few times per month:
 - a. In individuals with long-standing disease, brief (seconds to minutes) episodes of sudden bilateral loss of muscle tone with maintained consciousness that are precipitated by laughter or joking.
 - b. In children or in individuals within 6 months of onset, spontaneous grimaces or jaw-opening episodes with tongue thrusting or a global hypotonia, without any obvious emotional triggers.
2. Hypocretin deficiency, as measured using cerebrospinal fluid (CSF) hypocretin-1 immunoreactivity values (less than or equal to one-third of values obtained in healthy subjects tested using the same assay, or less than or equal to 110 pg/mL). Low CSF levels of hypocretin-1 must not be observed in the context of acute brain injury, inflammation, or infection.
3. Nocturnal sleep polysomnography showing rapid eye movement (REM) sleep latency less than or equal to 15 minutes, or a multiple sleep latency test showing a mean sleep latency less than or equal to 8 minutes and two or more sleep-onset REM periods.

Specify whether:

G47.411 Narcolepsy with cataplexy or hypocretin deficiency (type 1): Criterion B1 (episodes of cataplexy) or Criterion B2 (low CSF hypocretin-1 levels) is met.

G47.419 Narcolepsy without cataplexy and either without hypocretin deficiency or hypocretin unmeasured (type 2): Criterion B3 (positive polysomnography/multiple sleep latency test) is met, but Criterion B1 is not met (i.e., no cataplexy is present) and Criterion B2 is not met (i.e., CSF hypocretin-1 levels are not low or have not been measured).

G47.421 Narcolepsy with cataplexy or hypocretin deficiency due to a medical condition

G47.429 Narcolepsy without cataplexy and without hypocretin deficiency due to a medical condition

Coding note: For the subtype narcolepsy with cataplexy or hypocretin deficiency due to a medical condition and the subtype narcolepsy without cataplexy and without hypocretin deficiency due to a medical condition, code first the underlying medical condition (e.g., G71.11 myotonic dystrophy; G47.429 narcolepsy without cataplexy and without hypocretin deficiency due to myotonic dystrophy).

Specify current severity:

Mild: Need for naps only once or twice per day. Sleep disturbance, if present, is mild. Cataplexy, when present, is infrequent (occurring less than once per week).

Moderate: Need for multiple naps daily. Sleep may be moderately disturbed. Cataplexy, when present, occurs daily or every few days.

Severe: Nearly constant sleepiness and, often, highly disturbed nocturnal sleep (which may include excessive body movement and vivid dreams). Cataplexy, when present, is drug-resistant, with multiple attacks daily.

Subtypes

A diagnosis of narcolepsy, type 1 (NT1; i.e., with cataplexy or hypocretin deficiency) is most often based on the presence of recurrent sleepiness and cataplexy (given the limited use of cerebrospinal fluid [CSF] hypocretin determinations). However, cataplexy can emerge years following onset of sleepiness. Therefore, some individuals may be initially assigned a diagnosis of narcolepsy, type 2 (NT2; i.e., without cataplexy and either without hypocretin deficiency or with hypocretin unmeasured), based on sleepiness and positive multiple sleep latency test (MSLT) findings, only to be reassigned to a diagnosis of NT1 following emergence of cataplexy. NT1 established by demonstration of low CSF hypocretin levels may manifest without evidence of clear cataplexy. Other explanations for excessive daytime sleepiness (e.g., sleep deprivation, shift work, other sleep disorders) and episodes of sudden loss of muscle tone (e.g., seizures, falls of other origin, functional neurological symptom disorder [conversion disorder]) should be ruled out. NT2 is established on the basis of chronic sleepiness and characteristic nocturnal sleep polysomnography findings (e.g., short REM sleep latency) or MSLT findings showing short mean sleep latency and two or more sleep-onset REM periods (SOREMPs).

NT1 and NT2 can result from other neurological, infectious, metabolic, and genetic conditions. Inherited disorders, tumors, and head trauma are the most common causes of secondary narcolepsy. In other cases, the destruction of hypocretin neurons may be secondary to trauma or hypothalamic surgery. Head trauma or infections of the central nervous system can, however, produce transitory decreases in CSF hypocretin-1 levels without hypocretin cell loss, complicating the diagnosis.

Other etiologies include inflammatory lesions due to multiple sclerosis and acute disseminated encephalomyelitis, vascular disorders such as stroke, and encephalitis. Autosomal dominant cerebellar ataxia, deafness, and narcolepsy, or ADCA DN, is a familial degenerative disorder due to missense mutations of the DNA methyltransferase (DNMT1) gene. Cataplexy with some degree of sleepiness can be caused by other neurological conditions, including Prader-Willi syndrome, Niemann-Pick disease type C, Möbius syndrome, and Norrie disease. Hypocretin deficiency has been reported in Parkinson's disease as well. NT2-like physiology has been reported in myotonic dystrophy and Prader-Willi syndrome.

Diagnostic Features

The essential features of narcolepsy are recurrent daytime naps or lapses into sleep that occur typically daily but that must occur at a minimum of three times a week for at least 3 months (Criterion A), and are accompanied by one or more of the following: cataplexy (Criterion B1), hypocretin deficiency (Criterion B2), or characteristic abnormalities on a nocturnal polysomnogram or on the MSLT (Criterion B3). In most individuals with NT1, the first symptom to manifest is sleepiness or increased sleep need, followed by cataplexy. Sleepiness is worse in sedentary circumstances and typically is relieved by brief (10–20 minutes) naps.

NT1 generally manifests with *cataplexy*, typically brief episodes (seconds up to 2 minutes) of